



GE VERNOVA

ADMS

16.25.1

Switching Order Client Configuration Editor User Guide

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Change History

Date	Change Description
May 8, 2023	ADMS 3.16: • Updated Print/Export Settings. • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Apr 16, 2024	ADMS 16.7.0: • Updated document format and styling.
Jul 15, 2024	ADMS 16.10.0: • Updated Device Navigation Options.
Dec 10, 2024	ADMS 16.15.0: • Updated Other Settings.
Apr 29, 2025	ADMS 16.17.0: • Updated Switch Steps Options.

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1. Introduction

Switching operations are the formal processes of performing switching changes on the network for the purposes of maintenance, network reconfiguration, restoration of unplanned outages, and augmentation of the distribution network. For more information about switching operations, refer to the *Switching Operations User Guide*.

This document describes the Switching Order Client Configuration Editor, which is used to modify the Switching Order application appearance and behavior.

For more information about configuring the Switching Order application workflow, refer to the *Switching Order Server Configuration Editor User Guide*.

2. General Usage

The switching operations user interface and functions are configurable by using the Switching Order Client Configuration Editor. This tool is used to modify an XML configuration file that is used by the ADMS client.

When the Configuration Editor is opened, the configuration file `SwitchOrderClientConfig.xml` can be loaded using the File menu.

After changes have been made to the configuration file, it should be saved and then distributed to each client by using the File Updater.

3. General Tab

The General tab is used to set the user interface appearance, switching order server connections, and the type of network view that is used for switching operations.

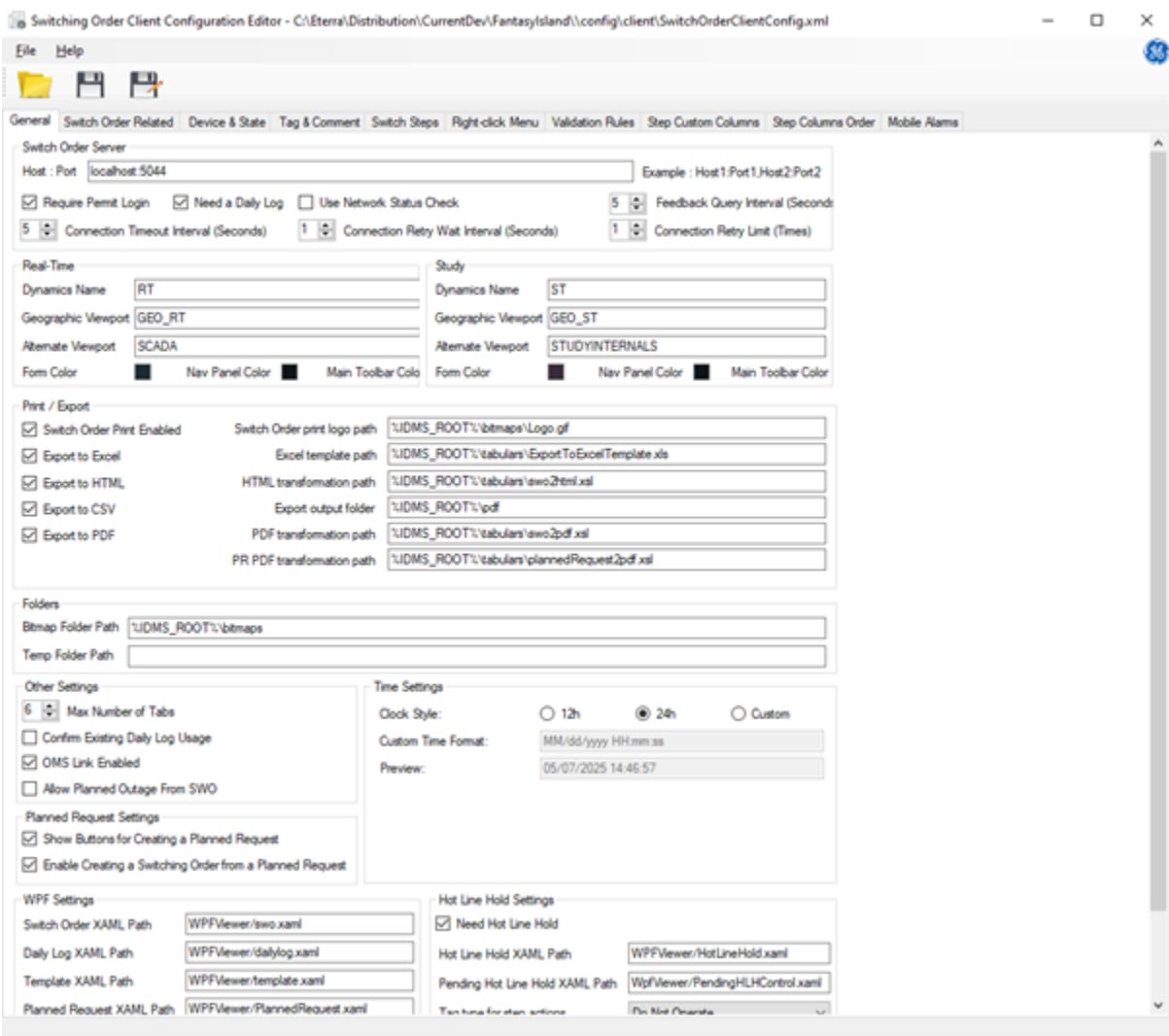


Figure 1. Configuration - General Tab

3.1. Switch Order Server Settings

Switch Order Server

Host : Port localhost:5044 Example : Host1:Port1,Host2:Port2

☒ Require Permit Login

☒ Need a Daily Log

☐ Use Network Status Check

5

Feedback Query Interval (Seconds)

5

Connection Timeout Interval (Seconds)

1

Connection Retry Wait Interval (Seconds)

1

Connection Retry Limit (Times)

10

Request Timeout (Seconds)

Host Port: The host name and port number (for example, Host1:7044, Host2:5044, Host3:5067) for each switching order server that the client connects to.

Require Permit Login: If this option is selected, the Switching Order application requires a valid Permit authentication session to operate.

If this option is not selected, the Switching Order application can run without a valid Permit authentication session (for example, if the real-time server is not running, if using invalid login and permission settings, and if the Permit server is not running). In this case, even if the real-time server is disconnected, a user can successfully work with Switching Order in study mode.

Need a Daily Log: If this option is selected, a daily log of switching operations is automatically created each day.

Use Network Status Check: This option prevents connections from being maintained between the Switch Order Server and client when network outage issues occur at the client side. If True, the Switch Order Server terminates its connection with the client 10 seconds after its connection with the client is lost. If False, an old connection between the server and client is maintained even after network connectivity is lost, and a new connection is established when the network is restored.

Note

To enable this option, the connection status check must be enabled both in the server and client configuration files. (The UseNetworkStatusMessage option must be enabled in the SwitchOrderServerConfig.txt file.)

Feedback Query Interval (Seconds): The feedback query interval for issued switch order steps. By default, the client issues a query for each issued switch order step every five seconds. The minimum query interval is one second. There is no upper limit.

Connection Timeout Interval (Seconds): If a server connection is not established, the time period in seconds before initiating another server connection attempt. The allowed range is 1–20 seconds.

Connection Retry Wait Interval (Seconds): The time period in seconds between server connection attempts. The allowed range is 1–60 seconds.

Connection Retry Limit (Times): The number of connection attempts that need to be accomplished before trying to connect to another server. The allowed range is 0–20.

Request Timeout (Seconds): The time period in seconds before issuing a failure message for a client operation if a server response is not received. The allowed range is 1–100 seconds.

3.2. Real-Time and Study Settings

The screenshot displays the 'Switching Order Server' configuration window. At the top, there is a 'Host : Port' field with an example 'Host1:Port1,Host2:Port2'. Below this are several checkboxes: 'Require Permit Login' (checked), 'Need a Daily Log' (checked), and 'Use Network Status Check' (unchecked). To the right of these are three spinners for 'Feedback Query Interval (Seconds)' (set to 5), 'Connection Timeout Interval (Seconds)' (set to 5), and 'Connection Retry Wait Interval (Seconds)' (set to 1). Further right are two more spinners for 'Connection Retry Limit (Times)' (set to 1) and 'Request Time' (set to 10). The window is divided into two main sections: 'Real-Time' and 'Study'. Each section has three text fields: 'Dynamics Name' (RT for Real-Time, ST for Study), 'Geographic Viewport' (GEO for Real-Time, STUDYGEO for Study), and 'Alternate Viewport' (SCADA for Real-Time, STUDYINTERNALs for Study). At the bottom of each section are three color selection boxes for 'Form Color', 'Nav Panel Color', and 'Main Toolbar Color'.

Dynamics Name: Shows the dynamics names for the Real-Time and Study modes of the Switching Order application.

Geographic Viewport: Contains the viewport name for the Real-Time and Study modes of the Switching Order application.

Alternate Viewport: Sets the name of the alternate viewport for the Real-Time and Study modes of the Switching Order application.

Form Color: The switching order form background color for the Real-Time and Study modes of the Switching Order application. Clicking the color box opens the Color dialog box to set the color.

Nav Panel Color: The switching order navigation panel background color for the Real-Time and Study modes of the Switching Order application. Clicking the color box opens the Color dialog box to set the color.

Main Toolbar Color: The switching order main toolbar background color for the Real-Time and Study modes of the Switching Order application. Clicking the color box opens the Color dialog box to set the color.

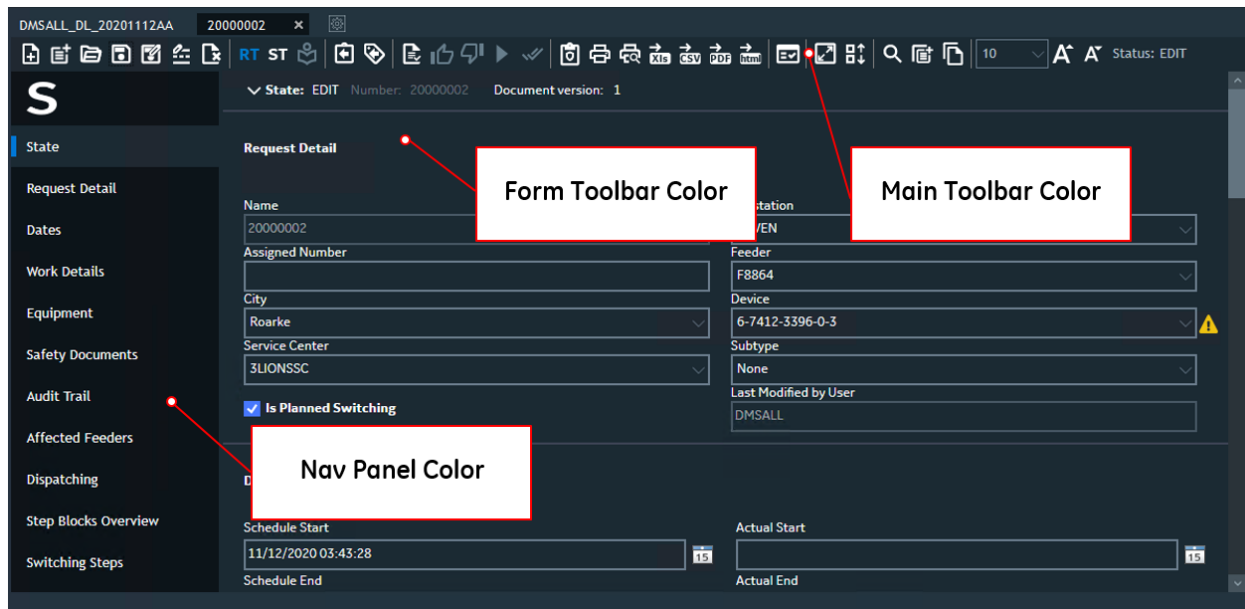


Figure 2. Configuring Switching Order Form Colors

3.3. Print/Export Settings

Print / Export		
☒ Switch Order Print Enabled	Switch Order print logo path	%IDMS_ROOT%\bitmaps\Logo.gif
☒ Export to Excel	Excel template path	%IDMS_ROOT%\tabulars\ExportToExcelTemplate.xls
☒ Export to HTML	HTML transformation path	%IDMS_ROOT%\tabulars\swo2html.xml
☒ Export to CSV	Export output folder	%IDMS_ROOT%\
☒ Export to PDF	PDF transformation path	%IDMS_ROOT%\tabulars\swo2pdf.xml
	PR PDF transformation path	%IDMS_ROOT%\tabulars\plannedRequest2pdf.xml

Switch Order Print Enabled: When selected, this option allows the switching orders to be printed.

Export to Excel: When selected, this option enables exporting switching orders to an Excel file.

Export to HTML: When selected, this option enables exporting switching orders to an HTML file.

Export to CSV: When selected, this option enables exporting switching orders to a CSV file.

Export to PDF: When selected, this option enables exporting switching orders to a PDF file.

Switch Order Print Logo Path: Sets the file path to the logo image, which will be added to the switching order in PDF.

Excel template path: Sets the path to the folder where the Excel templates are stored.

HTML transformation path: Sets the path to the HTML conversion template file.

Export output folder: Sets the path to the folder where the switching order files will be exported.

PDF transformation path: Sets the path to the PDF conversion template file for switching orders.

PR PDF transformation path: Sets the path to the PDF conversion template file for planned requests.

A switch order can be printed from the Switch Order form regardless of its state. The printed form is customizable.

You can export a switch order to the following formats:

- Excel (XML file with Excel-specific schema)
- CSV
- PDF
- HTML

During the export, an XML file with all switching order data is created and then transformed into the required format using the specified template file.

Exporting to PDF is done by transforming XML data to the intermediate XSL-FO format. The below image shows the general switching order to PDF export workflow.

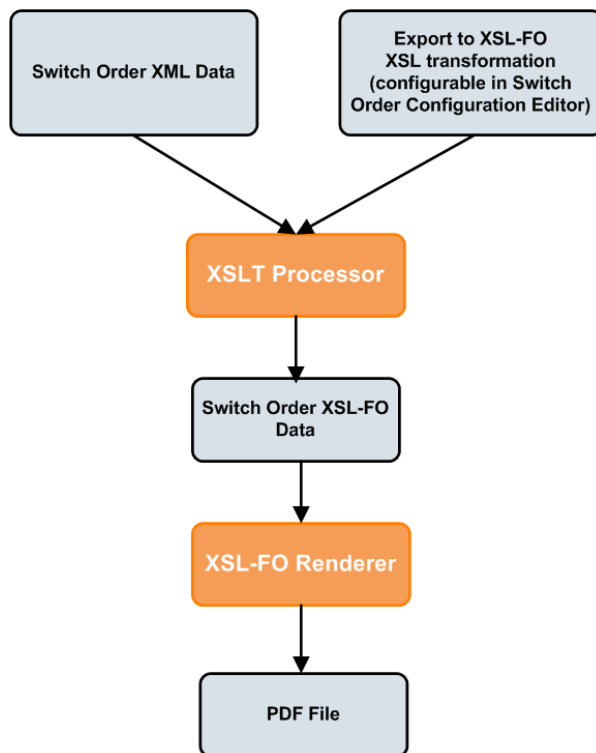


Figure 3. Switching Order to PDF Export Customization Diagram

For more details about creating and modifying an XSLT file to transform XML data to the XSL-FO format, refer to the following websites:

- For general reference, refer to <https://www.xml.com/pub/a/2001/01/17/xsl-fo/index.html>
-

For information about formatting tables, refer to <https://www.xml.com/pub/a/2001/01/24/xsl-fo/index.html#tables>

- For a tutorial, refer to <http://www.cafeconleche.org/books/bible2/chapters/ch18.html>

Exporting to HTML is done by transforming XML data to the XHTML format. The below image shows the general process of the switching order to HTML export workflow.

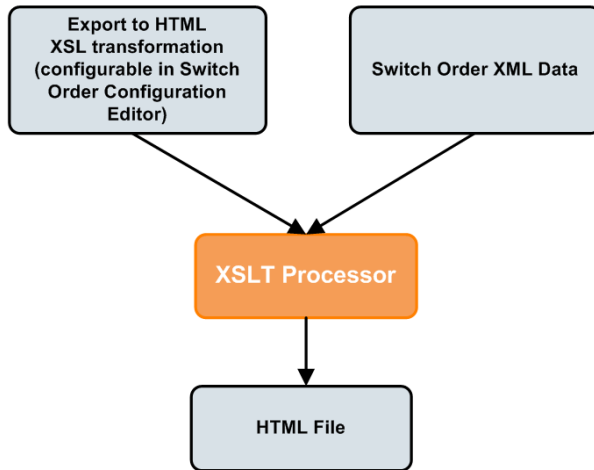


Figure 4. Switching Order to HTML Export Customization Diagram

For more details about creating and modifying an XSLT file to transform XML data to XHTML format, refer to the following websites:

- For general reference, refer to <http://www.w3.org/TR/xslt/>
- For a tutorial, refer to https://www.w3schools.com/xml/xsl_transformation.asp

3.4. Folders

Folders	
Bitmap Folder Path	%IDMS_ROOT%\bitmaps
Temp Folder Path	

Bitmap Folder Path: Sets the path to the folder where the bitmaps are stored. This path is used when assigning icons for right-click menu options.

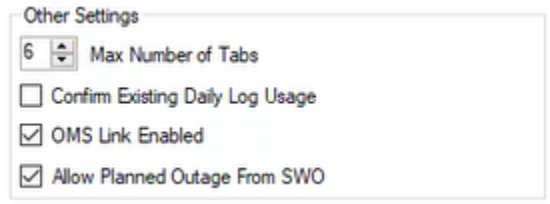
Temp Folder Path: Sets the path to a folder where the temporary switching order lists and temporary employee lists are saved. When blank, the RecentEmployees file is not created and the RecentSwitchOrder file is saved to a folder related to the Windows user (for example: C:\Users\<User>\AppData\Roaming\General Electric Technology GmbH\Advanced Presentation Framework\<Version>).

Note

The RecentEmployees file is not created if the Use Employee File option is selected on the

Switch Steps tab (see [Employee List for Issued to Field](#)).

3.5. Other Settings



Other Settings

6 Max Number of Tabs

☐ Confirm Existing Daily Log Usage

☒ OMS Link Enabled

☒ Allow Planned Outage From SWO

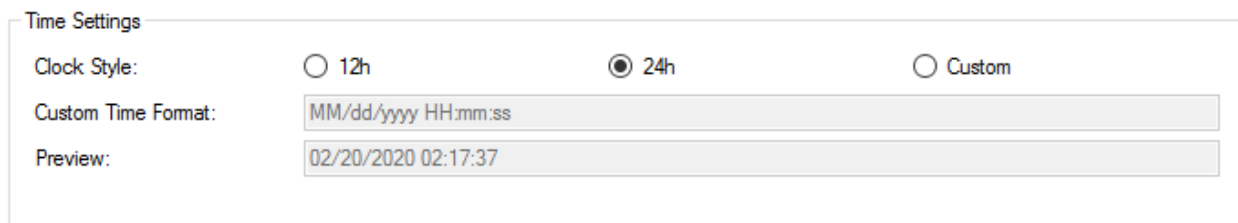
Max Number of Tabs: Sets the maximum number of tabs to be open at one time in the Switching Order application.

Confirm Existing Daily Log Usage: If selected, the system prompts the operator about whether to keep the Daily Log that was not completed during the previous shift, or to complete it and create a new log, as appropriate.

OMS Link Enabled: If selected, switching orders can be linked to incident and planned outages.

Allow Planned Outage From SWO: If selected, the system allows creating a planned outage from a switch order.

3.6. Time Settings



Time Settings

Clock Style: ☐ 12h ☒ 24h ☐ Custom

Custom Time Format: MM/dd/yyyy HH:mm:ss

Preview: 02/20/2020 02:17:37

Clock Style: Sets 12-hour, 24-hour, or a custom time format style to use in the Switching Order application.

Custom Time Format: Allows you to define a custom date and time format.

Preview: Shows a preview of the date and time in the defined format.

3.7. WPF Settings

WPF Settings	
Switch Order XAML Path	WPFViewer/swo.xaml
Daily Log XAML Path	WPFViewer/dailylog.xaml
Template XAML Path	WPFViewer/template.xaml
Planned Request XAML Path	WPFViewer/PlannedRequest.xaml

The Switching Order application uses WPF user interface that can be customized for project needs. For more information about customizing the Switching Order application, see [Customizing Switching Order Appearance](#).

Switch Order XAML Path: Path to the XAML file that defines the switching order form. By default, XAML configuration files are stored in the Distribution\Client\bin\WpfViewer folder.

Daily Log XAML Path: Path to the XAML file that defines the daily log form.

Template XAML Path: Path to the XAML file that defines the switching order template form.

Planned Request XAML Path: Path to the XAML file that defines the planned request form.

Note

If the XAML file paths point to missing or invalid XAML files, the Switching Order application uses the default XAML files that are provided with the application.

Note

The legacy WinForms user interface is not supported.

3.8. Planned Request Settings

Planned Request Settings	
☐	Show Buttons for Creating a Planned Request
☐	Enable Creating a Switching Order from a Planned Request

Show Buttons for Creating a Planned Request: Specifies whether options to create a planned request document are shown in the Switching Order application.

Enable Creating a Switching Order from a Planned Request: Specifies whether creating a switching order from a planned request is enabled.

3.9. Hot Line Hold Settings

Hot Line Hold Settings

☒ Need Hot Line Hold

Hot Line Hold XAML Path

WPFViewer/HotLineHold.xaml

Pending Hot Line Hold XAML Path

WpfViewer/PendingHLHControl.xaml

Tag type for step actions

Info

Action name for manual step

Measurement Types

Automatic Reclosing Status

HLH Tag

Do Not Operate

Section

☒ State Section

☒ Request Section

☒ Dates Section

☒ Device Contact Tag Section

☒ Attachment Section

☒ Work Details Section

☒ Equipment Section

☒ Safety Doc Section

☒ Audit Section

☒ Affected Feeder Section

☐ Area Details Section

☒ Step Grid Section

☒ Step Block Section

☒ Dispatching Section

Need Hot Line Hold: Specifies whether hot line hold document options (New HLH and Pending HLH) are shown on the Options page in the Switching Order application.

Hot Line Hold XAML Path: Path to the XAML file that defines the hot line hold document template form.

Pending Hot Line Hold XAML Path: Path to the XAML file that defines the Current Pending Hot Line Hold display template form.

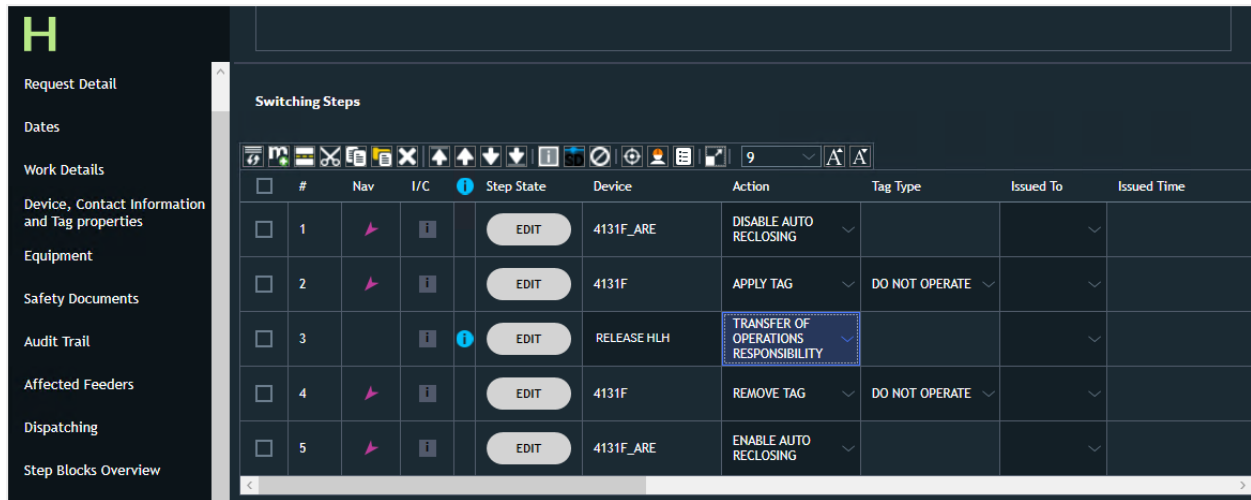
Tag Type for Step Actions: Specifies the tag type that is used for tagging actions for hot line hold documents that are created for devices from the network view.

Action Name for Manual Step: Name for the Release HLH step action.

Measurement Types: Automatic reclosing status of the protective reclosing device.

HLH Tag: Blue tag or do not operate tag that can be applied on the protective reclosing device.

Section: Used to Hide/Show different sections available in HLH document.



	#	Nav	I/C	Step State	Device	Action	Tag Type	Issued To	Issued Time
☐	1	↗	i	EDIT	4131F_ARE	DISABLE AUTO RECLOSING		✓	
☐	2	↗	i	EDIT	4131F	APPLY TAG	DO NOT OPERATE	✓	
☐	3		i	EDIT	RELEASE HLH	TRANSFER OF OPERATIONS RESPONSIBILITY		✓	
☐	4	↗	i	EDIT	4131F	REMOVE TAG	DO NOT OPERATE	✓	
☐	5	↗	i	EDIT	4131F_ARE	ENABLE AUTO RECLOSING		✓	

Figure 5. Hot Line Hold Document Switching Steps

4. Switch Order Related Tab

The Switch Order Related tab allows configuring the starting state of the switching order user interface, switching order form colors, integration with the Safety Documents function, and properties of the Employee Information popup.

The screenshot shows the 'Switching Order Client Configuration Editor' window with the 'Switch Order Related' tab selected. The window title is 'Switching Order Client Configuration Editor - C:\Eterra\Distribution\CurrentDev\FantasyIsland\config\client\SwitchOrderClientConfig.xml'. The interface includes a menu bar (File, Help) and a toolbar with icons for opening, saving, and printing. The 'Switch Order Related' tab is active, showing several configuration sections:

- Starting UI state:** Four checkboxes for collapsing sections: 'Collapsed Daily Log State section', 'Collapsed Switch Order State section', 'Collapsed Daily Log Properties section', and 'Collapsed Switch Order Properties section'. The 'Collapsed Switch Order State section' and 'Collapsed Switch Order Properties section' are checked.
- Email Notifications:** Checkboxes for 'Enable Email Notifications' (unchecked), 'Use Full Names' (checked), and 'Collapsed Email Notification...' (checked).
- Safety Doc Configuration:** Fields for 'User' (SAFETY_EMS), 'Scada' (MAIL/USER=%SAFETYUSER "ADD_EQUIP POINT/%COMPOSITEKEY/%DEVICENAME/%ID/%TOPOSUBNAME/%FEEDERNAME/%SRVCNTR/%ENERGYCAT/%GEOMETRY/%DEVICEX/%DEVICEY/%DNOMDEVICETYPE/%ALTGEOMETRY"), and 'Non-Scada' (MAIL/USER=%SAFETYUSER "ADD_EQUIP NOMODEL/%DMSID/%DEVICENAME/%ID/%TOPOSUBNAME/%FEEDERNAME/%SRVCNTR/%ENERGYCAT/%GEOMETRY/%DEVICEX/%DEVICEY/%DNOMDEVICETYPE/%ALTGEOMETRY"). A checkbox for 'Safety Doc Integration Enabled' is checked. A field for 'Safety Doc Real User Mail' contains MAIL/USER=%SAFETYUSER "CREATEDOCFROMTYPE %TAGTYPE".
- Safety Doc Types:** A dropdown for 'Select Safety Doc Type' is set to 'Hot Line Hold', and a field for 'Scada Tag Type' is set to 'HOTLINE'. Below, 'Hot Line Hold Resulting Devices' section has dropdowns for 'Upstream Device(s)' (Next upstream recloser/breaker) and 'Downstream Device(s)' (None). Checkboxes for 'Include DER Breakers', 'Include Direct Connected DER', and 'Include Bypass Switches' are all unchecked.
- Employee Information Popup:** Checkboxes for 'Use Employee Popup' and 'Use OMS for IssuedTo validation' are checked. A field for 'Employee Popup URL' contains NETVIEW://SO.RT/POPUP/TABULAR/SWOCrewsFromOms?Title=Crew List. A checkbox for 'Use Add New Employee Right-Click Option in Dispatching Section' is unchecked.
- Enable Gantt Chart:** A checkbox is checked. Below it, 'Gantt Chart colors' are listed with color swatches: Created Status (dark blue), Validated Status (medium blue), Approved Status (light blue), Executing Status (yellow), and Completed Status (green).
- Mobile Options:** A checkbox for 'Enable document dispatching' is checked.
- WFM Options:** A checkbox for 'Enable WFM interface' is unchecked.
- Associated Documents Validation Options:** A checkbox for 'Enable Associated Document...' is unchecked.

Figure 6. Configuration - Switch Order Related Tab

4.1. Starting UI State

The screenshot shows the 'Starting UI state' section of the configuration. It contains four checkboxes, all of which are checked:

- ☒ Collapsed Daily Log State section
- ☒ Collapsed Switch Order State section
- ☒ Collapsed Daily Log Properties section
- ☒ Collapsed Switch Order Properties section

Collapsed Daily Log Section: Select this option to collapse the Daily Log State section when the Switching Order application is started.

The screenshot shows the top section of the ADMS 16.25.1 Switching Order Client Configuration Editor. A dropdown menu for 'State' is expanded, showing 'EXECUTING' as the selected option. Other options visible are 'Number: 20000001' and 'Document version: 1'. Below the dropdown, there are tabs for 'State', 'Request Detail', and 'Equipment'. The 'Request Detail' tab is active, showing a 'Name' field with the value 'DMSALL_DL_20200729AA'.

Collapsed Daily Log Properties Section: This option is deprecated.

Collapsed Switch Order State Section: Select this option to collapse the Switch Order State section when the Switching Order application is started.

Collapsed Switch Order Properties Section: This option is deprecated.

4.2. Email Notifications

Enable Email Notifications: Select this option to enable the Email Notifications section in the Switching Order application.

Collapsed Email Notifications section: Select this option to collapse the Email Notifications section when the Switching Order application is started.

Use Full Names: Select this option to show recipient names in the notification list in the Email Notifications section. Not selecting this option shows recipient emails.

4.3. Safety Document Configuration

The screenshot shows the 'Safety Doc Configuration' form. It includes fields for 'User' (SAFETY_EMS), 'Scada' (MAIL/USER=%SAFETYUSER "ADD_EQUIP POINT/%COMPOSITEKEY/%DEVICENAME/%ID/%TOPOSUBNAME/%FEEDERNAME/%SRVCNTR/%ENERGYCAT/%GEOMETRY/%DEVICEX/%DEVICEY/%DNOMDEVICETYPE/%ALTGEOMETRY"), 'Non-Scada' (MAIL/USER=%SAFETYUSER "ADD_EQUIP NOMODEL/%DMSID/%DEVICENAME/%ID/%TOPOSUBNAME/%FEEDERNAME/%SRVCNTR/%ENERGYCAT/%GEOMETRY/%DEVICEX/%DEVICEY/%DNOMDEVICETYPE/%ALTGEOMETRY"), a checkbox for 'Safety Doc Integration Enabled' (checked), and 'Safety Doc Real User Mail' (MAIL/USER=%SAFETYUSER "CREATEDOCFROMTYPE %TAGTYPE").

User: The safety document user ID.

SCADA: URL for adding safety documents for SCADA devices.

Non-SCADA: URL for adding safety documents for non-SCADA devices.

Safety Doc Integration Enabled: If selected, allows creating safety documents from the switching order tagging steps.

Safety Document Real User Mail: The safety document real user mail.

4.4. Safety Doc Types

Safety Doc Types

Select Safety Doc Type: Hot Line Hold Scada Tag Type: HOTLINE

Hot Line Hold Resulting Devices

Upstream Device(s): Next upstream recloser/breaker

Downstream Device(s): None

☐ Include DER Breakers ☐ Include Direct Connected DER ☐ Include Bypass Switches

Select Safety Doc Type: Select the required safety doc type.

Scada Tag Type: Displays the tag type based on the selected safety doc type.

Note

When a safety doc type is selected, the values of the relevant fields are updated automatically.

Upstream Device(s): Select the upstream device relevant to the safety doc.

Downstream Device(s): Select the downstream device relevant to the safety doc.

Include DER Breakers: When selected, DER breakers are included in the safety doc.

Include Direct Connected DER: When selected, directly connected DERs are included in the list of devices added to the safety doc.

Include Bypass Switches: When selected, bypass switches are included in the list of devices added to the safety doc.

4.5. Employee Information Pop-Up

Note

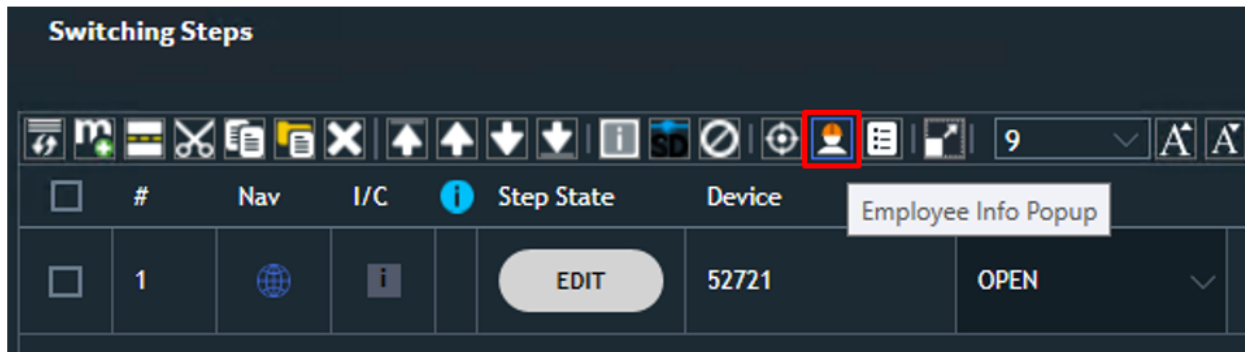
For more information about configuring the Employee Information popup (Employee Info display), see "Configuring the Employee Info Display" in the *Switching Order Configuration Editor User's Guide*.

Employee Information Popup

☒ Use Employee Popup ☐ Use OMS for IssuedTo validation

Employee Popup URL: NETVIEW://SO.RT/POPUP/TABULAR/SWOEmployee?Title=Employee Info

Use Employee Pop-Up: When this option is selected, the Employee Info Pop-Up button appears on the Switching Steps toolbar.



Employee Pop-Up URL: Sets the URL called by the Employee Info Pop-Up button and specifies the Employee Info display title.

Use OMS for Issued To Validation: When this option is selected, crew names in the Issued To fields are validated against crew names that are defined in the OMS crew model. This option is designed for utilities that use crew information from the crew model and assign crews (not employees) to switching steps.

- If selected, the tool prompts you to change the Employee Popup URL to one that calls the Crew List display, which lists crews from the OMS crew model and allows an operator to assign a crew for a switching step.

URL: NETVIEW://SO.RT/POPUP/TABULAR/SWOCrewsFromOms?Title=Crew List

- If cleared, the tool prompts you to change the Employee Popup URL to one that calls the Employee Info display, which contains detailed employee data, and allows an operator to assign an employee to a switching step. This display is populated from the OMS crew model and Employee Plug-In files, as configured by the Employee Info Pop-Up parameters in the Switching Order Server configuration file (see chapter "Switching Order Server Configuration File" in the *Switching Order Server Configuration Editor User Guide*).

URL: NETVIEW://SO.RT/POPUP/TABULAR/SWOEmployee?Title=Employee Info

Employees					
ID	Employee Name	Assign to Switching Step	Group Name	Job Title	Qualifications Navigation
FI4121	John Smith		West Alpha	Local Lineman	
FI4122	Jim Brown		West Alpha	Local Lineman	
FI1500	Q Jeff Carino				
FI1501	Ryan Eric Thom...				
FI1502	Daniel Jeffrey ...				
FI1503	Roger Brian Pat...				
FI1504	Kevin Robert T...				
FI1505	David Ed Bailey				

Figure 7. Employee Info Display

ID	Crew Name	Assign to Switching Step
FI2512	Adams (5288)	
FI2034	Adams [4432] - AUTO	
FI1972	Adams [5298] - AUTO	
FI1589	Addison (1173)	
FI2032	Addison (1431)	
FI1927	Amoroso (4069)	
FI2559	Amoroso (5250)	

Figure 8. Crew List Display

4.6. Gantt Chart

☒ Enable Gantt Chart

Gantt Chart colors

- Created Status
- Validated Status
- Approved Status
- Executing Status
- Completed Status

Enable Gantt Chart: If selected, the Gantt Chart button appears on the Switching Order Search display to show a Gantt chart with timeline for the searched switching orders.

Gantt Chart Colors: Allows you to configure colors for switching orders in different states on the Gantt chart.

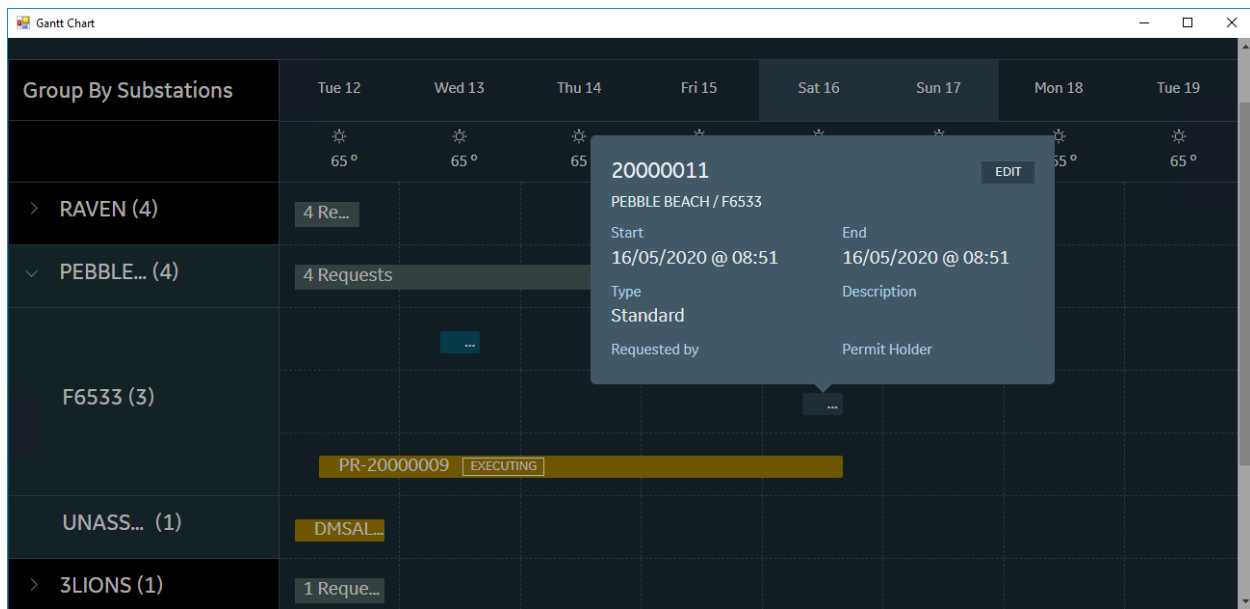


Figure 9. Switching Order Gantt Chart

4.7. Mobile Options

Mobile Options

☐ Enable document dispatching

Enable Document Dispatching: Select this option to enable the Document Dispatching section in the Switching Order application.

For more information about configuring the mobile dispatching feature, refer to the *Switching Order Server Configuration Editor User Guide*.

4.8. WFM Options

WFM Options

☐ Enable WFM interface

Enable WFM Interface: Select this option to enable the interface to the Workforce Management System (WFM). If this option is enabled, the Send to WFM button appears on the Switching Order toolbar. In addition, to enable sending documents to WFM, you must specify the `SharedWFMFolder` parameter in the `SwitchOrderServerConfig.txt` file, and the `ApiWfmUrl` parameter in the `DmsInterfaceServerConfig.xml` file.

5. Device & State Tab

This tab is used to configure the types of devices and their respective actions as displayed in a switching order step. The device types are linked to the main DNOM device types so that the switching actions can be checked and verified.

Switching Order Configuration Editor - D:\Eterra\Distribution\FantasyIsland\config\client\SwitchOrderClientConfig.xml

File Help

General Switch Order Related **Device & State** Tag & Comment Switch Steps Right-click Menu Validation Rules Step Custom Columns Step Columns Order Header Custom Fields

DNOM Type and Sub-type

dg	DNOM Type	Sub Type	Action Type
☒	COMPSW	AB SWITCH	TYPEABSWITCH
☐		ALARM	TYPEALARM
☐	TF	DMSAUTOTF	TYPETAPCHR
☐	SWT	DMSBREAKER	TYPEBREAK

Add Delete

Action Types

Select Action Type: TYPEABSWITCH Delete Create Action Type: Add

dg	State Name	Function	State Value	Phase Modifier	BackOut State	Tag Value	Copy To Safety	Device Name	Instruction Text
☒	A	Status	State1	All	B	None	False	%DEVICENAME%	SET %DEVICENA...
☐	B	Status	State2	All	A	None	False	%DEVICENAME%	SET %DEVICENA...
☐	AB	Status	State3	All	A	None	False	%DEVICENAME%	SET %DEVICENA...
☐	OPEN	Status	State4	All	A	None	False	%DEVICENAME%	SET %DEVICENA...
☐	CHECK NORMAL	None	CheckNormal	All		None	False		CHECK %DEVICE...

Copy Paste Add Delete

Figure 10. Configuration - Device & State Tab

For each device, there is a defined set of actions, including the inverse for each action as used in the go back (return to normal) orders.

The Add to Switch Order commands, which are used to add steps to the switching order, are configured using the DNOM Type, Sub Type, and Action Type parameters (see [STEPADD Command](#)).

If a DNOM Type, Sub Type, or an Action Type is missing, the Switching Order application issues an error message in red that indicates the missing device or action type. The operator forwards this information to an analyst. The analyst then uses the Device & State tab to add the missing data from the context error message to the switching operations configuration file.

5.1. DNOM Type and Sub-Type

DNOM Type and Sub-type

dg	DNOM Type	Sub Type	Action Type
☒	COMPSW	AB SWITCH	TYPEABSWITCH
☐		ALARM	TYPEALARM
☐	TF	DMSAUTOTF	TYPETAPCHR
☐	SWT	DMSBREAKER	TYPEBREAK

Add Delete

DNOM Type: Includes DNOM types.

Sub Type: Includes device subtypes for each DNOM type.

Action Type: Allows assigning an action type to a device subtype. The drop-down list is populated with all existing action types.

Action Type

TYPEDISTTF

TYPESWT

TYPESWT

TYPEBREAK

TYPECAP

TYPETAPCHR

TYPETF

TYPERECLOSER

TYPECUT

TYPEJUMPER

TYPEDISTTF

TYPEREGBYPASS

TYPEGRAL

TYPEALARM

TYPEGTBREAKER

TYPEABSWITCH

TYPETHROWOVER

TYPEESCOTTHROWOVER

TYPEFAULTINDICATOR

TYPEREACTOR

TYPECOMPSWT

NODEVICEACTIONS

TYPESECTIONALIZER

TYPEDMSPROTECTOR

TYPECOMPSWEARTHEDGROUNDED

TYPEMOBILESUBSTATION

TYPETMPCUT

TYPETMPGROUND

TYPETMPJUMPER

Note

One action type can be assigned to several device sub types. For example, the TYPESWT action type is assigned to switches and fuses (SW, DMSSWITCH, FUSE, and DMSFUSE).

Add: Click this button to add a new DNOM Type string.

Delete: Click this button to delete selected strings.

5.2. Action Types

Action Types

Select Action Type:

Create Action Type:

dg	State Name	Function	State Value	Phase Modifier	BackOut State	Tag Value	Copy To Safety	Device Name	Instruction Text
☒	CLOSE	Status	Close	All	OPEN	None	False	%DEVICENAME...	CLOSE %DEVICE...
☐	OPEN	Status	Open	All	CLOSE	None	False	%DEVICENAME...	OPEN %DEVICE...
☐	CHECK NORMAL	Status	None	All		None	False		CHECK %DEVICE...
☐	APPLY TAG	Status	None	All	REMOVE TAG	Apply	False		APPLY TAG ON ...
☐	OPEN AND TAG	Status	Open	All	REMOVE TAG A...	Apply	True		OPEN %DEVICE...
☐	REMOVE TAG A...	Status	Close	All	OPEN AND TAG	Remove	True		REMOVE TAG %...
☐	CHECK CLOSE	Status	None	All		None	False		CHECK %DEVICE...
☐	CHECK OPEN AN...	Status	None	All	REMOVE TAG	Apply	True		CHECK %DEVICE...
☐	REMOVE TAG	Status	None	All	APPLY TAG	Remove	True		REMOVE TAG %...
☐	OPEN PHASE A	Status	Open	A	CLOSE PHASE A	None	False	%DEVICENAME...	OPEN PHASE A ...

Select Action Type: Select an action type from the drop-down list to populate the grid with the action type states.

Delete: Click this button to delete all states of the selected action type.

Create Action Type: To add a new action type, enter the name in the text field and click the Add button.

State Name: Contains the name of a state or function.

When a device is added to a switching order step, its Action drop-down list is populated with all state names that pertain to the device's action type.

#	Nav	I/C	Step State	Device	Action	Tag Type
1			EDIT	52721	OPEN	

Dropdown menu options for Action:

- OPEN
- CHECK NORMAL
- APPLY TAG
- OPEN AND TAG
- REMOVE TAG AND CLOSE
- CHECK CLOSE
- CHECK OPEN AND TAG

Function: Allows selecting a function type for a state name from a drop-down list.

Function
Status ▼
None
Status
RecloseForGroundFa
ReclosingEnabled
BlockClose
SupervisoryCutOut
Ground Trip Enabled
Local Remote
TransferSchemeEnab
RecloserAs Switch
Auto Manual
Place Remove

State Value: Sets a state or function of the state name to be queried by the Switching Order.

The available state values depend on the selected Function type.

Phase Modifier: Sets phases to be queried by the Switching Order.

BackOut State: Sets an inverse action to the current device's state.

The back out state values are used to create go back (return to normal) switching order steps.

Also, when a device is added to a switching order, the application sets the Action based on the device current state's BackOut State value.

Tag Value: Sets a tag action to be queried by the Switching Order.

Copy to Safety Documents: A True/False indicator of whether the step can be copied to the Safety Documents.

Device Name Formatting: Allows defining the information to be displayed in the Device field of a switching order step.

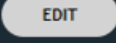
Instruction Text: Allows defining the information to be displayed in the Instruction field of a switching order step.

The Device and the Instruction fields of a switching order step are formatted based on the Device Name Formatting and Instruction Text configuration settings. The formatted values are set not only when the switching order step is added, but also when the Action selection is changed.

The following substitution indicators can be added to the above fields to display the device-specific information:

- %SUBSTN%: Shows the substation name of the device.
- %DEVICETYPE%: Shows the device type.
- %DEVICENAME%: Shows the device name.
- %POINTID%: Shows the measurement point ID.
- %ALTID%: Shows the alternative ID of the device name.
- %STATENAME%: Shows the currently selected Action.

For example, if the Instruction Text field contains "OPEN %DEVICENAME%" and the device name is 52304, the Instruction field in the Switching Order will display "OPEN 52304".

Switching Steps								
								
☐	#	Nav	I/C	 Step State	Device	Action	Issued To	Instruction
☐	1			 EDIT	52304	OPEN		OPEN 52304

 Note

DNOMTYPE state/action type mapping must be unique. For example, SWT DNOMTYPE is configured to have TYPESWT as an action. Attempting to assign the TYPESWT action to SHIMP DNOMTYPE will result in an error message.

6. Tag & Comment Tab

This tab is used to configure the tag types and pre-defined comments for the switching order steps.

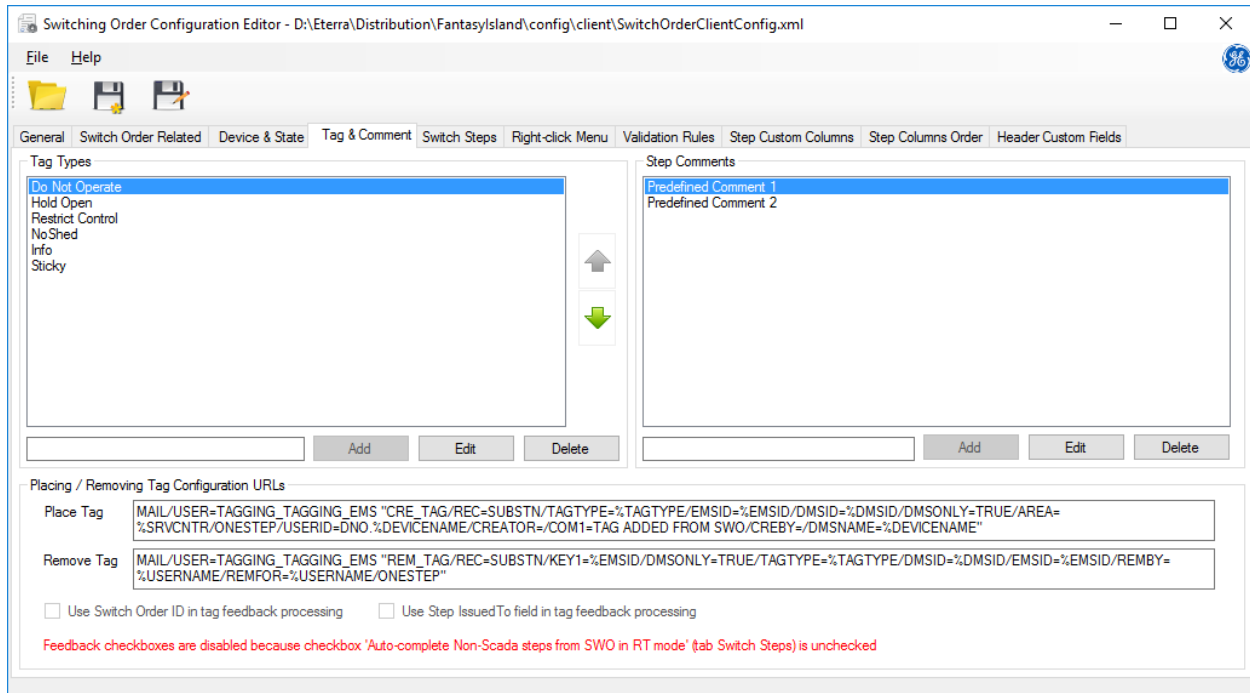


Figure 11. Configuration - Tag & Comment Tab

6.1. Tag Types

The switching orders do not directly communicate with the tagging system, so it is necessary to define each tag type that exists in the tagging system.

To define a tag type, enter the name of the tag type in the blank field, and then click Add. You can edit and remove tag types by using the Edit and Delete buttons.

The Placing/Removing Tag Configuration URLs grid includes the following options:

- **Place Tag/Remove Tag:** These fields define commands for placing and removing tags. The switching order application uses these commands to automatically add or remove tags on the geographic view when the Auto-Complete Non-SCADA Steps from SWO in RT Mode option on the Switch Steps tab is enabled.

In the Place Tag field, you can specify a comment that will be automatically added when a tag is automatically placed by the switching order application. The comments are configured by the /COM1 and /COM2 parameters. The default comment is "Safety Document. Holder.".

Along with text, you can include the following fields:

- %STEP.FIELDNAME (for example, %STEP.DEVICESUB) to show substation name

- %DOC.FIELDNAME (for example, %DOC.CREATED) to show the switching order creation date
- %SWONAME to show the switching order name
- **Use Switch Order ID in Tag Feedback Processing:** Use this option to bind a tag to a switching order (by ID).

Note

To use this feature, the Auto-Complete Non-SCADA Steps from SWO in RT Mode option on the Switch Steps tab must be enabled.

This feature allows a switching order to work only with its own tags, regardless of the tags that already exist in the system. For example, a switching order can add a tag, even if the device is already tagged with the same tag. However, a switching order can only remove tags that it created.

When a tag is automatically added from the switching order, the switching order's ID is automatically inserted in the tag's Comments field.

To determine whether a tag is associated with a switching order, the Switching Order application checks the tag's Comment field for the switching order's ID.

- **Use Step Issued To Field in Tag Feedback Processing:** Use this option to bind a tag to the responsible person (by the Issued To field value).

Note

To use this feature, the Auto-Complete Non-SCADA Steps from SWO in RT Mode option on the Switch Steps tab must be enabled.

This feature allows a tag to be removed only by the person who applied it.

If this option is selected, a switching order can add several tags of the same type if the tagging steps are issued to different people. If this option is not selected, a switching order cannot add several tags of the same type to a device. When a tag is already applied, another similar tagging step is considered as already completed and therefore cannot be implemented.

When a tag is automatically added from the switching order, the tagging step's Issued To value is automatically inserted in the tag's Comments field.

To determine whether a tag is associated with a person, the Switching Order application checks the tag's Comment field for the person's name.

#	Nav	I/C	Step State	Device	Action	Tag Type	Issued To	Issued Time	Completed Time
1			COMPLETED	52721	APPLY TAG	HOLD OPEN	John Smith	07/29/2020 03:46:00	07/29/2020 04:03:44
2			ISSUED	52721	APPLY TAG	HOLD OPEN	John Smith	07/29/2020 03:46:26	15
3			ISSUED	52721	APPLY TAG	HOLD OPEN	Andy Garcia	07/29/2020 03:46:28	15

Green indicates that the step is already implemented and marked as completed. The tag is applied by the Issued To person.

Pink indicates that the step is already implemented but waiting to be marked as completed. If a similar tag is already applied, the same tag cannot be applied if the Issued To person is the same.

Cyan indicates that the step is not yet implemented. Although a similar tag is already applied to the device, another tag can be applied if the Issued To person is different.

TAGGING
Tag Details

Tag: TEMS4E33E50E
Order #: DNO.52721
Tag Type: HOLD OPEN

Site: EMS
Device: RAVEN
52721

Created By: DMSALL
Created For: DMSALL

Creator: B038DB6C
Created: 29-Jul-2020 04:03:42

Modified By:
Modified For:

Modified:

URL:

Comments

Safety Document
Holder

JOHN SMITH

OK

Note

When both the "Use switch Order ID in tag feedback processing" and "Use Step Issued To in tag feedback processing" options are selected, their functionalities are combined. In this case, similar tags can be applied both via different switching orders and by different employees.

6.2. Step Comments

Use the Step Comments grid to create a list of pre-defined comments that will be available in the

Comments field of switching steps.

To add a predefined comment, enter the text in the blank field, and then click Add. You can edit and remove comments by using the Edit and Delete buttons.

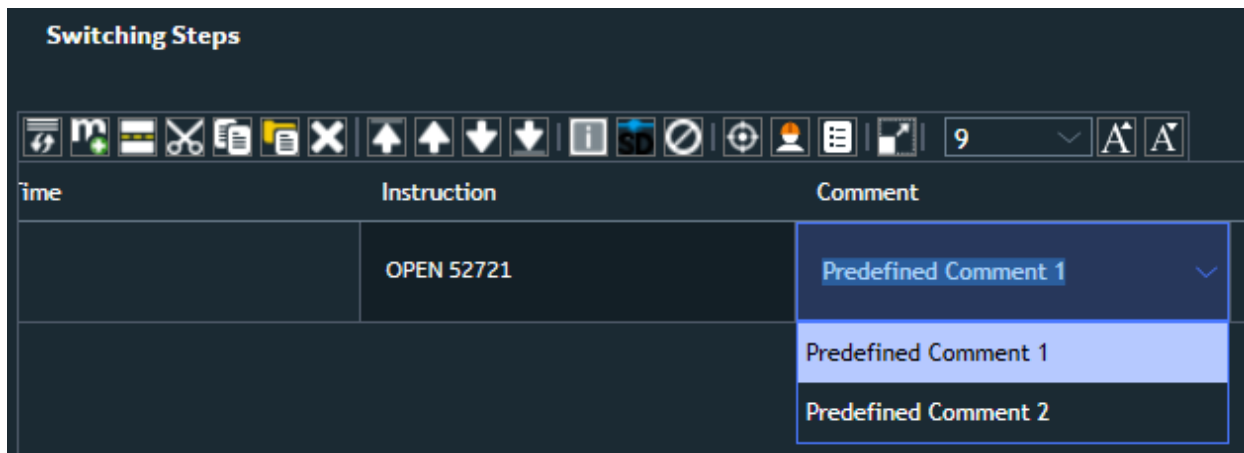


Figure 12. Selecting a Predefined Comment

7. Switch Steps Tab

The Switch Steps tab is used to configure the Switch Steps panel appearance and behavior.

File Help

General Switch Order Related Device & State Tag & Comment Switch Steps Right-click Menu Validation Rules Step Custom Columns Step Columns Order Mobile Alarms

Switch Steps

- ☒ SD Column Visible
- ☒ Warn about problems before issuance of a step
- ☒ Auto-fill completion time for SCADA devices
- ☒ Disable 'Complete Step Confirm' popup
- ☒ Auto Complete Steps
- ☒ Auto-complete Non-Scada steps from SWO in RT mode
- ☒ Copy the Issued To Field when Copying a Step
- ☐ Disable 'Issue Step Confirm' popup
- ☒ Disable Comment editing after completion
- ☐ Auto complete Non Controllable Scada points from...

Device Navigation

Default Locate: Geographic

Default Zoom: 10

Zoom level for Containers: 100

Zoom level: 10

Zoom level: 100

☐ Show PIP for Non-Container Objects

☐ Show PIP for Container Objects

☐ Scada Displays Not Available

☒ Use geo navigation for URD devices

☐ Use KEEPLAYERSETTINGS

Isolation Switch Navigation URL: FGCMDDISPLAY/VIEWPORT=Geo_/_DYN/URL "NETVIEW://_DYN/VIEW/DETAIL/%STATIONNAME?LOCATE=%D

Ground Navigation URL: FGCMDDISPLAY/VIEWPORT=Geo_/_DYN/URL "NETVIEW://_DYN/VIEW/DETAIL/%STATIONNAME?LOCATE=%D

Customer Navigation URL: FGCMDDISPLAY/VIEWPORT=Geo_/_DYN/URL "NETVIEW://_DYN/VIEW/DETAIL/%STATIONNAME?LOCATE=CUS

State Name Localization

Open State Name:

Close State Name:

Employee List for Issued To field

☒ Use Employee File

Employee list file path: %IDMS_ROOT%\config\server\SWOEmployee.bt

Manual Step Action/State Type

Action/State Type: NODEVICEACTIONS

Completion Time-Out

Time to wait for step...: 10

Hover Over

Tooltips visibility mode: Show always

Safety Instructions

☒ Enable Safety Instructions

Safety Instructions File Path: %IDMS_ROOT%\config\client\SWOMultilineInstructionsList.bt

Switch Steps Separator

Switching Step Divider Text	Header
☒ A to B	Move Load from Moss Point Elder Ferry Btk. 6598 to 5478
☐ B to A	Move Load from Moss Point Elder Ferry Btk. 5478 to 6598

Figure 13. Configuration - Switch Steps Tab

7.1. Switch Steps Options

Switch Steps

- ☐ SD Column Visible
- ☒ Warn about problems before issuance of a step
- ☒ Auto-fill completion time for SCADA devices
- ☒ Disable 'Complete Step Confirm' popup
- ☒ Auto Complete Steps
- ☒ Auto-complete Non-Scada steps from SWO in RT mode
- ☒ Copy the Issued To Field when Copying a Step
- ☐ Disable 'Issue Step Confirm' popup
- ☒ Disable Comment editing after completion
- ☒ Auto complete Non Controllable S...

SD Column Visible: Select this option to show the Safety Document column in the Switching Steps panel.

Auto Complete Steps: Select this option to allow automatic completion of the steps where the device state in the Switching Order matches the device state on the network view.

Note

This option does not apply to issued steps that are dispatched to the issued employee in the Mobile Enterprise system.

Warn About Problems Before Issuance of a Step: Select this option to show a warning that the current step de-energizes customers.

Auto-Complete Non-SCADA Steps from SWO in RT Mode: Select this option to enable auto-implementing steps in the real-time mode.

Non-SCADA devices can be operated directly from the Switching Order document in real-time mode. When the Complete button on the Switch Order form is clicked, the steps are automatically implemented and completed.

Note

This option does not apply to SCADA devices.

Auto-Fill Completion Time for SCADA Devices: Select this option to auto-fill the completion time when the device state in the Switching Order matches the device state on the network view.

For switching steps with SCADA controlled devices, the operator completes the switching step via the appropriate SCADA control and/or tagging displays. In the Switching Order, the completion time for this step is automatically set after the device status changes or after a successful tag placement. If the step involves both a SCADA control and tag placement, the completion time will be set when the second of the two actions is completed.

Copy the Issued To field when Copying a Step: Select this option to copy the Issued To, Issued Time, and Comment fields when copying a switching step to a Daily Log or a Switching Order. In addition, the Issued To and Comment fields are copied when creating return to normal steps.

Note

For completed steps, the Issued Time field is not copied.

Disable the "Complete Step Confirmation" Pop-Up: Disables the confirmation prompt that appears when trying to automatically complete a switching step for a non-SCADA device.

Disable the "Issue Step Confirmation" Pop-Up: Disables the confirmation prompt that appears when you try to issue a switching step for a device that is already in the desired state.

Disable Comment Editing After Completion: Disables the ability to edit comments for completed switching steps.

Auto complete Non Controllable Scada points from SWO in RT mode: Enables automatic completion of switching steps for non-controllable SCADA points when the **C** button is clicked in real-time mode.

7.2. Device Navigation Options

Device Navigation

Default Locate

Geographic

Default Zoom

10

☐ Show PIP for Non-Container Objects

Follow

☐ Show PIP for Container Objects

Follow

☐ Scada Displays Not Available

☒ Use geo navigation for URD devices

☒ Use KEEPPLAYERSSETTINGS

Zoom level for Containers

100

Zoom level

10

Zoom level

100

Isolation Switch Navigation URL

FGCMD DISPLAY/VIEWPORT=GEO_%DYN/URL "NETVIEW://%DYN/VIEW/DETAIL/%STATIONNAME?L

Ground Navigation URL

FGCMD DISPLAY/VIEWPORT=GEO_%DYN/URL "NETVIEW://%DYN/VIEW/DETAIL/%STATIONNAME?L

Customer Navigation URL

FGCMD DISPLAY/VIEWPORT=GEO_%DYN/URL "NETVIEW://%DYN/VIEW/DETAIL/%STATIONNAME?L

Default Locate: To set the type of network view that is used for the device navigation from the switching order, select the required option from the drop-down menu.

The term "Geographic" means that the associated network view for the device navigation is a geographic network view. The term "Schematic" means that the associated network view is a schematic network view.

	#	Nav	I/C	Step State	Device	Action	Tag T
	1			EDIT	52721	OPEN	
	2			EDIT	52722_DAST	OPEN	

Default Zoom: Select the zoom level at which the located device is shown on the network view.

Zoom Level for Containers: Select the zoom level at which the located device is shown on the container schematic display.

Show PIP for Non-Container Objects: Select this option to enable the picture-in-picture window on the network view when the operator uses the Locate function from the Switching Order for non-container objects (for example, switches, fuses, and so on).

- **Zoom Level:** Select the zoom level at which the located non-container object is shown on the PIP window.

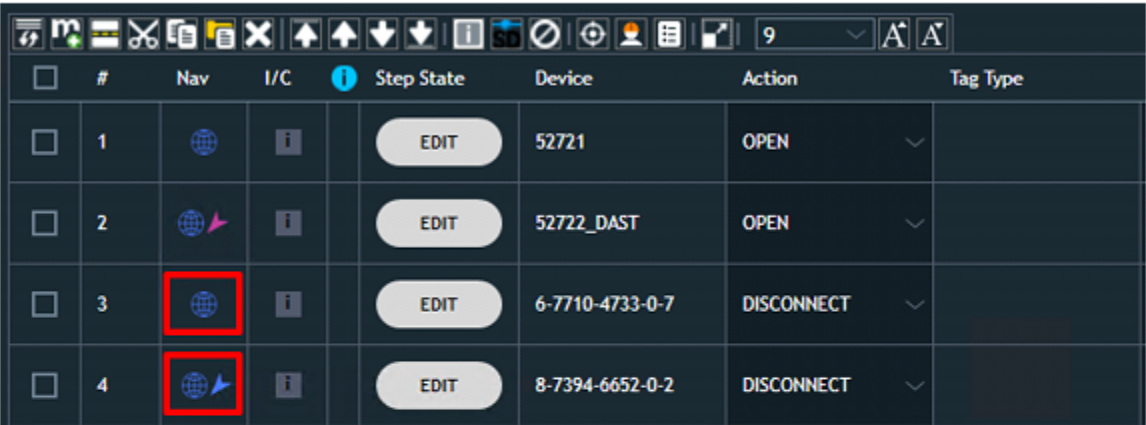
Show PIP for Container Objects: Select this option to enable the picture-in-picture window on the network view when the operator uses the Locate function from the Switching Order for container objects (for example, switch cabinets, vaults, reclosers, and so on).

- **Zoom Level:** Select the zoom level at which the located container object is shown on the PIP window.

SCADA Displays Not Available: If this option is selected, the automated switching devices are navigated on not SCADA displays (for example, loop schematic or station internal displays); the alternate navigation does not attempt to call up a SCADA display. This option should be used for customers that do not have SCADA displays to switch automated devices from.

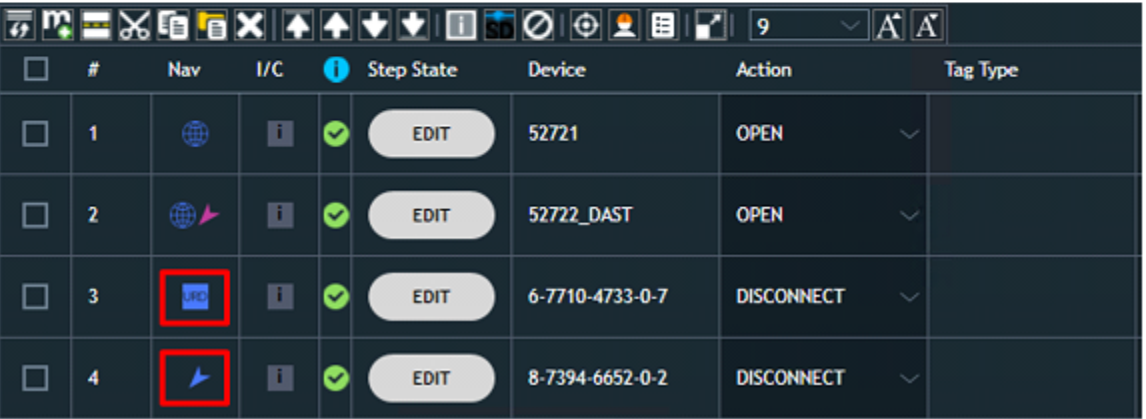
Use Geo Navigation for URD devices: Select this option to enable locating underground and container devices on the network view. When this option is not selected, the devices are located only on the URD schematic or container internal views.

Geo navigation for schematic devices is enabled:



☐	#	Nav	I/C	Step State	Device	Action	Tag Type
☐	1			EDIT	52721	OPEN	
☐	2			EDIT	52722_DAST	OPEN	
☐	3			EDIT	6-7710-4733-0-7	DISCONNECT	
☐	4			EDIT	8-7394-6652-0-2	DISCONNECT	

Geo navigation for schematic devices is disabled:



☐	#	Nav	I/C	Step State	Device	Action	Tag Type
☐	1			EDIT	52721	OPEN	
☐	2			EDIT	52722_DAST	OPEN	
☐	3			EDIT	6-7710-4733-0-7	DISCONNECT	
☐	4			EDIT	8-7394-6652-0-2	DISCONNECT	

Use KEEPLAYERSETTINGS: If this option is selected, the current layer settings are maintained during navigation to geographic views. If unchecked, all layer settings revert to their default states, including Laterals, Non-Reconfigurable Sections, Future, Labels, and so on.

Isolation Switch Navigation URL: A URL that is used when navigating to an isolation switch from a planned request document.

Ground Navigation URL: A URL that is used when navigating to a ground from a planned request

Customer Navigation URL: A URL that is used when navigating to a customer from a planned request document.

Switch Step Grid Colors	
Real-time	
Edit (Even Rows)	Aa
Edit (Odd Rows)	Aa
Selected	Aa
Issuing	Aa
Issued (Manual)	Aa
Issued (State Feedback)	Aa
Issued (State Matches)	Aa
Completed	Aa
Cancelled	Aa
Read-Only Cell	Aa
Error	Aa
Validate With No Change	Aa
Validate With Change	Aa
Validate With Warnings	Aa
Study Mode	
Edit (Even Rows)	Aa
Edit (Odd Rows)	Aa
Selected Step	Aa
Implementing	Aa
Implemented	Aa
Read-Only Cell	Aa
Error	Aa

Click the color box to set the field background color, and click the "Aa" letters next to each box to choose the text color.

Switching Steps									
#	Nav	I/C	Step State	Device	Action	Issued To	Instruction		
0.1			ISSUED		COMMENT BY OPERATOR	Al I Owen	Comment by Operator		
1			ISSUED	52304	OPEN	Eugene B	OPEN 52304		
1.1			EDIT	52304	CLOSE		CLOSE 52304		

7.4. Switch Steps Separator

Switch Steps Separator

	Switching Step Divider Text	Header
☐	A to B	Move Load from Moss Point Elder Ferry Brk 6598 to 5478
☒	B to A	Move Load from Moss Point Elder Ferry Brk 5478 to 6598

☐ Use Custom Text for Switching Step Separator

Add Delete

The **Switch Steps Separator** section defines how separator lines appear within a Switching Order. At the bottom of this section is the **Use Custom Text for Switching Step Separator** option.

When this option is enabled:

- Free-form text can be entered in the **Switching Step Divider Text** field.
- The entered text is used directly as the separator in exported documents.

When this option is disabled:

- A list of predefined separator pairs can be configured, with each containing a **Divider Text** and a corresponding **Header**.
- These pairs are added to the table and are available for selection from a dropdown when creating switching steps, supporting consistent and reusable separator labels across switching orders.

This configuration supports both standardized workflows using predefined separator pairs and cases that require custom separator text.

7.5. State Name Localization

State Name Localization

Open State Name:

Close State Name:

State Name Localization fields are used for configuring OPEN and CLOSED switch states in local languages. OPEN or CLOSED appears for switches in corresponding states in the Action field of the Switching Order Step pane.

- **Open State Name:** Type "Open" in the local language to display in the Action field of the Switching Steps section.
- **Close State Name:** Type "Closed" in the local language to display in the Action field of the Switching Steps section.

7.6. Employee List for Issued to Field

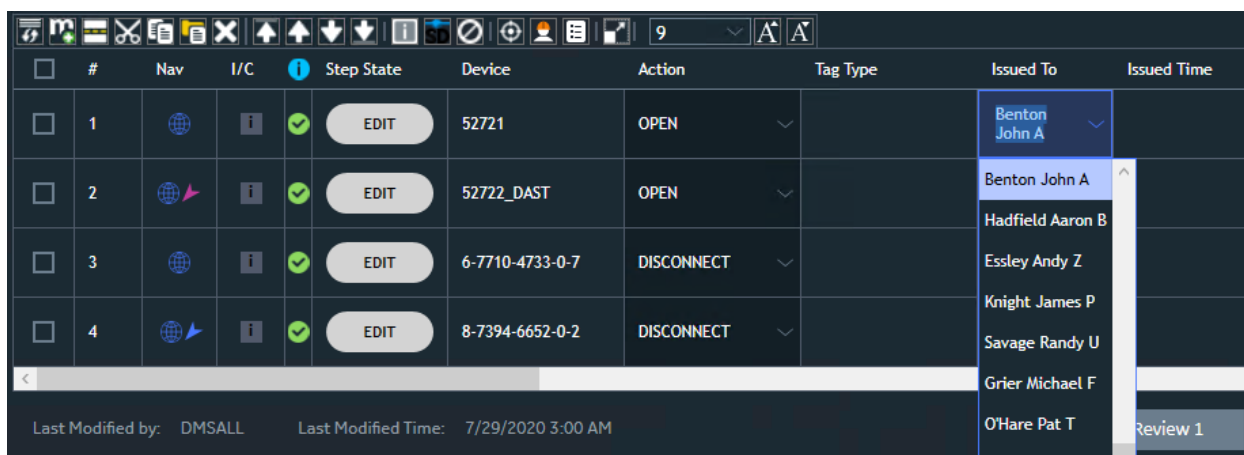
Employee List for Issued To field

☒ Use Employee File

Employee list file path:

%IDMS_ROOT%\config\server\SWOEmployee.txt

This section defines whether the Issued To drop-down list on the Switching Order Step pane is populated with names from an Employee file or if names will be manually entered.



#	Nav	I/C	Step State	Device	Action	Tag Type	Issued To	Issued Time
1			EDIT	52721	OPEN		Benton John A	
2			EDIT	52722_DAST	OPEN		Benton John A	
3			EDIT	6-7710-4733-0-7	DISCONNECT		Hadfield Aaron B	
4			EDIT	8-7394-6652-0-2	DISCONNECT		Essley Andy Z	

Last Modified by: DMSALL Last Modified Time: 7/29/2020 3:00 AM

Review 1

Figure 14. Issued To List

The "Employee List for Issued To field" section contains the following fields:

- **Use Employee File:** If selected, an Employee file is used to populate the Issued To drop-down list .
- **Employee List File Path:** Specifies the path and name of the Employee file to be used to generate the employee list.

If you do not select the Use Employee File check box, you can specify employee names directly in the Issued To field. When you do so, The entered employee names are saved to the RecentEmployees temporary file, located in the folder specified by the Temp Folder Path field on the Folders tab (see [Folders](#)). The saved employee names are available for selection from the Issued To drop-down list.

7.7. Manual Step Action/State Type

Use the Action/State Type configuration option on the Switch Steps tab to assign an Action Type to the Add Manual Step button  on the Switching Steps.

The default action type of a manual step is NODEVICEACTIONS.

Manual Step Action/State Type

Action/State Type: NODEVICEACTIONS v

A manual step is a step that includes no actions. For example, it can be created to add an operator's comment or transfer operation responsibility. The Field Device name, Instruction, Notes, and so on are manually entered for the step in the Switching Order.

#	Nav	I/C	Step State	Device	Action	Tag Type	Issued To
1		i	EDIT		TRANSFER OF OPERATIONS RESPONSIBILITY		
2		i	EDIT		COMMENT BY OPERATOR		
3		i	EDIT		TRANSFER OF OPERATIONS RESPONSIBILITY		

To modify manual step properties, such as state names and instructions, select the NODEVICEACTIONS Action/Sub Type on the Device & State tab (for more information, see [Device & State Tab](#)).

Switching Order Configuration Editor - D:\Eterra\Distribution\FantasyIsland\config\client\SwitchOrderClientConfig.xml

File Help

General Switch Order Related Device & State Tag & Comment Switch Steps Right-click Menu Validation Rules Step Custom Columns Step Columns Order Header Custom Fields

DNOM Type and Sub-type

DNOM Type	Sub Type	Action Type
☐ COMPSW	DMSCOMPOSITESWITCH	TYPECOMPSWT
☒	NODEVICEACTIONS	NODEVICEACTIONS
☐ SWT	DMSSECTIONALIZER	TYPESECTIONALIZER

Add Delete

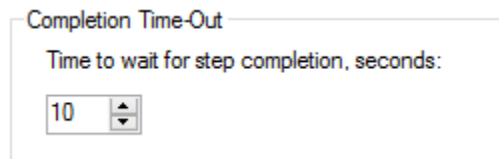
Action Types

Select Action Type: NODEVICEACTIONS Delete Create Action Type: Add

State Name	Function	State Value	Phase Modifier	BackOut State	Tag Value	Copy To Safety Doc	Device Name	Instruction Text
☒ Transfer of operatio...	None	None	All		None	False		Transfer of operatio...
☐ Comment by Operator	None	None	All		None	False		Comment by Operator

Copy Paste Add Delete

7.8. Completion Time-Out

A configuration dialog titled "Completion Time-Out". It contains a label "Time to wait for step completion, seconds:" followed by a numeric spinner control set to the value "10".

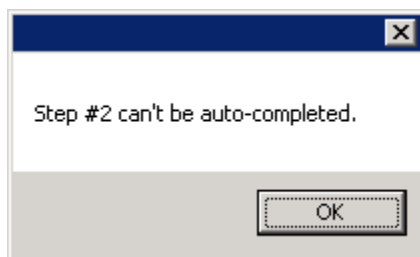
Completion Time-Out

Time to wait for step completion, seconds:

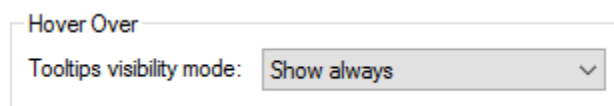
10

This option is used when the Auto-Implement Steps from SWO in RT option is selected (see [Switch Steps Options](#)).

This option configures the time to wait for step completion when the initial auto-completion attempt failed (for example, due to a forbidden tag). If the step is not completed in the specified time period, the error message appears.



7.9. Hover Over

A configuration dialog titled "Hover Over". It contains a label "Tooltips visibility mode:" followed by a dropdown menu currently showing "Show always".

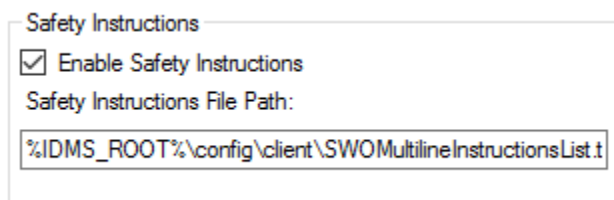
Hover Over

Tooltips visibility mode: Show always

Tooltips visibility mode: Configures the visibility of tooltips when pointing to a switching step.

- **Show always:** Tooltips are always shown when pointing to a switching step.
- **Hide always:** Tooltips are never shown when pointing to a switching step.
- **Show when error:** Tooltips are shown only when a switching step contains validation errors, or if it was automatically changed during validation (for example, the Action field was blank, and then automatically filled during validation).

7.10. Safety Instructions

A configuration dialog titled "Safety Instructions". It contains a checked checkbox labeled "Enable Safety Instructions" and a text field labeled "Safety Instructions File Path:" with the value "%IDMS_ROOT%\config\client\SWOMultilineInstructionsList.t".

Safety Instructions

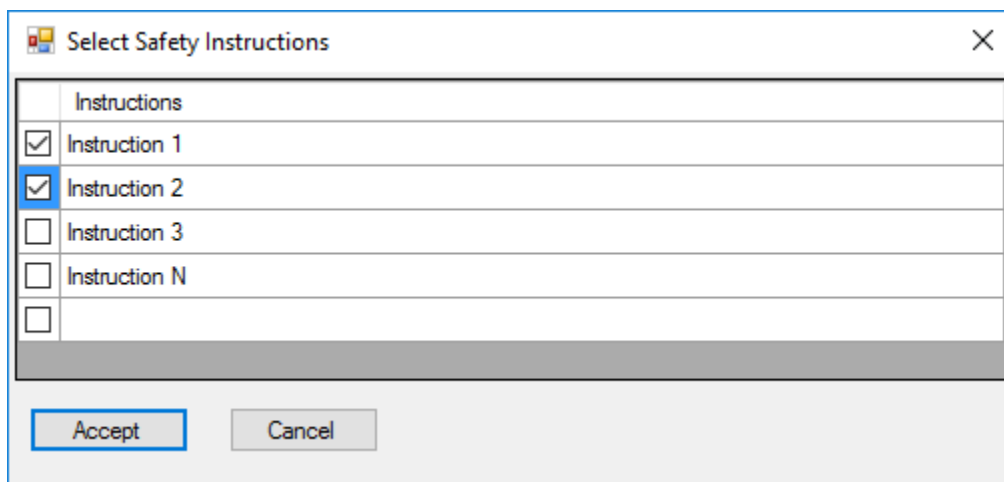
☒ Enable Safety Instructions

Safety Instructions File Path:

%IDMS_ROOT%\config\client\SWOMultilineInstructionsList.t

Enable Safety Instructions: If selected, the Safety Instructions field appears in the Switching Steps section. Multiple-line text can be entered in the Safety Instruction field using the Shift+Enter key combination.

Safety Instructions File Path: Path to the file that contains a list of pre-defined safety instructions for the Safety Instructions field in the Switching Steps section. Each line in the file is a separate instruction. These instructions can be selected from the Select Safety Instructions display, which can be called from the Switching Step toolbar.



8. Right-Click Menu Tab

The Right-click Menu tab is used to configure the context menu for the Switching Steps section in real-time and study modes.

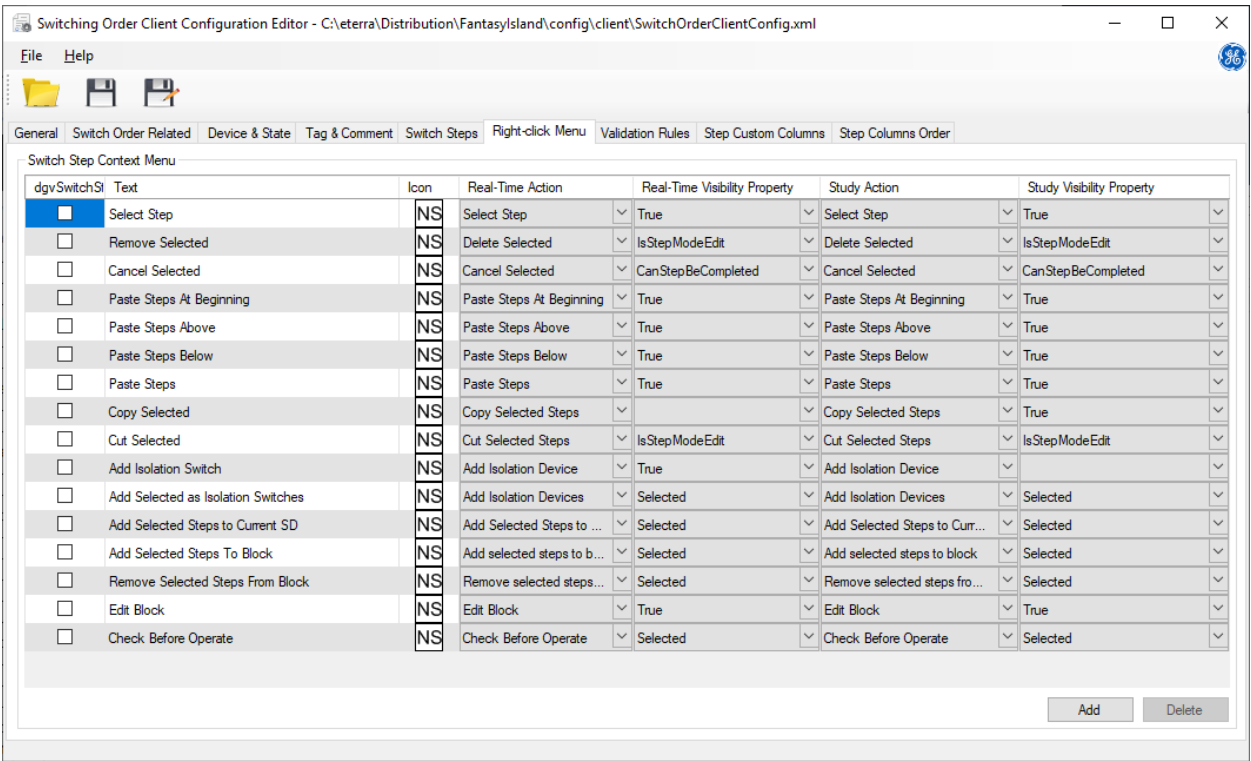


Figure 15. Configuration - Right-Click Menu Tab

- Text:** Sets the text for the context menu option.
- Icon:** Shows a preview of the image for the context menu item. Clicking an image opens a dialog box to select another image. The image folder is specified on the General Properties tab (see [Folders](#)).
- Real-Time/Study Action:** Allows assigning an action to the context menu option. Applies to the context menu in both real-time and study mode.
- Real-Time/Study Visibility Property:** Configures the condition when the right-click menu option is visible on the network view. If no Visibility property is selected, the right-click menu option is visible and enabled at all times.
- Add:** Click to add a context menu option.
- Delete:** Click to permanently remove the selected context menu option.

9. Validation Rules Tab

The Validation Rules tab is used to configure the validation rules and messages that are applied to switching steps.

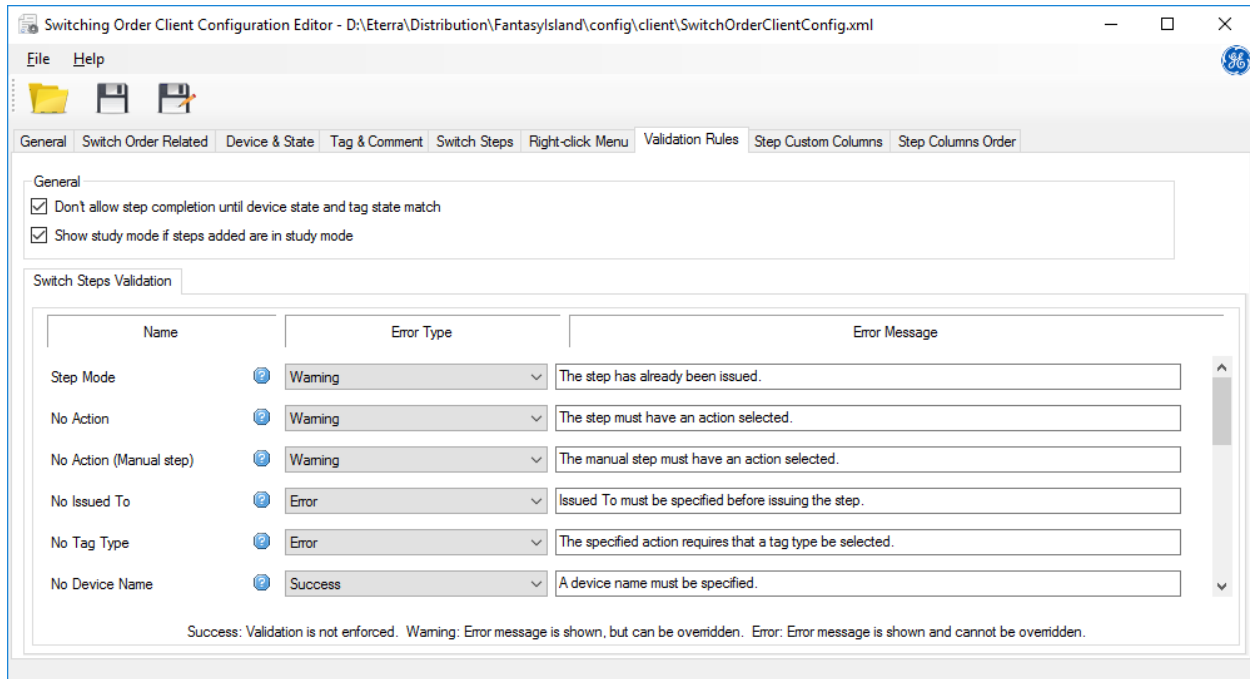


Figure 16. Configuration - Switch Steps Validation Tab

General properties include the following options:

- Do not allow step completion until the device state and the tag state match.
- Show study mode if the added steps are in study mode.

The Validation Rules grid includes the following fields:

- **Name:** Represents the validation scenario for which you can configure the error type and error message.

Hover over the name property to view the scenario description.

- **Error Type:** Error type to be applied to the validation scenario.
 - **Success:** Validation completes successfully. Error message is not shown.
 - **Warning:** Validation completes with a warning. Error message is shown, but can be skipped.
 - **Error:** Validation completes with an error. Error message is shown and cannot be skipped.
- **Error message:** Custom message to be displayed when the validation scenario occurs.

The switching step validation scenarios are as follows:

- **Step Mode:** Check that the step has not been issued or completed already.

- **No Action:** Check that the Action field is not empty.
- **No Action (Manual Step):** Check that the Action field is not empty for manual steps.
- **No Issued To:** Check that the Issued To field is not empty.
- **No Tag Type:** Check that the Tag Type field is not empty if the action includes tagging.
- **No Device Name:** Check that the Device field is not empty.
- **No Device Name (Manual):** Check that the Device field is not empty for manual steps.
- **No Device Found:** Check that the device exists in the system (by ID).
- **Similar Step:** Checks that a switching order does not contain similar steps.
- **Phase not Modeled:** Checks that a switching order does not contain actions on non-modeled phases.
- **Unknown Crew in Issued To:** Checks that a switching order is issued to a valid crew.
- **Non-existing component is referenced in a step:** Checks whether devices that are associated with switching order steps exist in the model. For example, a step may have an action to place a cut on a line which no longer exists in the model.
- **Step components do not have required phases:** Checks whether devices that are associated with switching order steps have required phases. For example, a step may have an action to place an ABC jumper on a line with only A phase modeled.
- **Device is in Use:** When trying to add a step for placing a mobile substation, checks whether the mobile substation is already placed and connected. This option applies when recording operator's actions as switching steps.
- **Mobile Substation Not Found:** When trying to add a step for placing a mobile substation, checks whether the mobile substation exists in the system. This option applies when recording operator's actions as switching steps.
- **HLH: Issue step with Apply Tag:** For the execution phase 2, checks that Apply Tag switching steps are not issued, because Release Tag switching steps are expected. Applies to Hot Line Hold documents only.
- **HLH: Issue step with Remove Tag action:** For the execution phase 1, checks that Remove Tag switching steps are not issued, because Apply Tag switching steps are expected. Applies to Hot Line Hold documents only.
- **Complete Step with Assigned Crew Online (by Non-Supervisor):** When an operator without supervisor privileges attempts to complete a switching step that is dispatched to a crew, which is currently online in the Mobile Enterprise system, a warning appears to suggest that the operator should notify the crew about the step completion. The default error type for this rule is Error.
- **Complete Step with Assigned Crew Online (by Supervisor):** When an operator with supervisor privileges attempts to complete a switching step that is dispatched to a crew, which is currently online in the Mobile Enterprise system, a warning appears to suggest that the operator should notify the crew about the step completion. The default error type for this rule is Warning.

- **Cancel Step with Assigned Crew Online (by Non-Supervisor):** When an operator without supervisor privileges attempts to cancel a switching step that is dispatched to a crew, which is currently online in the Mobile Enterprise system, a warning appears to suggest that the operator should notify the crew about the step cancellation. The default error type for this rule is Error.
- **Cancel Step with Assigned Crew Online (by Supervisor):** When an operator with supervisor privileges attempts to cancel a switching step that is dispatched to a crew, which is currently online in the Mobile Enterprise system, a warning appears to suggest that the operator should notify the crew about the step cancellation. The default error type for this rule is Warning.
- **Take Back a Switching Order with Issued Switching Steps from the Assigned Mobile Crew (by Non-Supervisor):** When an operator without supervisor privileges attempts to take back a switching order with issued switching steps from the crew assigned in the Mobile Enterprise system, a warning appears to suggest that the operator should cancel the issued switching steps before taking back the switching order. The default error type for this rule is Error.
- **Take Back a Switching Order with Issued Switching Steps from the Assigned Mobile Crew (by Supervisor):** When an operator with supervisor privileges attempts to take back a switching order with issued switching steps from the crew assigned in the Mobile Enterprise system, a warning appears to suggest that the operator should cancel the issued switching steps before taking back the switching order. The default error type for this rule is Warning.
- **Unassign from mobile which has issued steps:** When an operator without supervisor privileges attempts to unassign a crew from a switching order with issued switching steps from the crew assigned in the Mobile Enterprise system, a warning appears to suggest that the operator should cancel the issued switching steps for this crew, before unassigning the crew. The default error type for this rule is Warning.
- **Unassign by Supervisor from mobile which has issued steps:** When an operator with supervisor privileges attempts to unassign a crew from a switching order with issued switching steps from the crew assigned in the Mobile Enterprise system, a warning appears to suggest that the operator should cancel the issued switching steps for this crew, before unassigning the crew. The default error type for this rule is Warning.

10. Step Custom Columns

The Step Custom Columns tab is used to configure and add custom columns to the Switching Steps section. The custom columns can show values returned from the server for the associated device.

SWOC	Header Name	Serialize Key	Queried Value	RT Visibility	ST Visibility	PDF/HTML Exportable & Printable	Editable
☐	DNOMX	DNOMX	%DNOMX	☒	☒	☒	☒
☐	SITE	SITE	%SITE	☒	☒	☒	☐
☐	LOCATION	LOCATION	%LOCATION	☒	☒	☒	☒
☐	DYNFEEDERNAME	DYNFEEDERNAME	%DYNFEEDERNAME	☒	☒	☒	☒
☐	TOPOSUBNAME	TOPOSUBNAME	%TOPOSUBNAME	☒	☒	☒	☒
☐	SWITCHCUSTCNT	SWITCHCUSTCNT	%SWITCHCUSTCNT	☒	☒	☒	☒

Add Custom Column Delete

Figure 17. Configuration - Step Custom Columns Tab

- **Header Name:** The header name of the custom column.
- **Serialize Key:** The internal name of the custom column header.
- **Queried Value:** The name of the query object (query helper) to query the device data from the DNOM model. Examples of query helpers: ALTID, LOCATION, DYNFEEDERNAME, TOPOSUBNAME, and SWITCHCUSTCNT.

For more information about querying the DNOM model, refer to the *ADMS Modeling Reference Guide*. For the list of device query helpers, refer to the *ADMS Modeling Reference Tables*.

Note

The percent ("%") symbol must be placed before the text to recognize it as a query helper.

- **RT Visibility:** When selected, the column is visible when switching orders are in the real-time mode.
- **ST Visibility:** When selected, the column is visible when switching orders are in the study mode.
- **PDF/HTML Exportable:** When selected, enables exporting to PDF/HTML files.
- **Editable:** When selected, allows editing of the custom column content.
- **Add Custom Column:** Adds a new blank row.
- **Delete:** Deletes the selected row.

Tag Type	Issued To	Instruction	Comments	DNOMX	SITE	SWITCHCUSTCNT	ALTID	LOCATION	DYNFEEDERNA...	TOPOSUBNAME
-		OPEN 52721		937733.437500	6-7612-2617-0-8	2827	6-7612-2617-0-8	937733 811005	F8863	RAVEN
-		Disconnect Transformer		937628.375000	8018 West St RAVEN	1	8018 West St RAVEN	937628 809454	F8863	RAVEN
-		OPEN 10051-1		937645.625000	6-7611-2788-0-1-E	3	6-7611-2788-0-1-E	937646 809597	F8863	RAVEN

Figure 18. Switching Steps Custom Columns

11. Step Columns Order Tab

The Step Columns Order tab is used to customize the column order and width in the Switching Steps section.

	Header Name	Field Name	Custom	Column Width
☐		Selected	No	30
☐	Step	StepNumber	No	30
☐	Nav	NavImage	No	30
☐	AltNav	AltNavImage	No	30
☐	I	IssueImage	No	30
☐	C	CompleteImage	No	30
☐	Validation	Validation	No	100
☐	Step State	StepState	No	100
☐	Device	formattedDeviceName	No	110
☐	Action	SwitchStepActions	No	120
☐	Tag Type	SwitchStepTagTypes	No	100
☐	SD	isPartOfSafetyDoc	No	30
☐	Issued To	SwitchStepIssuedToRecent	No	180
☐	Issued Time	issuedtime	No	180
☐	Completed Time	actualtime	No	180
☐	Instruction	StepInstruction	No	300
☐	Safety Instruction	SafetyInstruction	No	300
☐	Comments	comment	No	300
☐	Issued By	issuedby	No	180
☐	DNOMX	DNOMX	Yes	120
☐	SITE	SITE	Yes	120

Figure 19. Configuration - Step Columns Order Tab

- **Mode:** Switching order mode (Real Time, Study, or Daily Log). This field allows customizing the Switching Steps grid for the selected mode.
- **Allow to Rearrange Column Order at Run-Time:** If selected, the user can rearrange columns at run time.

The column order user preferences are automatically stored in an XML file and are applied to all Switching Orders operated by the user. The XML files are stored in the C:\Users\<User>\AppData\Roaming\General Electric Technology GmbH\Advanced Presentation Framework\<Version> directory with the name SwitchingOrder_RecentColumnOrder.<UserName>.

- **Header Name:** The column header name as it appears in the Switch Step pane.
- **Field Name:** The column header name syntax.
- **Custom:** Indicates if the column is custom or default.
- **Column Width:** Sets the column width in pixels.

To change the column order, select a column and use the arrow buttons at the right side of the tab to move the column up or down.



Note

Study mode column preferences are also applied to switching order templates.

12. Mobile Alarms Tab

The Mobile Alarms tab allows you to configure messages for alarms that appear as toast notifications in the ADMS client to inform the operator about the switching order updates in the Mobile Enterprise system.

The information that can be captured in a message is limited by the allowable query fields, which are listed for the given message type in the Mobile Alarms Types and Arguments List table.

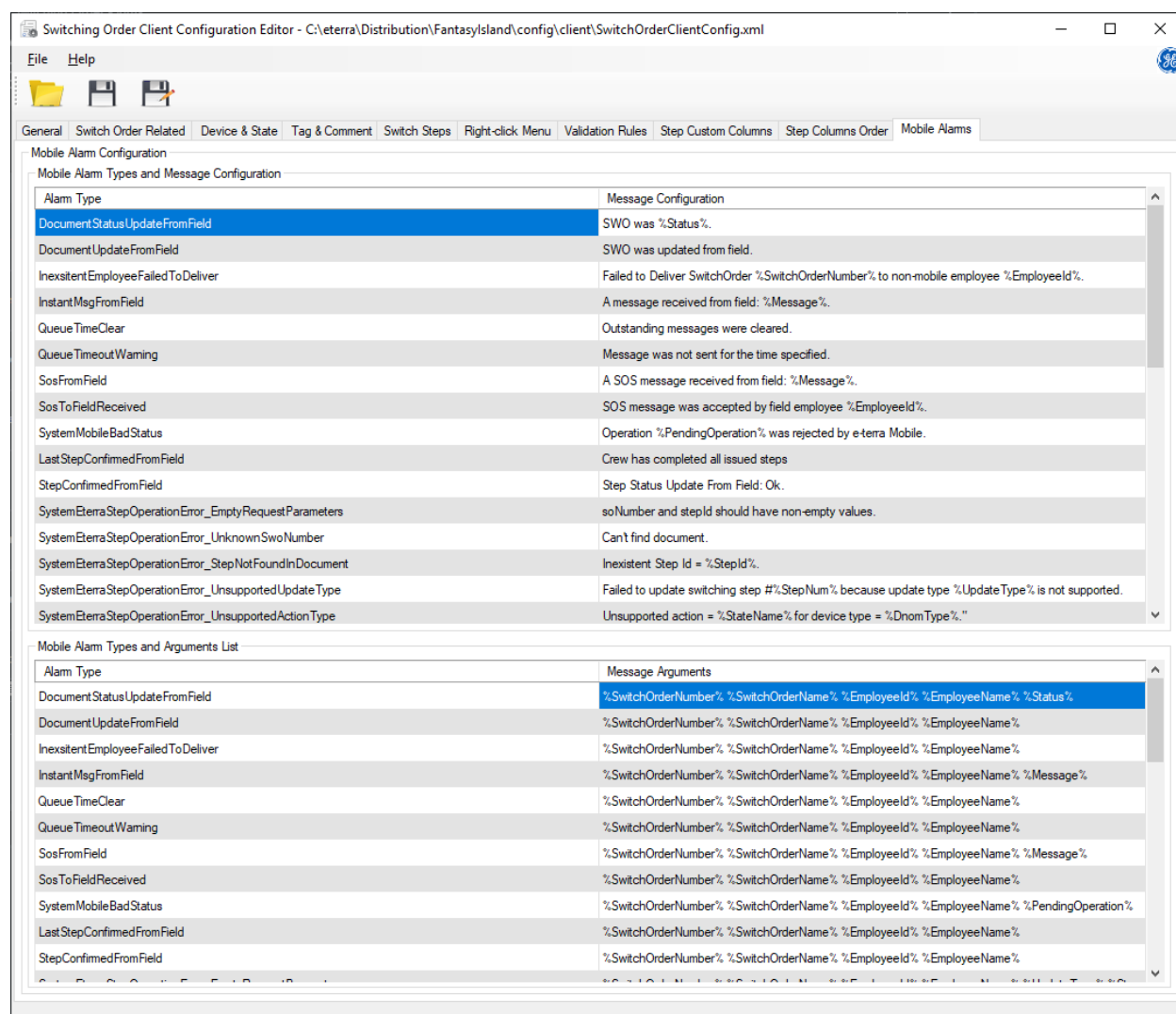


Figure 20. Configuration - Mobile Alarms Tab

The Mobile Alarms Types and Message Configuration table allows you to define a human-readable message for every message type to appear in the alarm.

The Mobile Alarms Types and Arguments List table contains all message arguments available for each message type. It serves as the analyst's reference.

To configure a chronology message:

1. Copy the needed arguments from the Mobile Alarm Types and Arguments List table into the corresponding message type fields in the Mobile Alarm Types and Message Configuration table, and then edit the messages as needed.
2. Save the changes via the File > Save menu.

The table below shows the available query fields for all message types.

Table 1. Available Query Fields for All Message Types

Message Type	Available Query Fields
DocumentStatusUpdateFromField	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, Status
DocumentUpdateFromField	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
InexistentEmployeeFailedToDeliver	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
InstantMsgFromField	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, Message
QueueTimeClear	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
QueueTimeoutWarning	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
SosFromField	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, Message
SosToFieldReceived	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
SystemMobileBadStatus	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, PendingOperation
LastStepConfirmedFromField	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
StepConfirmedFromField	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
SystemEterraStepOperationError_EmptyRequestParameters	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_UnknownSwoNumber	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_StepNotFoundInDocument	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_UnsupportedUpdateType	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
Failed to update switching step #%%StepNum% because update type %UpdateType% is not supported.	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName

Message Type	Available Query Fields
SystemEterraStepOperationError_UnsupportedActionType	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
Unsupported action = %StateName% for device type = %DnomType%."	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_UnsupportedDnomType	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
Unsupported device type %DnomType%."	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_ScadaIsNotSupported	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_DisConnectionBroken	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_DisReportsFail	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_EditStepsAreNotSupported	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_CompletedStepsAreNotSupported	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_CancelledStepsAreNotSupported	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_HistoricalStepsAreNotSupported	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_NotIssuedStepsAreNotSupported	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraStepOperationError_CrewIdNotMatched	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, UpdateType, StepId, StepNum, DnomType, StateName
SystemEterraEmployeeOperationError	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
SystemEterraSWOOperationError	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName, Status
SystemMobileFailedToDeliver_SWO	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName

Message Type	Available Query Fields
SystemMobileFailedToDeliver_HeaderUpdate	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
SystemMobileFailedToDeliver_StepUpdate	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
SystemMobileFailedToDeliver_Assign	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
SystemMobileFailedToDeliver_Unassign	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
SystemMobileFailedToDeliver_Dispatch	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
SystemMobileFailedToDeliver_Takeback	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
SystemMobileFailedToDeliver_SosToField	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
GenericInfo_SetAllDevicesOffline	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
GenericInfo_LogoffAllUsers	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName
GenericInfo_ClearDispatchStates	SwitchOrderNumber, SwitchOrderName, EmployeeId, EmployeeName

13. STEPADD Command

All switching order related commands start with one of the following:

```
netview://SO.RT
netview://SO.ST
```

Where:

- SO indicates "Switching Order" to the command handling function of Browser and ADMS.
- RT or ST indicates the active dynamics: RT for real-time mode or ST for study mode.

Adding the token "STEPADD" to the commands above creates a Step Add command.

The next part of the command identifies the device to be added to the switching order step. The syntax of the device identification depends on whether the step is added from the DMS or the SCADA display.

When the Switching Order client receives a STEPADD command, it issues queries to the NetServer:

- **Component ID Query:** This request queries for Component ID, which is needed to proceed with a Device Info query. Used only when the STEPADD command is issued from a SCADA display.

The Component ID query requires no configuration values. It is performed based on the composite key required to be supplied in the command.

- **Device Info Query:** Based on input values, this query returns all required data for the DNOM device associated with the switching step.

This query is performed for STEPADD commands initiated from both SCADA and DMS displays, for all devices.

STEPADD Command Syntax for SCADA Display

The only device identifier available when a device is selected on a SCADA display is the "composite key". All SCADA devices modeled in DMS have a DNOM representation; the SCADA composite key links the individual visual representation of a given device to its corresponding database record.

The composite key allows getting the device Component ID, Substation Name, Component Type, and Components Sub Type. Then, using Component ID, a regular query for all necessary DNOM device data can be issued, as if the user command had been performed from a DMS display.

When adding a switching order step from a SCADA display, the syntax of a STEPADD command is in the format:

```
netview://SO.RT/STEPADD?SCADACOMPKEY=<SCADA COMPOSITE KEY>
```

A command issued from a SCADA display looks like the following:

```
netview://SO.RT/STEPADD?SCADACOMPKEY=%ck%
```

In the following example, an actual composite key is substituted in the command:

```
netview://SO.RT/STEPADD?SCADACOMPKEY=SCADEK=SUBSTN SUBSTN=RAVEN DEVTYP=CB DEVICE=8863F MEAS=M  
POINT=STTS
```

STEPADD Command Syntax for DMS Displays

Adding switching order steps from a DMS display differs from the SCADA display procedure since more device-specific information is available when the device is selected. A switching order step can be added based on this device-specific information.

The simplest form of the STEPADD command issued from a DMS display requires the following four pieces of information:

- Active dynamics name
- Substation name the device belongs to
- Device ID
- DNOM object type (Viewer Configuration type) of the device

The syntax of a STEPADD command issued from a DMS display is:

```
netview://%DYN/SWITCHORDER/STEPADD//%TOPOSUBNAME/%ID?COMPTYPE=%DNOMOBJECTTYPE
```

The example below demonstrates the command to create a switching order for a device in a grid tabular:

```
NETVIEW://SO.%DYN/STEPADDNEW//%STATION/%DEVICEID?COMPTYPE=%DEVICETYPESTRING&COMPNAME=%DEVICENAME&  
mp;CUOPEN=FALSE
```

The example below demonstrates a command in a device-specific toolbar or context menu for issuing a switching step from a DMS display:

```
netview://%DYN/SWITCHORDER/STEPADD//%TOPOSUBNAME/%ID?COMPTYPE=%DNOMOBJECTTYPE
```

In the following example, actual arguments are substituted in the command:

```
netview://SO.RT/STEPADD//RAVEN/10348309?COMPTYPE=DMSSWITCH
```

STEPADD Command Syntax for Temporary Devices

You can also add switching order steps for creating temporary devices (cuts, jumpers, and temporary grounds). These commands are available from toolbars and context menus for cuts, jumpers, and temporary grounds.

The example below demonstrates a command in a cut toolbar or context menu for creating a cut

from the geographic network view:

```
NETVIEW://%DYN/SWITCHORDER/STEPADDCUT//%TOPOSUBNAME/%ID?DNOMTYPE=%DNOMRECORDNAME&COMPTYPE=%DNOMOBJECTTYPE|%CUTTYPE
```

13.1. Command Workflow

The sections below demonstrate the STEPADD command workflow.

13.1.1. Sub Type Lookup Based on the COMPTYPE Command Argument

To construct a Device Info query, the DNOM type of the device is required. The DNOM type can optionally be passed in the STEPADD command using the "DNOMTYPE=<DNOMTYPE>" command argument; however, in most cases the DNOM type is taken from the switching order configuration file.

The DNOM type is taken based on the Component Type. For example, the Component Type of a switch is DMSSWITCH, and the DNOM type is SWT. Thus, normally, if the STEPADD command is issued from a DMS display, it contains the command argument "COMPTYPE=%DNOMOBJECTTYPE".

As an example, after a user clicks on a switch and requests to add a step, the STEPADD's COMPTYPE command argument resolves to COMPTYPE=DMSSWITCH. When processing the command, the Switching Order client performs a configuration lookup in the Sub Type column, as shown in the image below.

DNOM Type and Sub-type		
DNOM Type	Sub Type	Action Type
☐ SWT	☐ DMSRECLOSE	TYPE RECLOSE
☐ TF	☐ DMSREGULATOR	TYPE TAPCHR
☒ SWT	☒ DMSSWITCH	TYPE SWT
☐ TMPSWT	☐ DMSTEMPSWITCH	TYPE JUMPER
☐ COMPSW	☐ ESCO THROWOVER	TYPE ESCO THROWOVER

In this example, it finds the row containing "Sub Type=DMSSWITCH" and returns all states configured to this its TYPESWT Action Type.

Action Types

Select Action Type:

Create Action Type:

State Name	Function	State Value	Phase Modifier	BackOut State	Tag Value	Copy To Safety	Device Name	Instruction Text
☒ CLOSE	Status	Close	All	OPEN	None	False	%DEVICENAME...	CLOSE %DEVICE...
☐ OPEN	Status	Open	All	CLOSE	None	False	%DEVICENAME...	OPEN %DEVICEN...
☐ CHECK NORMAL	Status	None	All		None	False		CHECK %DEVICE...
☐ APPLY TAG	Status	None	All	REMOVE TAG	Apply	False		APPLY TAG ON %...
☐ OPEN AND TAG	Status	Open	All	REMOVE TAG AN...	Apply	True		OPEN %DEVICEN...
☐ REMOVE TAG AN...	Status	Close	All	OPEN AND TAG	Remove	True		REMOVE TAG %T...
☐ CHECK CLOSE	Status	None	All		None	False		CHECK %DEVICE...
☐ CHECK OPEN AN...	Status	None	All	REMOVE TAG	Apply	True		CHECK %DEVICE...
☐ REMOVE TAG	Status	None	All	APPLY TAG	Remove	True		REMOVE TAG %T...
☐ OPEN PHASE A	Status	Open	A	CLOSE PHASE A	None	False	%DEVICENAME...	OPEN PHASE A O...
☐ CLOSE PHASE A	Status	Close	A	OPEN PHASE A	None	False	%DEVICENAME...	CLOSE PHASE A ...

After the Device Info query results are received from NetServer, the Switching Order client populates the Action combo box with all appropriate device actions and selects the device Action.

#	Nav	I/C	Step State	Device	Action	Tag Type
1				52721	CHECK NORMAL	
2				52722_DAST	CLOSE	
3				6-7710-4733-0-7	OPEN	
4				8-7394-6652-0-2	CHECK NORMAL	

In addition to specifying the DNOM type for the switching step, the Component Type lookup selects the State Name associated with the switching step. The State Name is used for loading the Action-related configuration, as well as device name and instruction formatting.

The switching step's State Name is selected based on the Sub Type. Using the Device Info query results, the step's Action is set based on the current state of the device at the time the switching step is added, if configured accordingly. This functionality aims to determine the Action, which is opposite of the device's current state.

For example, if the current state of a circuit breaker is Closed, the Device Info returns the current state as 1. The configuration lookup of state values finds a match in row 1 and uses the BackOut State value to select the Action, which in this case will be set to Open.

In addition to the selected Action, the current row of the State Name is used to format the Device Name and Instruction for the given step. If the Device Name Formatting field is empty, the Device Name is displayed with the modeled name. For example, if the modeled device name is "8862F", the point name is "STTS", and the configured formatting string is "%DEVICENAME%_%POINTID%", the

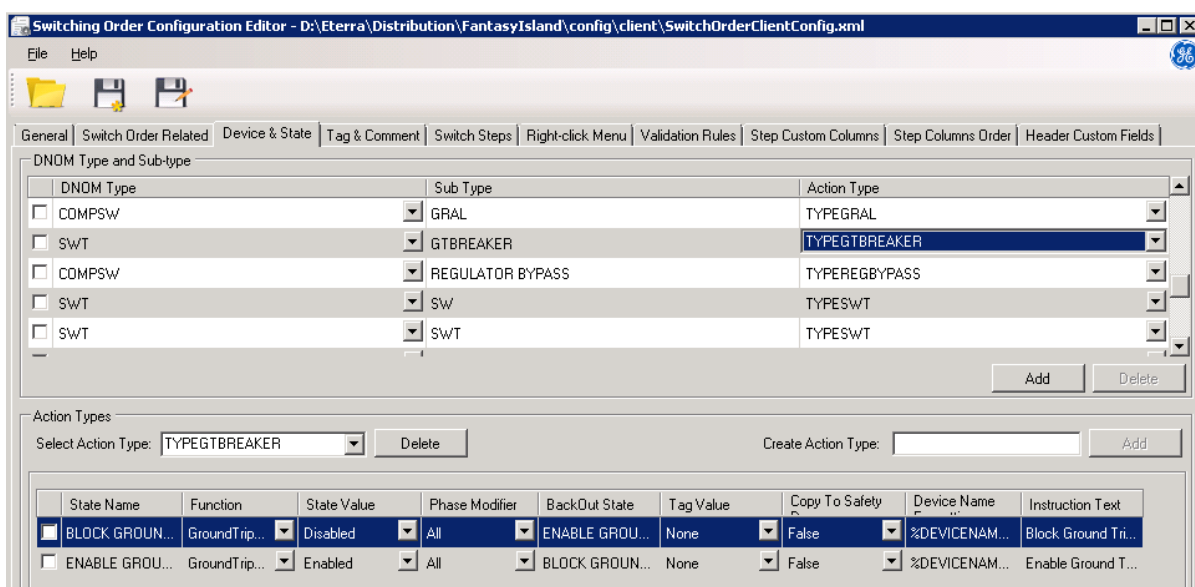
formatted Device Name is "8862F_STTS". Similarly, if the configured Instruction Text is set to "CLOSE %DEVICENAME%", the formatted Instruction is displayed as "OPEN 8862F_STTS".

13.1.2. Action Type Lookup Based on the SUBTYPE Command Argument

A Component Type can be associated with several Sub Types to use an Action Type different from the default value resulting from using the COMPTYPE argument as described above.

For example, consider a SCADA circuit breaker with a Ground Trip measurement. The default Component Type for the breaker device is DMSBREAKER, but the user may need to have a very narrow set of available actions related to a ground trip switching step. To accomplish this, the user configures a Sub Type specifically for this case and associates it with a custom Action Type, as described below.

- Associate "Sub Type=GTBREAKER" with the "Action Type=TYPEGTBREAKER", which has only two available states related to the Ground Trip measurement.



13.2. Command Arguments

The following STEPADD command arguments can be used to create switching order steps:

- **DAILYLOG:** If specified, the step is added to the daily log. If not specified, it is added to the active switching order. If no active switching order exists, a new switching order is created.
- **SCADACOMPKEY=<Composite Key>:** Used when adding steps from SCADA displays. Usually, the command is "SCADACOMPKEY=%ck%". It picks up the composite key from the instance of the picture where the command was issued. The following is a command example with the resolved composite key:

```
netview://SO.RT/STEPADD?SCADACOMPKEY=SCADEK=SUBSTN SUBSTN=RAVEN DEVTYP=CB DEVICE=8863F MEAS=M  
POINT=STTS
```

- **COMPTYPE=<DNOM object type>**: Used when steps are added from the DMS display's device-specific toolbar or menu in the following format: COMPTYPE=%DNOMOBJECTTYPE%.
- **DNOMTYPE=<DNOMTYPE>**: For example, DNOMTYPE=SWT is used to query for information related to a switch.
- **ACTION=<State Name>**: Command argument to set the Action value on the switching order step. Normally used for manual devices or in cases where the action should not be set based on the current state of the device at the time the step is added. For example, ACTION=ENABLE GT.
- **DISPLAY=<Display Name>**: Command argument to specify from which SCADA display the step was added. For example, if the command DISPLAY=%dis% is used, the proxy "%dis%" is resolved by Browser as the name of the display the step was added from. This value is used to navigate back from the Switching Order UI to the appropriate display.
- **SCADAONLY**: Command argument to indicate that the SCADA point the step is added for is not modeled in DNOM.
- **STEP.<PARAMETER_1>=<VALUE_1>**: Command argument to pass specific switching step parameters. A plain text or DNOM query can be used as a parameter value. Note that the passed values override the default column values. The following is an example command:

```
NETVIEW://DYN/SWITCHORDER/STEPADD//%TOPOSUBNAME/%ID?COMPTYPE=%DNOMOBJECTTYPE&STEP.ISSUEDTO=TEST_I  
SSUED_TO&STEP.COMMENT=%NAME and %ID
```

- **DOC.<PARAMETER_1>=<VALUE_1>**: Command argument to pass specific switching order document properties. A plain text or DNOM query can be used as a parameter value. Note that the passed values override the default column values. The following is an example command:

```
NETVIEW://SO.RT/STEPADDNEW//RAVEN/22239252?COMPTYPE=DMSSWITCH&COMPNAME=51984&CUROPEN=FALSE&DOC.sche  
dStartTime=2017-07-02T19:35:26+03:00&DOC.schedEndTime=2017-07-13T22:35:26+03:00
```

- **POID=<PlannedOutageID>**: Command argument to set the Planned Outage value for the switching order document. Normally used when creating a switching order from a planned outage.
- **SYNCPLOUTAGE**: Queries the OMS server for the associated switching order document name and then opens the switching order document. If the switching order does not exist, a new switching order is created similar to using the STEPADD command. This command also sets the PlannedOutageId field for a switching order.

The following is an example command:

```
NETVIEW://SO.DYN/SYNCPLOUTAGE?STATION=%STATION&DEVICE=%DEVICEID&COMPTYPE=%DEVICETYPESTRIN  
G&COMPNAME=%DEVICENAME&CUROPEN=FALSE&DOC.schedStartTime=%PRIMARYSTARTTIME&DOC.sche  
dEndTime=%PRIMARYENDTIME&DOC.nominalfeeder=%FEEDERNAME&DOC.device=%DEVICENAME&POID=%ID
```

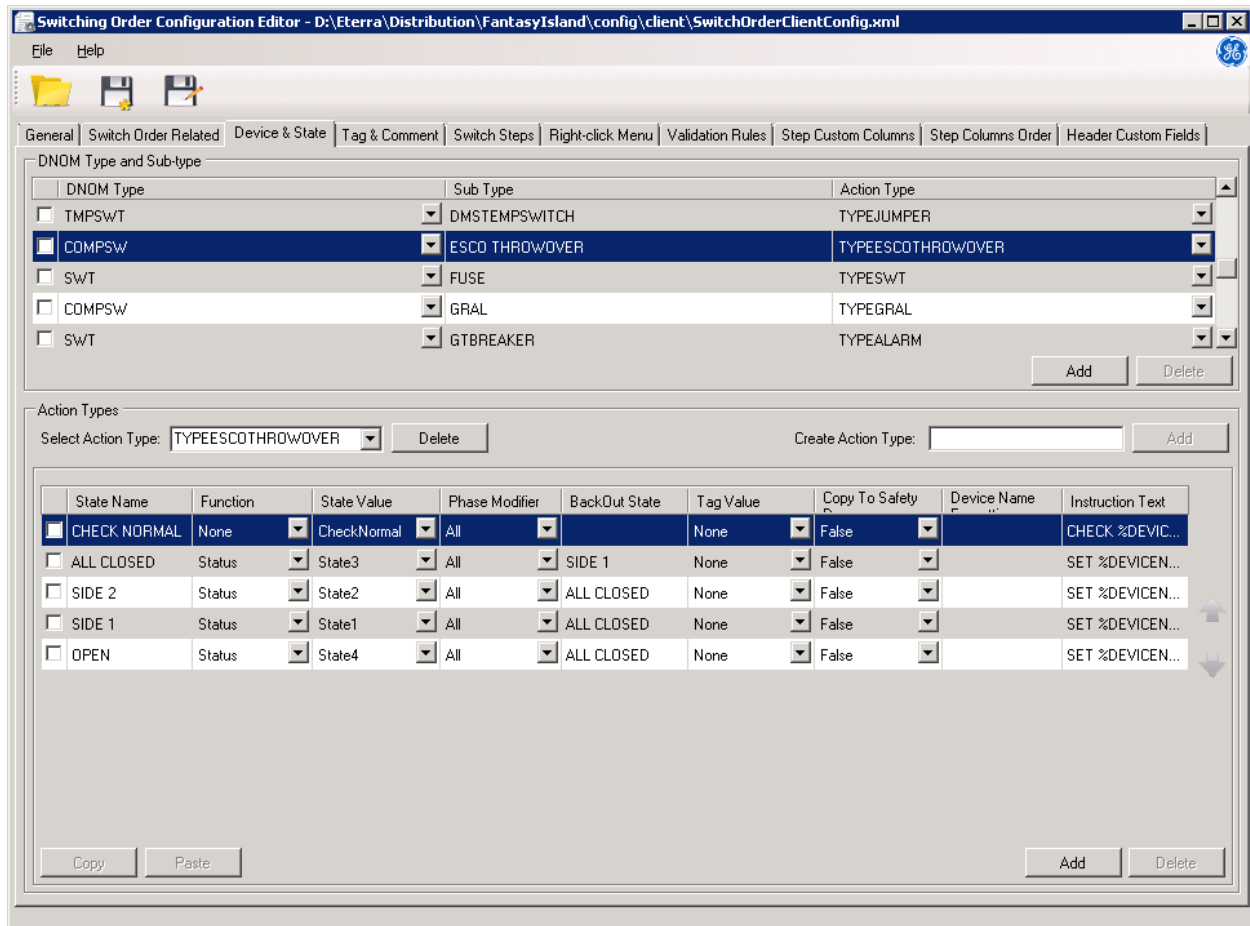
13.3. Adding a Step for a Composite Switch

Adding a Switching Order step for a composite switch requires a special approach. It is not enough to add a step for "COMPTYPE=DMSCOMPOSITESWITCH" because there are a number of distinct types of composite switches (for example, regulator bypass, GRAL, AB switch, throwover, and ESCO throwover).

For composite switches, the Switching Order client knows the specific composite switch type only when it receives the Device Info query results.

As described earlier, the DNOM Type is normally looked up based on the Sub Type. The DNOM Type is required to form the Device Info query. Normally, you need to have a "Sub Type=DMSCOMPOSITESWITCH" entry in the switching order configuration with the "DNOM Type=COMPSW". This configuration is useful to look up the DNOM Type, but it cannot specify a State Type because the type of composite switch to be used cannot be specified.

Therefore, rather than having to look up the DNOM Type, a better solution is to include the command argument "DNOMTYPE=COMPSW" directly on the STEPADD command for a composite switch. In this case, the Switching Order client is not looking up the DNOM type prior to constructing the Device Info query. Instead, it postpones the look up of the State Type until the query results are returned. It then looks up the State Type based on the composite switch type (for example, ESCO throwover or GRAL).



13.4. SCADA-Only Steps

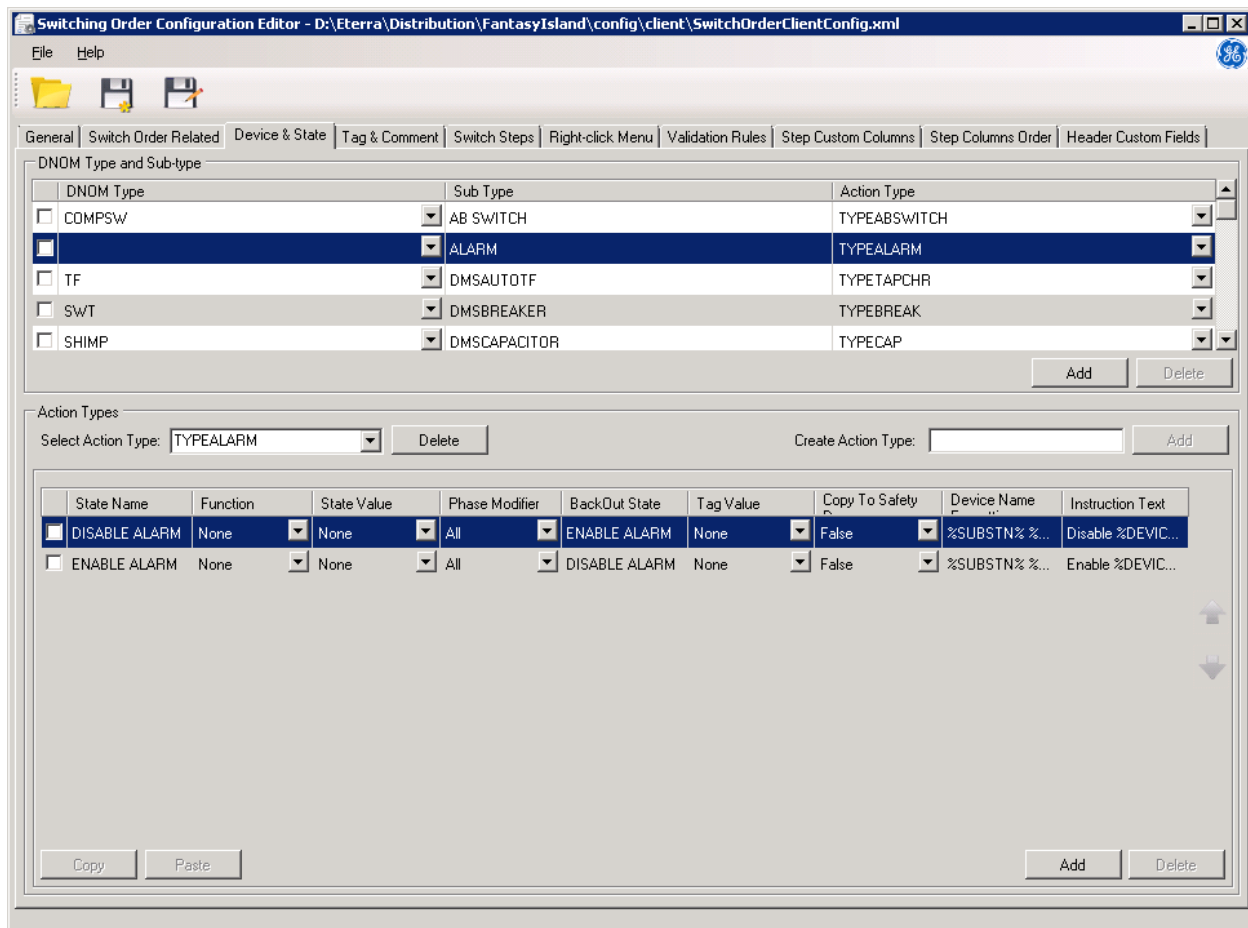
This section describes the commands for adding steps to SCADA-only points and devices.

13.4.1. Adding a Step for a SCADA-Only Point

SCADA displays may have references to SCADA points without a corresponding DNOM device. For example, a SCADA display might have a generic status point that shows whether ALARM is enabled or not. The operator may need to include a step to disable ALARM prior to other regular steps in a switching order. At the end of the switching order, the operator might include a step to enable ALARM to restore the normal state of the system.

As described earlier, during the processing of the STEPADD command, the Component ID query is normally issued to NetServer based on the Composite Key. If this is done for a SCADA-only point, an error occurs, as the device does not exist in DNOM.

To signal the STEPADD command's processing code to process a SCADA-only point, the command must contain the argument SCADAONLY. This instructs the command processing code to add the manual step and not perform any queries for device information.



The example below demonstrates a SCADA-only status point command:

```
netview://SO.RT/STEPADD?SCADACOMPKEY=%ck%&DISPLAY=%dis%&SCADAONLY&
SUBTYPE=ALARM&ACTION=DISABLE ALARM
NETVIEW://RT/SWITCHORDER/STEPADD?USER=DMSALL&SCADAONLY&
SUBTYPE=ALARM&STEP.SITE=NOSCADANOCRY&step.devicesub=NEWYOURK&step.devicename=DEVICENOSCADA&STEP.DNOMX=
012345.00000&STEP.COMMENT=TEST COMMENT&STEP.STATION=TEST&STEP.ISSUEDTO=VMEDVED
```

While processing the command above, the STEPADD command's processing code adds a manual step, looks for the appropriate Action Type based on the search results of "SUBTYPE=ALARM", and selects the Action based on the command argument "ACTION=DISABLE ALARM".

This command also includes the argument "DISPLAY=%dis%", which picks up the name of the display from where the step is added. The display name helps the user navigate from the Switching Order UI back to the display where the step was added.

13.4.2. Adding a Step for a SCADA-Only Device

You can add switching steps for SCADA-only devices that are not modeled in the DNOM model and do not exist in the Distribution Management System. In this case, all information that is usually

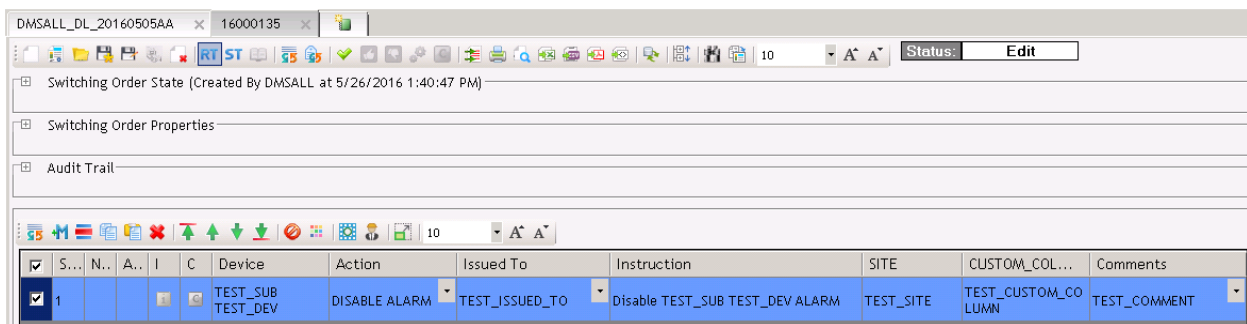
queried from the DNOM model (for example, substation name, device name, and location) needs to be passed into a switching order directly using appropriate URL fields.

The command syntax is as follows:

```
NETVIEW://RT/SWITCHORDER/STEPADD?USER=<PERMIT_USER_NAME>&SCADAONLY&SUBTYPE=<SUBTYPE>&STEP.<PARAMETER_1>=<VALUE_1>&STEP.<PARAMETER_2>=<VALUE_2>&STEP.<PARAMETER_N>=<VALUE_N>
```

For example:

```
NETVIEW://RT/SWITCHORDER/STEPADD?USER=DMSALL&SCADAONLY&SUBTYPE=ALARM&STEP.SITE=TEST_SITE&STEP.DEVICESUB=TEST_SUB&STEP.DEVICENAME=TEST_DEV&STEP.COMMENT=TEST_COMMENT&STEP.ISSUEDTO=TEST_ISSUED_TO&STEP.CUSTOM_COLUMN=TEST_CUSTOM_COLUMN
```



Because data for SCADA-only devices cannot be queried from the DNOM model, the switching order commands need to include the appropriate SCADA database queries. For more information, refer to the Scada product documentation.

13.5. OMS-Related Commands

The following commands can be used to create switching orders for incidents and planned outages:

- **CREATESWOFORINCIDENT:** Creates a switching order for an incident.

To create a switching order with switching steps for all of the incident's switching steps:

```
NETVIEW://SO.%DYN/CREATESWOFORINCIDENT?INCIDENTNAME=%NAME&INCIDENTID=%ID&HISTORICALOMS=TRUE
```

To create a switching order with switching steps for all of the incident's restoration steps:

```
NETVIEW://SO.%DYN/CREATESWOFORINCIDENT?INCIDENTNAME=%NAME&INCIDENTID=%ID&HISTORICALOMS=FALSE&RESTORATION=TRUE
```

- **SYNCINCIDENT:** Creates or opens an existing switching order for an incident.

To create or open a switching order with switching steps for all of the incident's switching steps:


```
NETVIEW://SO.%DYN/SYNCINCIDENT?INCIDENTNAME=%NAME&INCIDENTID=%ID&HISTORICALOMS=TRUE
```

To create or open a switching order with switching steps for all of the incident's restoration steps:

```
NETVIEW://SO.%DYN/SYNCINCIDENT?INCIDENTNAME=%NAME&INCIDENTID=%ID&HISTORICALOMS=FALSE&RESTORATION=TRUE
```

- **CREATESWOFORPLOUTAGE:** Creates a switching order with open and close switching steps for the planned outage device.

```
NETVIEW://SO.%DYN/SYNCPLOUTAGE?STATION=%STATION&DEVICE=%DEVICEID&COMPTYPE=%DEVICETYPESTRING&COMPNAME=%DEVICENAME&DOC.schedStartTime=%PRIMARYSTARTTIME&DOC.schedEndTime=%PRIMARYENDTIME&DOC.nominalfeeder=%FEEDERNAME&DOC.device=%DEVICENAME&POID=%ID
```

- **SYNCPLOUTAGE:** Creates or opens an existing switching order with open and close switching steps for the planned outage device.

```
NETVIEW://SO.%DYN/SYNCPLOUTAGE?STATION=%STATION&DEVICE=%DEVICEID&COMPTYPE=%DEVICETYPESTRING&COMPNAME=%DEVICENAME&DOC.schedStartTime=%PRIMARYSTARTTIME&DOC.schedEndTime=%PRIMARYENDTIME&DOC.nominalfeeder=%FEEDERNAME&DOC.device=%DEVICENAME&POID=%ID
```

13.6. Other Commands

The following commands can be used to manage switching orders:

- **API_STATUS_UPDATE:** Updates a switching order status. Supported status values are: REJECT, APPROVE, DELETE, EXECUTE, COMPLETE.

```
NETVIEW://SO.%DYN/API_STATUS_UPDATE?ACTION=<DocumentStatus>&NAME=<DocumentName>
```

Or:

```
NETVIEW://SO.%DYN/API_STATUS_UPDATE?ACTION=<DocumentStatus>&WORKREQUEST=<DocumentWorkRequestID>
```

For example:

```
NETVIEW:///SO.RT/API_STATUS_UPDATE?ACTION=EXECUTE&WORKREQUEST=DMSALL_AUTO_20210521AA
```

14. Adding Approver Columns to Switching Order Search

You can configure the Switching Order Search grid tabular display to show the first, second, and third approvers.

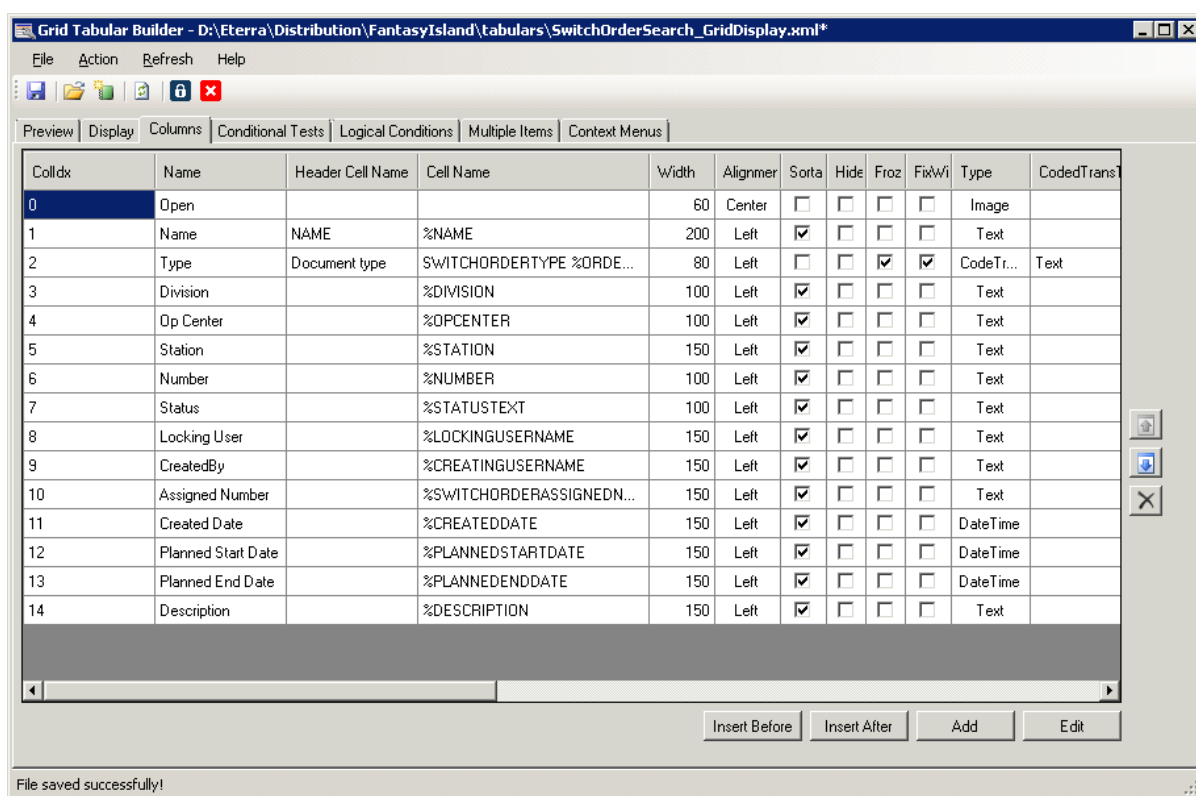
To add the approver column:

1. Open the Grid Tabular Editor.

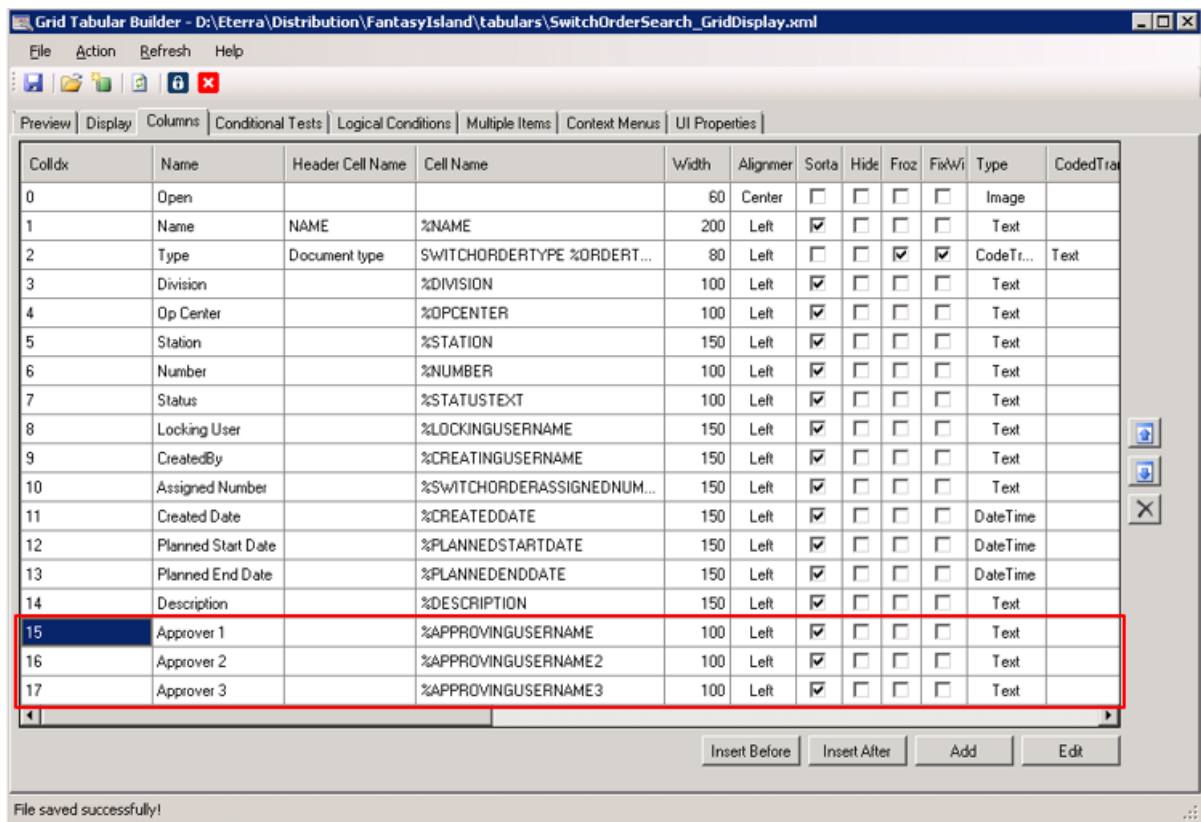
From the Windows Start menu, navigate to All Programs> Eterra> Distribution> Tools> Configuration Tools> Grid Tabular Editor.

The Grid Tabular Editor starts.

2. Select File> Open, and choose the SwitchOrderSearch_GridDisplay file from the list on the Open pop-up display.
3. Click the Columns tab.



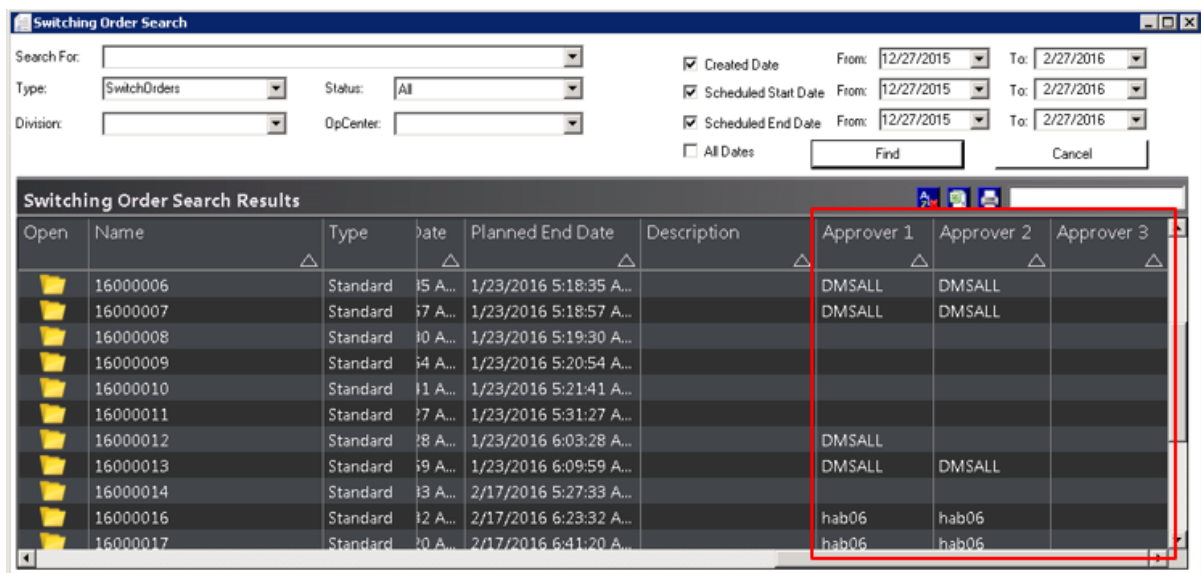
4. Add the three approver columns, as shown below:



The Cell Name fields should be specified as %APPROVINGUSERNAME, %APPROVINGUSERNAME2, and %APPROVINGUSERNAME3.

5. Save the changes via File> Save.
6. Open the Switching Order Search.

The new columns appear on the Switching Order Search display.



15. Customizing Switching Order Appearance

The Windows Presentation Foundation (WPF) UI implementation of the Switching Order application allows you to customize the appearance of the Switching Order application for specific project needs. You can define field labels, controls, and layouts by modifying the appropriate XAML configuration files.

The following pages can be customized using XAML configuration files. The other Switching Order pages (for example, Switching Order Search) can be customized using the Resource Editor tool. For more details, refer to the *Resource Editor User Guide*.

- Daily Log
- Switching Order
- Planned Request
- Hot Line Hold
- Pending Hot Line Hold
- Template

To apply custom Switching Order properties:

1. Configure XAML configuration files for the Daily Log, Switching Order, Planned Request, Hot Line Hold, Pending Hot Line Hold, and Template pages. Configure XAML configuration files that are referenced in the XAML configuration files (for example, configure Document.SharedSections.xaml, which defines sections that are common for all switching order documents.

By default, the XAML configuration files are stored in the D:\Eterra\Distribution\Client\bin\WpfViewer folder. You can manually edit these configuration files using a text editor, such as Notepad++, or using a configuration tool, such as Visual Studio Community to change field labels, controls, layouts, and other switching order property settings.

For more information about XAML, refer to [https://msdn.microsoft.com/en-us/library/hh852724\(v=pandp.51\).aspx](https://msdn.microsoft.com/en-us/library/hh852724(v=pandp.51).aspx).

2. On the General tab of the Switching Order Client Configuration Editor, set paths to XAML files.

WPF Settings		Hot Line Hold Settings	
Switch Order XAML Path	WPFViewer/swo.xaml	☒ Need Hot Line Hold	
Daily Log XAML Path	WPFViewer/dailylog.xaml	Hot Line Hold XAML Path	WPFViewer/HotLineHold.xaml
Template XAML Path	WPFViewer/template.xaml	Pending Hot Line Hold XAML Path	WpfViewer/PendingHLHControl.xaml
Planned Request XAML Path	WPFViewer/PlannedRequest.xaml	Tag type for step actions	Do Not Operate
		Action name for manual step	

Note

If XAML files are invalid or absent under the specified paths, the default page layout is applied.

15.1. Switching Order Elements in XAML Files

This section shows places where particular switching order elements are defined in the associated XAML configuration files. Configuration files that define the Switching Order document page are used as examples.

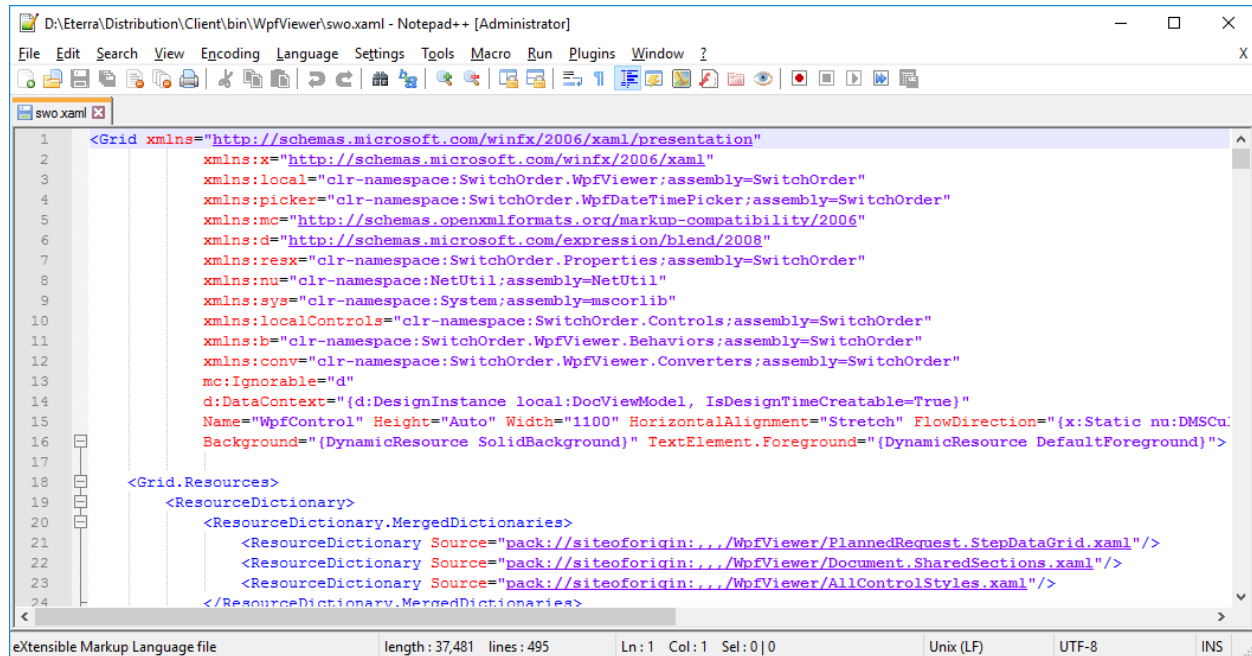
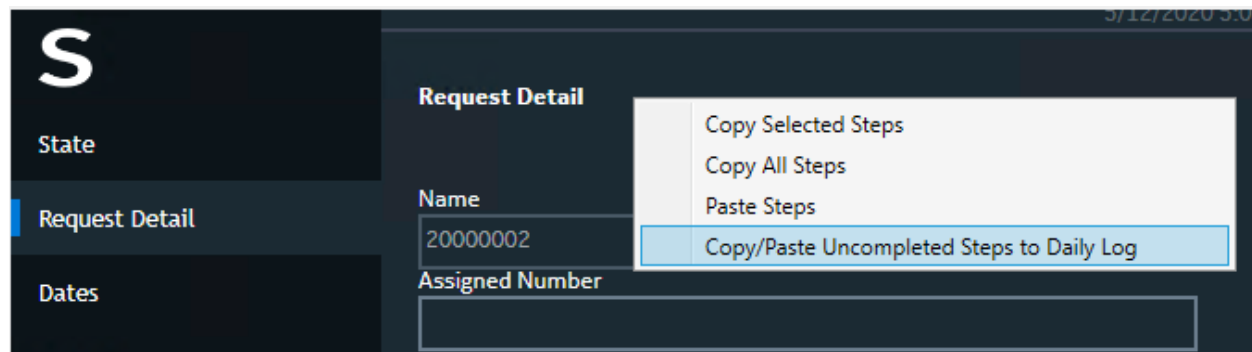
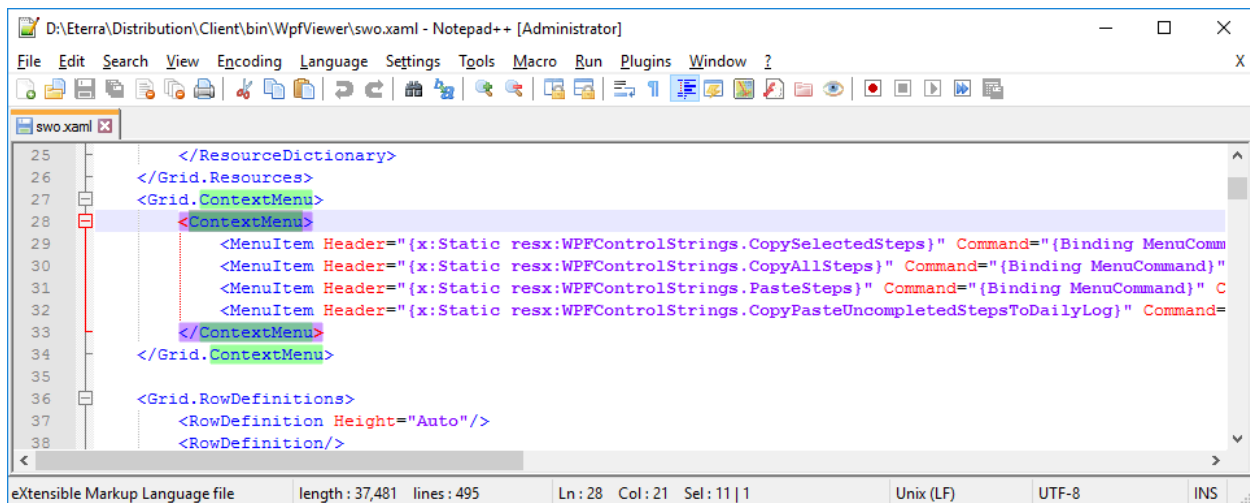


Figure 21. Switching Order XAML Configuration File

15.1.1. Context Menu



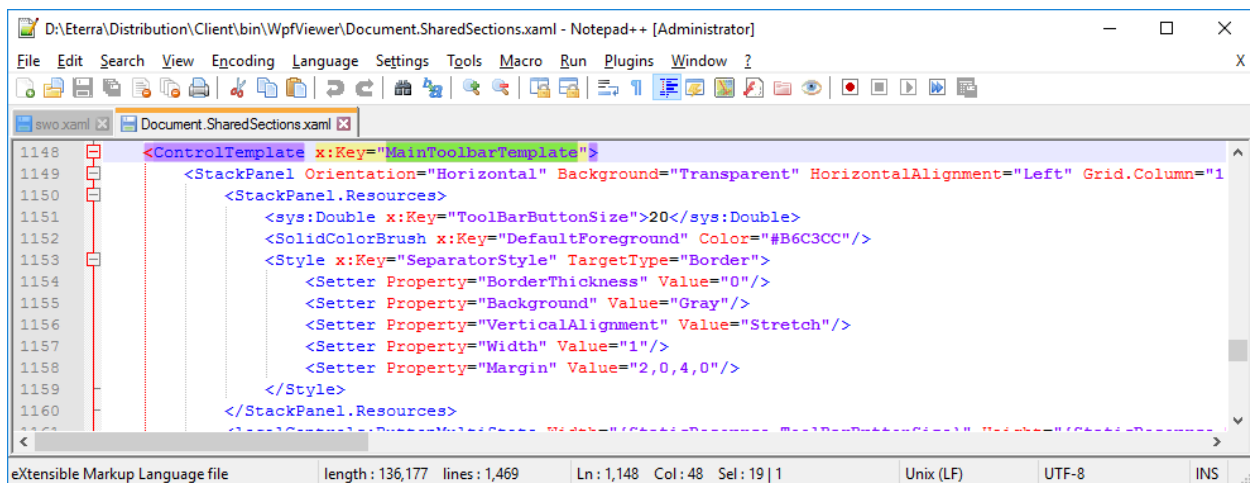
The Switching Order context menu is configured in the ContextMenu node in the swo.xaml file.



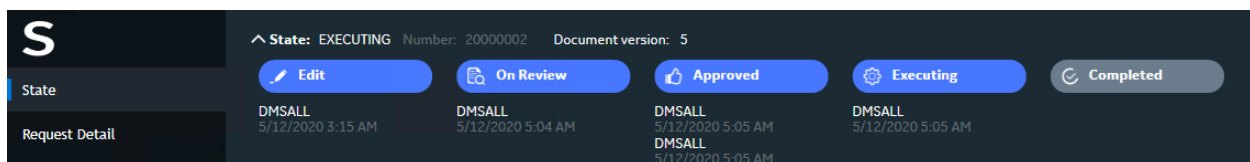
15.1.2. Toolbar



The Switching Order toolbar is configured in the MainToolbarTemplate node in the Document.SharedSections.xml file.



15.1.3. State



The Switching Order State section is configured in the StateSectionTemplate node in the Document.SharedSections.xml file.

```

D:\Eterra\Distribution\Client\bin\WpfViewer\Document.SharedSections.xaml - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
Document.SharedSections.xaml
72 <ControlTemplate x:Key="StateSectionTemplate">
73 <Expander HorizontalAlignment="Stretch" Template="{StaticResource ExpanderControlTemplate}" BorderThickness="0" Padding="0"
74 IsExpanded="{Binding UiStateExpanded}">
75 <Expander.Header>
76 <Grid Grid.ColumnSpan="2" Height="35" Margin="-3,0,0,0">
77 <StackPanel Orientation="Horizontal" VerticalAlignment="Center">
78 <TextBlock Grid.Column="0" FontWeight="Bold"
79 VerticalAlignment="Center" Margin="0,0,0,0" Style="{StaticResource DefaultTextBlockStyle}">
80 <Run Text="{x:Static resx:WPFControlStrings.SectionHeader_State}" />
81 </TextBlock>
82 <TextBlock Text="{Binding State}" Grid.Column="0" Name="SwoState" AutomationProperties.AutomationId="SwoState"
83 VerticalAlignment="Center" Margin="8,0,11,0" Style="{StaticResource DefaultTextBlockStyle}" />
84 </TextBlock>
85 <TextBlock Opacity="0.3" VerticalAlignment="Center" Style="{StaticResource DefaultTextBlockStyle}">
86 <Run Text="{x:Static resx:WPFControlStrings.Number}" />
87 </TextBlock>
88 <TextBlock Text="{Binding SwoNumber}" Name="SwoNumber" Grid.Column="3" AutomationProperties.AutomationId="SwoNum"
89 Opacity="0.3" VerticalAlignment="Center" Style="{StaticResource DefaultTextBlockStyle}" />
90 </TextBlock>
91 <TextBlock Margin="24,0,0,0" VerticalAlignment="Center" Style="{StaticResource DefaultTextBlockStyle}">
92 <Run Text="{x:Static resx:WPFControlStrings.DocumentVersion}" />
93 </TextBlock>
94 <TextBlock Text="{Binding DocVersion}" Name="DocVersion" Grid.Column="3" AutomationProperties.AutomationId="DocVer"
</TextBlock>
</Expander.Header>
</Grid>
</Expander>
</ControlTemplate>

```

length: 136,177 lines: 1,469 Ln: 72 Col: 49 Sel: 20 | 1 Unix (LF) UTF-8 INS

15.1.4. Request Detail

S

State

Request Detail

Dates

Work Details

Equipment

Safety Documents

Audit Trail

Affected Feeders

Request Detail

Name

20000006

Assigned Number

City

Roarke

Service Center

3LIONSSC

Is Planned Switching

☒

Related Incident

2760000010

Planned Outage

RAVEN 6-7512-4121-0-4

Substation

RAVEN

Feeder

F8864

Device

6-7512-4121-0-4

Subtype

None

Last Modified by User

DMSALL

The Switching Order Request Detail section is configured in the DetailsSectionTemplate node in the Document.SharedSections.xaml file.

```

336 <ControlTemplate x:Key="DetailsSectionTemplate">
337     <StackPanel>
338         <Grid Grid.Row="0" Grid.Column="0" Grid.ColumnSpan="2" HorizontalAlignment="Stretch" Height="Auto" >
339             <Grid.ColumnDefinitions>
340                 <ColumnDefinition/>
341                 <ColumnDefinition Width="{StaticResource InterColumnWidth}" />
342                 <ColumnDefinition/>
343                 <ColumnDefinition Width="20" />
344             </Grid.ColumnDefinitions>
345             <Grid.RowDefinitions>
346                 <RowDefinition Height="Auto" />
347                 <RowDefinition Height="Auto" />
348                 <RowDefinition Height="Auto" />
349                 <RowDefinition Height="Auto" />
350                 <RowDefinition Height="Auto" />
351                 <RowDefinition Height="Auto" />
352             </Grid.RowDefinitions>

```

15.1.5. Dates

Dates	
Schedule Start	Actual Start
10/03/2020 00:00:00	
Schedule End	Actual End
10/03/2020 00:05:00	

The Switching Order Dates section is configured in the DatesSectionTemplate node in the Document.SharedSections.xml file.

```

578 <ControlTemplate x:Key="DatesSectionTemplate">
579     <Grid>
580         <Grid.RowDefinitions>
581             <RowDefinition Height="Auto" />
582             <RowDefinition Height="Auto" />
583         </Grid.RowDefinitions>
584         <Grid.ColumnDefinitions>
585             <ColumnDefinition/>
586             <ColumnDefinition Width="{StaticResource InterColumnWidth}" />
587             <ColumnDefinition/>
588         </Grid.ColumnDefinitions>
589         <StackPanel>
590             <Label Content="{x:Static resx:WPFControlStrings.ScheduleStart}" Grid.Row="1" VerticalAlignment="Top" />
591             <Label Content="{x:Static resx:WPFControlStrings.ActualStart}" Grid.Row="1" VerticalAlignment="Top" />
592             <picker:CustomWpfDateTimePicker x:Name="SchedStartTimePicker" Height="Auto" Value="{Binding ScheduleStart}" FormatType="Custom" CustomFormat="{Binding DateTimeFormat}" />
593             <picker:CustomWpfDateTimePicker x:Name="ActualEndTimePicker" Height="Auto" Value="{Binding ActualEnd}" FormatType="Custom" CustomFormat="{Binding DateTimeFormat}" />
594         </StackPanel>

```


15.1.6. Work Details

The screenshot shows the 'Work Details' section of the application. On the left is a sidebar with a search icon 'S' and a list of sections: State, Request Detail, Dates, and Work Details (which is highlighted). The main area is titled 'Work Details' and contains two input fields: 'Work Request' with the value 'DMSALL_AUTO_20201002AA' and 'Search key words'. To the right of these fields is a large text area labeled 'Description'.

The Switching Order Work Details section is configured in the WorkDetailsSectionTemplate node in the Document.SharedSections.xml file.

The screenshot shows the XML file 'Document.SharedSections.xml' open in Notepad++. The file is an xAML document defining a 'ControlTemplate' for 'WorkDetailsSectionTemplate'. The code defines a grid with two rows and two columns. The first row contains a label for 'Work Request' and a text box with a button. The second row is empty. The status bar at the bottom indicates the file is an 'eXtensible Markup Language file' with a length of 136,177 and 1,469 lines.

```

620 <ControlTemplate x:Key="WorkDetailsSectionTemplate">
621 <Grid Grid.Row="1">
622 <Grid.RowDefinitions>
623 <RowDefinition Height="Auto"/>
624 <RowDefinition Height="Auto"/>
625 </Grid.RowDefinitions>
626 <Grid.ColumnDefinitions>
627 <ColumnDefinition/>
628 <ColumnDefinition Width="{StaticResource InterColumnWidth}" />
629 <ColumnDefinition/>
630 </Grid.ColumnDefinitions>
631 <StackPanel>
632 <Label Content="{x:Static resx:WPFControlStrings.WorkRequest}" />
633 <local:Controls:TextBoxWithButton Style="{StaticResource TextBoxWithButtonNullable}" Text="{Bindi
634 </local:Controls:TextBoxWithButton> IsReadOnly="{Binding IsHeaderReadOnly}" AutomationP
635
636

```

15.1.7. Equipment

The screenshot shows the 'Equipment' section of the application. On the left is a sidebar with a search icon 'S' and a list of sections: State, Request Detail, Dates, Work Details, and Equipment (which is highlighted). The main area is titled 'Equipment' and contains two tables: 'Isolating Devices' and 'Temporary Grounds'. Below each table is a 'Details' section showing the selected item's information.

Device	Station	Structure	Actions
51622	RAVEN	341213267	[Icons]

Isolating Devices Details
 [{"Id":"22239682","Name":"51622","Station":"RAVEN","DeviceType":"dmsSwitch"}]

Line	Station	Actions
224696032T	RAVEN	[Icons]

Temporary Grounds Details
 [{"Id":"E39690D9-D3AA-45D3-81B7-708A2AC43552","Name":"TEST_GROUND"}]

The Switching Order Equipment section is configured in the EquipmentSectionTemplate node in the Document.SharedSections.xml file.

```

667
668 <ControlTemplate x:Key="EquipmentSectionTemplate">
669     <Grid Grid.Row="1" Grid.ColumnSpan="2">
670         <Grid.ColumnDefinitions>
671             <ColumnDefinition Width="*" />
672             <ColumnDefinition Width="20" />
673             <ColumnDefinition Width="*" />
674         </Grid.ColumnDefinitions>
675
676         <Grid.RowDefinitions>
677             <RowDefinition Height="Auto" />
678             <RowDefinition Height="Auto" />
679             <RowDefinition Height="Auto" />
680             <RowDefinition Height="Auto" />
681         </Grid.RowDefinitions>
682
683

```

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

Document.SharedSections.xaml

eXtensible Markup Language file length: 136,177 lines: 1,469 Ln: 668 Col: 14 Sel: 0 | 0 Unix (LF) UTF-8 INS

15.1.8. Safety Documents

Safety Doc

Safety Doc #	Safety Doc Type	Safety Doc Holder	Safety Doc Second Holder
Other Doc #	Safety Doc Type	Safety Doc Holder	Safety Doc Second Holder

The Switching Order Safety Documents section is configured in the SafetyDocSectionTemplate node in the Document.SharedSections.xml file.

```

1035
1036 <ControlTemplate x:Key="SafetyDocSectionTemplate">
1037     <Grid HorizontalAlignment="Stretch" Height="Auto" Margin="0" VerticalAlignment="Top" Width="Auto">
1038         <Grid.Resources>
1039             <Style TargetType="{x:Type Label}" BasedOn="{StaticResource {x:Type Label}}">
1040                 <Setter Property="HorizontalAlignment" Value="Left" />
1041                 <!--<Setter Property="FontWeight" Value="Bold" />-->
1042                 <Setter Property="Margin" Value="0,5,0,2" />
1043             </Style>
1044         </Grid.Resources>
1045         <Grid.ColumnDefinitions>
1046             <ColumnDefinition Width="1*" />
1047             <ColumnDefinition Width="10" />
1048             <ColumnDefinition Width="1*" />
1049             <ColumnDefinition Width="10" />
1050             <ColumnDefinition Width="1*" />
1051             <ColumnDefinition Width="10" />

```

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Document.SharedSections.xaml

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15.1.9. Audit Trail

Audit Trail				
User Name	Date/Time	State	Version	Description
DMSALL	5/12/2020 5:04:57 AM	Created By	1	No tracked changes.
DMSALL	5/12/2020 5:04:57 AM	Validated By	2	No tracked changes.
DMSALL	5/12/2020 5:05:22 AM	Approved By	3	Changes: - Approver was set to 'DMSALL' - Time of approval was set to '05/12/2020 05:05:21' - Last modified user was set to 'DMSALL' - Document version changed from '2' to '3' Switching Step changed from '#2/51622/OPEN' to '#2/51622/CLOSE'.
DMSALL	5/12/2020 5:05:24 AM	Approved By	4	Changes: - Switch Order state changed from 'VALIDATED' to 'APPROVED' - Approver #2 was set to 'DMSALL' - Time of approval #2 was set to '05/12/2020 05:05:24' - Document version changed from '3' to '4' Changes: - Actual Start Time was set to '05/12/2020 05:05:25' - Switch Order state changed from 'APPROVED' to 'EXECUTING'

The Audit Trail section is configured in the AuditTrailSectionTemplate node in the Document.SharedSections.xml file.

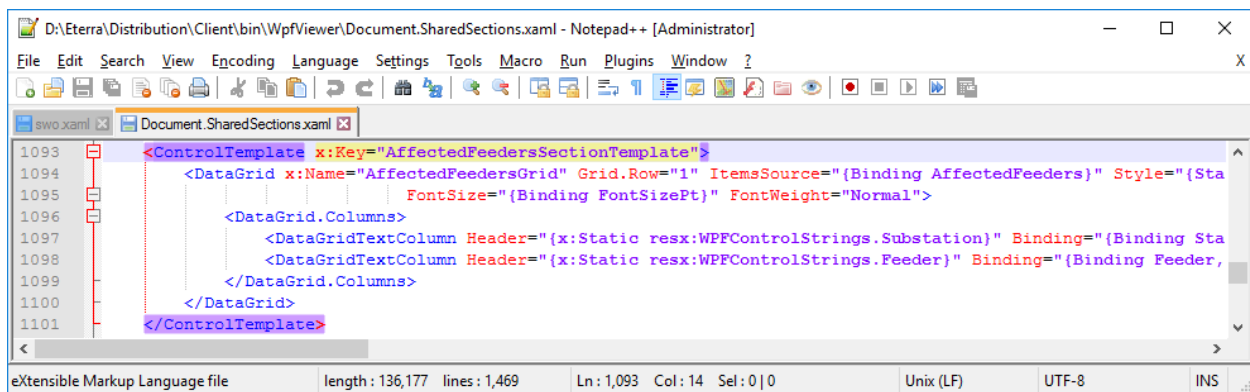
```

985 <ControlTemplate x:Key="AuditTrailSectionTemplate">
986 <DataGrid x:Name="AuditTrailsGrid" ItemsSource="{Binding AuditTrails}" IsReadOnly="{Binding IsHeaderRea
987 Style="{StaticResource GridStyleForDocumentheader}" FontSize="{Binding FontSizePt}
988 <DataGrid.Columns>
989 <DataGridTextColumn Header="{x:Static resx:WPFControlStrings.UserName}" Binding="{Binding UserNa
990 <DataGridTextColumn Header="{x:Static resx:WPFControlStrings.DateTime}" Binding="{Binding TimeSt
991 <DataGridTextColumn Header="{x:Static resx:WPFControlStrings.StateOfSwitchingOrder}" Binding="{B
992 <DataGridTextColumn Header="{x:Static resx:WPFControlStrings.DocumentVersion2}" Binding="{Bindin
993 <DataGridTextColumn Header="{x:Static resx:WPFControlStrings.Description}" Binding="{Binding Agg
994 </DataGrid.Columns>
995 </DataGrid>
996 </ControlTemplate>
  
```

15.1.10. Affected Feeders

Affected Feeders	
Substation	Feeder
RAVEN	F8864

The Affected Feeders section is configured in the AffectedFeedersSectionTemplate node in the Document.SharedSections.xml file.



15.1.11. Step Blocks

Step Blocks Overview									
Block Name	Block Color	Block Dependency					Dependent Blocks		
Block1							Block2		
Block2		Block1							

Switching Steps									
12									
	#	Nav	I/C	Step State	Device	Action	Issued To	Instruction	Tag
☐	1				EDIT	413403	OPEN		OPEN 413403
☐	2				EDIT	413403	CLOSE		CLOSE 413403
☐	3				EDIT	6-8007-1522-0-1	DISCONNECT		Disconnect Transformer
☐	4				EDIT	6-8007-1522-0-1	CONNECT		Connect Transformer

The Step Blocks section is configured in the StepBlocksSectionTemplate node in the Document.SharedSections.xaml file.

```

997
998 <ControlTemplate x:Key="StepBlocksSectionTemplate">
999   <DataGrid x:Name="StepBlocksSectionGrid" ItemsSource="{Binding StepBlocksOverview.StepBlocks}" Selected
1000     SelectionMode="Single" Style="{StaticResource GridStyleForDocumentheader}"
1001   >
1002     <DataGrid.Columns>
1003       <DataGridTextColumn Header="{x:Static resx:WPFControlStrings.AddBlockDialog_BlockName}" Binding
1004       <!--<DataGridTextColumn Header="{x:Static resx:WPFControlStrings.AddBlockDialog_StepNumberMin}"
1005       <DataGridTextColumn Header="{x:Static resx:WPFControlStrings.AddBlockDialog_Ste
1006       <DataGridTemplateColumn Header="{x:Static resx:WPFControlStrings.AddBlockDialog_BlockColor}" x:
1007     <DataGridTemplateColumn.CellTemplate>
1008       <DataTemplate>
1009         <Grid Focusable="False" Height="16" Margin="0,0,10,0">
1010           <!--<Grid.Background
1011         </Grid>
1012       </DataTemplate>
1013     </DataGridTemplateColumn.CellTemplate>
1014   </DataGrid.Columns>
1015 </DataGrid>
1016 </ControlTemplate>

```

Document.SharedSections.xaml

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15.1.12. Email Notifications

Email Notifications

On Validate	On Reject
On Approve	On Execute
On Complete	On Delete
testuser@testcompany.com	

The Email Notifications section is configured in the EmailNotificationTemplate node in the Document.SharedSections.xaml file.

```

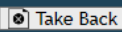
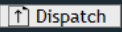
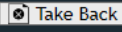
757 <ControlTemplate x:Key="EmailNotificationTemplate">
758   <Grid Margin="0">
759     <Grid.Resources>
760       <Style TargetType="{x:Type Label}" BasedOn="{StaticResource {x:Type Label}}">
761         <Setter Property="HorizontalContentAlignment" Value="Right" />
762         <Setter Property="Width" Value="Auto" />
763       </Style>
764       <Style TargetType="{x:Type TextBox}" BasedOn="{StaticResource {x:Type TextBox}}">
765         <Setter Property="Width" Value="350" />
766       </Style>
767     </Grid.Resources>
768     <Grid.RowDefinitions>
769       <RowDefinition Height="Auto" />
770     </Grid.RowDefinitions>
771   </Grid>
772 </ControlTemplate>

```

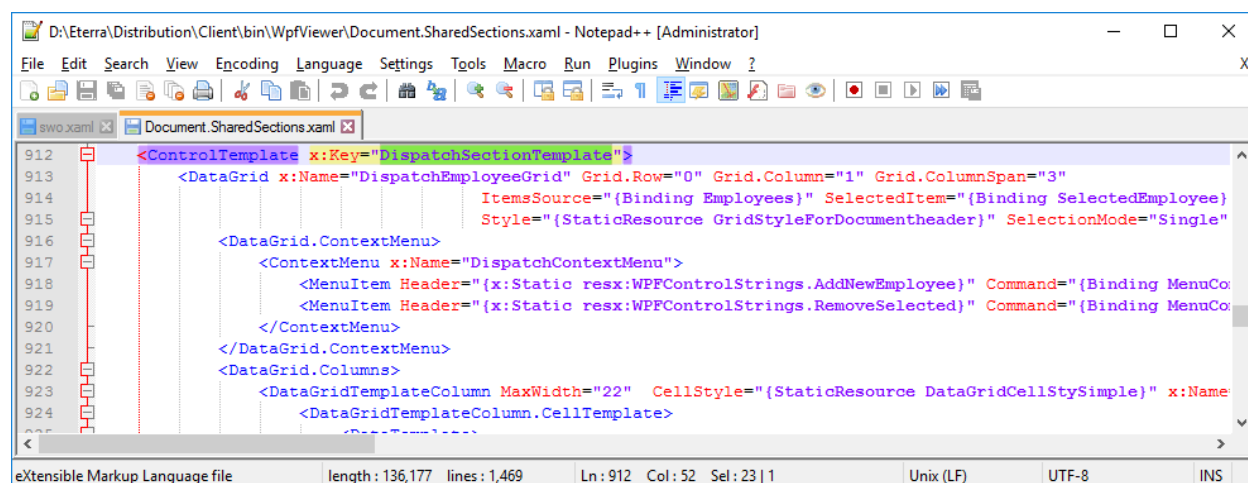
Document.SharedSections.xaml

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15.1.13. Dispatching

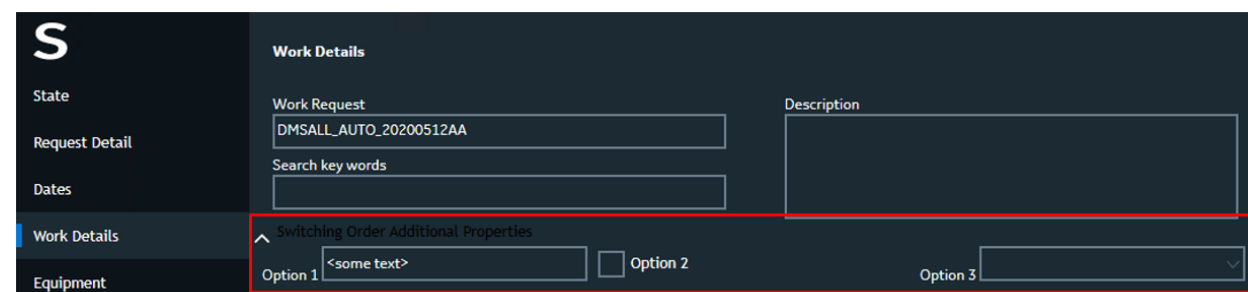
Dispatching				
	Id	Name	Dispatch Status	Actions
	FI2685	Bonnie L	Accepted	 Take Back  SOS
	FI1502	Daniel Jeffrey	None	 Dispatch  SOS
	FI1515	Eugene B	Dispatched	 Take Back  SOS

The Dispatching section is configured in the DispatchSectionTemplate node in the Document.SharedSections.xaml file.



```
<ControlTemplate x:Key="DispatchSectionTemplate">
    <DataGrid x:Name="DispatchEmployeeGrid" Grid.Row="0" Grid.Column="1" Grid.ColumnSpan="3"
        ItemsSource="{Binding Employees}" SelectedItem="{Binding SelectedEmployee}"
        Style="{StaticResource GridStyleForDocumentheader}" SelectionMode="Single"
        <DataGrid.ContextMenu>
            <ContextMenu x:Name="DispatchContextMenu">
                <MenuItem Header="{x:Static resx:WPFControlStrings.AddNewEmployee}" Command="{Binding MenuCo
                <MenuItem Header="{x:Static resx:WPFControlStrings.RemoveSelected}" Command="{Binding MenuCo
            </ContextMenu>
        </DataGrid.ContextMenu>
        <DataGrid.Columns>
            <DataGridTemplateColumn MaxWidth="22" CellStyle="{StaticResource DataGridCellStySimple}" x:Name=
            <DataGridTemplateColumn.CellTemplate>
```

15.1.14. Custom Properties



The Additional Properties section is configured using the Switching Order Client Configuration Editor. The XAML schema is automatically updated with the custom fields. The fields are taken directly from the client configuration file at run time.

15.1.15. Switching Steps

Switching Steps									
	#	Nav	I/C	Step State	Device	Action	Issued To	Instruction	
☐	0.1		C	ISSUED		COMMENT BY OPERATOR	Al I Owen	Comment by Operator	
☐	1		C	ISSUED	52304	OPEN	Eugene B	OPEN 52304	
☐	1.1		i	EDIT	52304	CLOSE		CLOSE 52304	

The Switching Steps section is configured in the `<ControlTemplate x:Key="StepsGridTemplate">` node in the `PlannedRequest.StepDataGrid.xaml` file.

```

D:\Eterra\Distribution\Client\bin\WpfViewer\PlannedRequest.StepDataGrid.xaml - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
PlannedRequest.StepDataGrid.xaml
540 </Style>
541 <ControlTemplate x:Key="StepsGridTemplate">
542 <Grid Margin="0,0,0,0" MinHeight="200">
543 <Grid.RowDefinitions>
544 <RowDefinition Height="Auto"/>
545 <RowDefinition Height="*/"/>
546 </Grid.RowDefinitions>
547 <FrameworkElement x:Name="dummyElement" Visibility="Collapsed" />
548 <Grid>
549 <StackPanel Orientation="Horizontal">
550 <StackPanel.Resources>
551 <Style TargetType="Button" BasedOn="{StaticResource toolbarButtonStyle}"
eXtensible Markup Language file length: 126,658 lines: 1,461 Ln: 541 Col: 15 Sel: 0 | 0 Unix (LF) UTF-8 INS

```

15.2. Switching Order Fields

The following tables provide fields that are currently supported in a Switching Order XAML model.

General

Name	Type
NavigateCommand	ICommand
NavigateToOMSCommand	ICommand
AddressBookCommand	ICommand
ClearSelfCommand	ICommand
DispatchCommand	ICommand
TakebackCommand	ICommand
AssignCommand	ICommand
UnassignCommand	ICommand
SOSCommand	ICommand
ChatCommand	ICommand

Name	Type
MenuCommand	ICommand
NavigateToIsolationDeviceCommand	ICommand
NavigateToDeviceCommand	ICommand
NavigateToTempGroundDeviceCommand	ICommand
NavigateToCustomerCommand	ICommand
RemoveTempGroundDeviceCommand	ICommand
RemoveIsolatingDeviceCommand	ICommand
RemoveCustomerCommand	ICommand
UpdOmsDocCommand	ICommand
ActionCommand	ICommand
StepActionCommand	ICommand
Regions	RegionList
IsDocumentReadOnly	bool
IsDailyLog	bool
IsHeaderEnabled	bool
IsHeaderReadOnly	bool
IsValidateEnabled	bool
IsApproveEnabled	bool
IsExecuteEnabled	bool
IsRejectEnabled	bool
IsCompleteEnabled	bool
IsNotFirstSave	bool
MobileActionAllowed	bool
MobileAssignAllowed	bool
IsNameFieldEnabled	bool
AnyStepsSelected	bool
AnySteps	bool
AnyUncompletedSteps	bool
AnyActionsInClipboard	bool
_editableRegionDropDowns	bool
EditableRegionDropDowns	bool
Divisions	DummyRegion[]
ServiceCenters	ObservableCollection<DummyRegion>
Subtypes	ObservableCollection<CodedTranslationEntry>
Worktypes	ObservableCollection<WorkTypeComboboxItem>
StepsGridViewModel	StepsGridViewModel

Name	Type
AffectedFeeders	ObservableCollection<AffectedFeederViewModel>
AuditTrails	ObservableCollection<AuditTrailModelView>
TempGroundDevices	ObservableCollection<TempGroundDeviceViewModel>
IsolatingDevices	ObservableCollection<IsolatingDeviceViewModel>
IsolatingDevicesDetails	string
TempGroundsDetails	string
StepBlocksOverview	StepBlocksOverviewViewModel
Stations	ObservableCollection<string>
Feeders	ObservableCollection<string>
EquipmentList	ObservableCollection<string>
Equipment	string
EquipmentId	string
DateTimeFormat	string
UseNewToolbar	bool
IsPickerKeyboardInputEnded	bool
FontSize	float
StepsFontSize	float
FontSizePt	float
UiStateCollapsed	bool
UiStateExpanded	bool
UiPropsCollapsed	bool
UiPropsExpanded	bool
UiDispatchingCollapsed	bool
UiDispatchingExpanded	bool
UiEmailExpanded	bool
UiFeedersExpanded	bool
EmailEnabled	bool
DispatchingEnabled	bool
EmailSectionVisible	Visibility
PlannedOutageIconVisible	Visibility
PlannedRequestIconVisible	Visibility
AssociatedPlannedRequest	string
PlannedOutageIconInverseVisible	Visibility
DispatchSectionVisible	Visibility
PlannedOutageVisible	Visibility
RelatedIncVisible	Visibility

Name	Type
UnlinkDocEnabled	Visibility
AreGridsVisible	bool
GridsVisibility	Visibility
StepBlockVisibility	Visibility
SafetyDocVisibility	Visibility
Name	String
AssignedNumber	String
SwoNumber	String
SwoRelatedInc	String
SwoPlannedOutage	String
WorkRequest	String
Division	String
Substation	String
Feeder	String
FeederSelectionChanged()	void
ServiceCenter	String
Device	String
Subtype	CodedTranslationEntry
SelectedWorkType	WorkTypeComboboxItem
HasWorkTypeOptions	Visibility
WorkTypeSelectionEnabled	bool
Description	String
DocVersion	uint
IsPlannedSwitching	bool
SearchTags	string
SafetyDocName1	string
SafetyDocHolder1	string
SafetyDocSecondHolder1	string
SafetyDocType1	string
SafetyDocName2	string
SafetyDocHolder2	string
SafetyDocSecondHolder2	string
SafetyDocType2	string
SafetyTagTypes	List<string>
NotifyOnValidate	String
NotifyOnApprove	String

Name	Type
NotifyOnComplete	String
NotifyOnDelete	String
NotifyOnReject	String
NotifyOnExecute	String
SchedStartTime	DateTime
SchedEndTime	DateTime
ActualStartTime	DateTime
ActualEndTime	DateTime
ModifiedTime	DateTime
LastModifiedUser	string
Mode	String
IsEditing	bool
IsValidated	bool
IsApproved	bool
IsExecuting	bool
IsVisible_ExecuteButton	bool
IsVisible_CompleteButton	bool
CreatedBy	String
CreatedTime	DateTime
ValidatedBy	String
ValidatedTime	DateTime
ExecutedBy	String
ExecutedTime	DateTime
ApprovedBy	String
ApprovedTime	DateTime
ApprovedTime2	DateTime
ApprovedTime3	DateTime
ApprovedBy2	String
ApprovedBy3	String
NumberOfApprovers	int
NumberOfReviewers	int
CompletedBy	String
CompletedTime	DateTime
ValidatedProgress	int
ApprovedProgress	int
ExecutingProgress	int

Name	Type
IsCompleted	bool
CompletedProgress	int
MaxGridHeight	double
State	string
DispatcherEmployee	SelectedEmployee
CustomProperties	CustomPropertyCollection
ExpandState	TypedCollection<bool>
SelectedEmployee	DispatcherEmployee
Employees	ObservableCollection<DispatcherEmployee>
AnyEmployeeSelected	bool
ShowFeederWarning	bool
ShowStationWarning	bool
ShowEquipmentWarning	bool
EquipNavVisibility	Visibility
EquipWarnVisibility	Visibility
TrackDevicesVisible	bool
UntrackDevicesVisible	bool
SetSafetyInstructionBtnVisible	bool
IsStudy	bool
ImplementSelectedTsbVisible	bool
StopImplementingTsbVisible	bool
IssueSelectedTsbVisible	bool
AssignCrewTsbVisible	bool
StepsOpTsbEnabled	bool
StepsSafeOpTsbEnabled	bool
IssueStepsEnabled	bool
IsStepsGridExpanded	bool
IsCustomHeaderSectionVisible_SWO	bool
IsCustomHeaderSectionVisible_DailyLog	bool

ObservableCollection<DummyRegion>

Name	Type
Id	String
Name	String

StepsGridViewModel

Name	Type
IncreaseFontButtonCommand	RelayCommand<string>
DecreaseFontButtonCommand	RelayCommand<string>
ToolBarButtonCommand	RelayCommand<string>
ReturnToNormalCommand	RelayCommand<string>
AddManualStepCommand	RelayCommand<string>
SafetyInstructionCommmand	RelayCommand<object>
SwitchingSteps	ObservableCollection<StepModelView>
ColumnsDisplayIndex	TypedCollection<int>
PreventStepsIssue	bool
IsStudy	bool
IsReadOnly	bool
UserPermissions	SwoUserPermissions
DispatchEmployeeList	HashSet<string>
FontSize	double
FontSizePt	float
IsSDVisible	bool
IsSafetyInstructionVisible	bool
CanIncreaseFont	bool
CanDecreaseFont	bool
AllStepsSelected	bool
EmployeeList	ObservableCollection<string>

ObservableCollection<StepModelView>

Name	Type
Action<EStepAction>	event
Selected	bool
IsSeparator	bool
StepNumberStr	string
StepNumber	string
DeviceColumnControlType	ECellControlType
FormattedDeviceName	string
StateName	string
TagType	string
SD	bool
IssuedTo	string
IssuedTime	DateTime

Name	Type
IssuedTimeStr	string
SystemCompletedTime	DateTime
ActualCompletedTimeStr	string
ActualCompletedTime	DateTime
TimeFormat	string
IsActualCompletedTimeEditable	bool
Instruction	string
Ownership	EStepOwnership
IsOwnedByMobileEmployee	bool
IsGenericColumnReadOnly	bool
IsCommentReadOnly	bool
Comment	string
IssuedBy	string
SwitchStepActions	List<string>
ActionColumnControlType	ECellControlType
TagColumnControlType	ECellControlType
SwitchStepTagTypes	List<string>
HasTagOptions	bool
StepBlockColor	SolidColorBrush
SafetyInstruction	string
IsReadOnly	bool
IsStepManual	bool
IsStepDeviceNameReadOnly	bool
IsStepModeEdit	bool
IsStepModeAtLeastIssued	bool
IsStepModeCompleted	bool
IsStepModeHistorical	bool
IsStepModeCancelled	bool
IsStepReadOnly\{	bool
IsStepImplemented	bool
IsStepModeReadOnly	bool
IsStepModeIssuedOrReadOnly	bool
IsSafetyInstructionReadOnly	bool
CanIssueStep	bool
CanCompleteStep	bool
IsStepModeEditAndCompleted	bool

Name	Type
IsStepActionEditable	bool
Failed	bool
StateBackgroundBrush	SolidColorBrush
StateForegroundBrush	SolidColorBrush
CanStepBeIssued	bool
CanStepNotBeIssued	bool
CanStepBeCompleted	bool
CanStepNotBeCompleted	bool
MainNavigationType	ENavigationType
AltNavigationType	ENavigationType
ShowNavImage	bool
HasAltGeometry	bool
HasGeometry	bool
ValidationTooltipVisibility	Visibility
ValidationTooltip	string
ValidStatus	ESwitchStepValidStatus
AggregatedValidStatus	ESwitchStepValidStatus
StepMode	string
this[string key] (Custom Field Key)	string
IsCustomPropReadOnlyProp	bool
CommentsList	ObservableCollection<string>
EmployeeList	ObservableCollection<string>
DevicesForMobileSubstation	ObservableCollection<string>
Model	StepsGridViewModel
SwitchStepCommand	RelayCommand<string>
SafetyInstructionCommand	RelayCommand

ObservableCollection<AffectedFeederViewModel>

Name	Type
Station	String
Feeder	String

ObservableCollection<AuditTrailModelView>

Name	Type
Id	string
UserName	string

Name	Type
TimeStamp	DateTime
DocVersion	uint
Description	string
StateName	string
Changes	string
AggregatedDescription	string
TooltipVisibility	Visibility
PropertyChangedEventHandler	event

ObservableCollection<TempGroundDeviceViewModel>

Name	Type
Line	string
DeviceId	string
StationName	string
Name	string
X	string
Y	string

ObservableCollection<IsolatingDeviceViewModel>

Name	Type
DeviceId	string
DeviceName	string
StationName	string
StructureId	string
StructureName	string

ObservableCollection<DispatcherEmployee>

Name	Type
IsIssued	bool
IsMobile	bool
IsEditable	bool
Id	string
Name	string
ETA	DateTime
DispatchBtnVisible	Visibility
TakebackBtnVisible	Visibility

Name	Type
SosBtnVisible	Visibility
CanDispatch	bool
CanTakeback	bool
IsSosEnabled	bool
DispatchStatusText	string
DispatchStatus	ESwoEmployeeState
NetStatus	EmployeeNetStatus
IsUnknownVisible	Visibility
IsLogonOnlineVisible	Visibility
IsLogonOfflineVisible	Visibility
IsLogoffVisible	Visibility
DeviceId	string
Online	bool
SOSText	string
PropertyChangedEventHandler	event

DocViewModelPlannedRequest

Name	Type
RequestTypes	ObservableCollection<RequestTypeComboboxItem>
SelectedRequestType	RequestTypeComboboxItem
RequestType	string
Address	string
AssociatedSWO	string
HasAssociatedSwitchOrder	bool
HasSWOButton	bool
ReasonForWork	string
CausesOutage	bool
IsClearanceNeeded	bool
RequestorComments	string
ReviewedBy1	String
ReviewTime1	DateTime
PriorityNumber	int
IsUrgent	bool
ReasonForUrgency	string
CustomerNotificationMessage	string
Customers	ObservableCollection<TracedCustomer>

Name	Type
AllCustomersView	ICollectionView
CriticalCustomersView	ICollectionView
CustomerMessageFormat	string
AffectedCustomersList	ObservableCollection<string>
AffectsCriticalCustomers	bool
IsPlannedOutageProcessing	bool
string	override
CreateEditSwoCommand	ICommand
CreatePlannedOutageCommand	ICommand
AddTracedCustomersRTCommand	ICommand
AddTracedCustomersSTCommand	ICommand

RequestTypeComboboxItem

Name	Type
Key	string
Text	string
CTE	CodedTranslationEntry

15.3. Editing XAML Files

This section provides basic instructions for manually editing XAML files in a text editor to customize the switching order properties. Alternatively, you can edit XAML files using a configuration tool, such as Visual Studio Community.

15.3.1. Customizing Labels

By default, field labels are automatically taken from the resource file in the code. You can customize field labels directly in the associated XAML files. Alternatively, you can customize field labels using the Resource Editor tool. For more information, refer to the *Resource Editor User Guide*.

To modify a field label in a XAML file, change the Text value in the TextBlock node for the associated field in the XAML configuration file.

```
<StackPanel Orientation="Horizontal" VerticalAlignment="Center">
  <TextBlock Grid.Column="0" FontWeight="Bold"
    VerticalAlignment="Center" Margin="0,0,0,0" Style="{StaticResource DefaultTex
  <Run Text="{x:Static resx:WPFControlStrings.SectionHeader_State}"/>:
</TextBlock>
  <TextBlock Text="{Binding State}" Grid.Column="0" Name="SwoState" AutomationProperties.A
    VerticalAlignment="Center" Margin="8,0,11,0" Style="{StaticResource DefaultTe
  <TextBlock Opacity="0.3" VerticalAlignment="Center" Style="{StaticResource DefaultTextBl
    <Run Text="{x:Static resx:WPFControlStrings.Number}"/>:
  </TextBlock>
  <TextBlock Text="{Binding SwoNumber}" Name="SwoNumber" Grid.Column="3" AutomationPropert
    Opacity="0.3" VerticalAlignment="Center" Style="{StaticResource DefaultTe
```

15.3.2. Customizing Parameter Bindings

To modify a parameter control, change the associated Binding and Name values. For a binding, you can use any of the fields listed in [Switching Steps](#).

```
<TextBlock Text="{Binding SwoNumber}" Name="SwoNumber" Grid.Column="3" AutomationPropert
  Opacity="0.3" VerticalAlignment="Center" Style="{StaticResource DefaultTe
  <TextBlock Margin="24,0,0,0" VerticalAlignment="Center" Style="{StaticResource DefaultTe
    <Run Text="{x:Static resx:WPFControlStrings.DocumentVersion}"/>:
  </TextBlock>
  <TextBlock Text="{Binding DocVersion}" Name="DocVersion" Grid.Column="3" AutomationPrope
    VerticalAlignment="Center" Style="{StaticResource DefaultTextBlockStyle}"
  </StackPanel>
```

15.3.3. Adding and Removing Fields

To remove a field or section from a switching order document page:

- Remove the field or section definition from the associated XAML file.

To add a field or section to a switching order document page:

- Add the field or section definition to the associated XAML file.

```
D:\Eterra\Distribution\Client\bin\WpfViewer\Document.SharedSections.xaml - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
swo.xaml Document.SharedSections.xaml
88 <TextBlock Text="{Binding SwoNumber}" Name="SwoNumber" Grid.Column="3" AutomationProperty
89 Opacity="0.3" VerticalAlignment="Center" Style="{StaticResource DefaultTe
90
91 <TextBlock Margin="24,0,0,0" VerticalAlignment="Center" Style="{StaticResource DefaultTe
92 <Run Text="{x:Static resx:WPFCtrlStrings.DocumentVersion}"/>:
93 </TextBlock>
94 <TextBlock Text="{Binding DocVersion}" Name="DocVersion" Grid.Column="3" AutomationPrope
95 VerticalAlignment="Center" Style="{StaticResource DefaultTextBlockStyle}"
96 </StackPanel>
eXtensible Markup Language file length: 136,177 lines: 1,469 Ln: 88 Col: 50 Sel: 7 | 1 Unix (LF) UTF-8 INS
```

Sections are typically defined using ControlTemplate nodes. Fields are defined using TextBlock, CheckBox, and ComboBox nodes.

15.3.4. Changing the Row and Column Order

To modify the order of columns and rows in the grid, edit the Grid.Row and Grid.Column values for parameters.

```
D:\Eterra\Distribution\Client\bin\WpfViewer\Document.SharedSections.xaml - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
swo.xaml Document.SharedSections.xaml
553 <!--Doesn't exist on mockup-->
554 <StackPanel Visibility="Collapsed">
555 <Label Content="{x:Static resx:WPFCtrlStrings.DocumentVersion}" Grid.Row="6" Grid.Column="2"/>
556 <Label Content="{Binding DocVersion}" Name="DocVersionProp" Grid.Row="6" Grid.Column="3" AutomationPrope:
557 </StackPanel>
558
559 <!--Doesn't exist on mockup-->
560 <StackPanel Grid.Row="6" Grid.Column="2" Visibility="{Binding IsCompleted, Converter={StaticResource Boolean'
561 <Label Content="{x:Static resx:WPFCtrlStrings.CompletedByUser}"/>
562 <localControls:TextBoxWithButton Style="{StaticResource TextBoxWithButtonNullable}" Text="{Binding Compl
563 </StackPanel>
eXtensible Markup Language file length: 136,177 lines: 1,469 Ln: 555 Col: 105 Sel: 4 | 1 Unix (LF) UTF-8 INS
```

16. Customizing the PDF Export Format

You can export switching orders, daily logs, and templates as a PDF.

The PDF generation process is as follows:

1. Switching order XML and XSLT documents are fed to an XSLT engine.
2. Using the rules defined in the XSLT document, the XSLT engine transforms the XML document to an XSL-FO (XSL Formatting Objects) document.
3. The XSL-FO document, which is usually stored only in memory, is fed to an FO engine.
4. The FO engine uses formatting rules defined in the XSL-FO document to render a printable document as a PDF file.

The below image shows the general switching order to PDF export workflow.

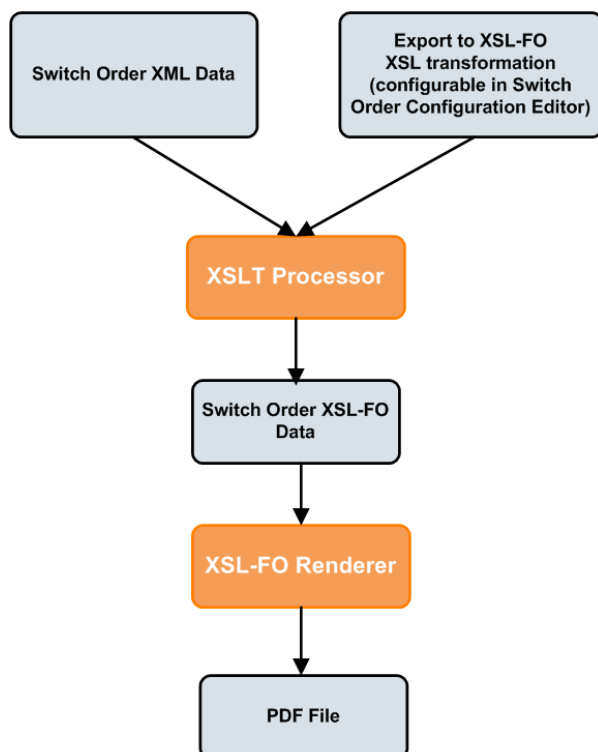


Figure 22. Switching Order to PDF Export Customization Diagram

The switching order PDF conversion template is defined in the `swo2pdf.xsl` configuration file. You can create or edit XSL files using text editors or XSL and XSL-FO graphical editors with "What You See is What You Get" (WYSIWYG) functionality.

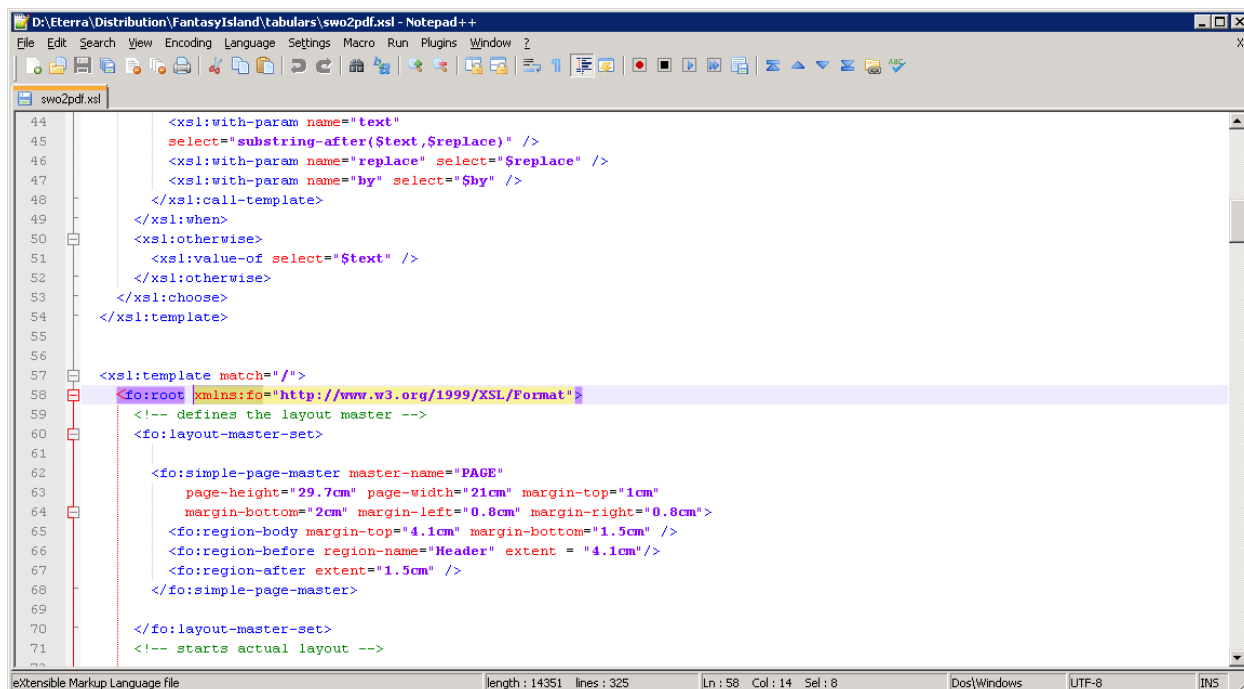


Figure 23. The swo2pdf.xml Configuration File

In the swo2pdf.xml configuration file, elements in the xmlns:fo namespace define the PDF layout. For example:

- **fo:layout-master-set:** Specifies the name, height, and width of the master page, and specifies the margins and orientation (portrait or landscape) of the entire page. (For more information, refer to <https://www.webucator.com/xml-training/course/introduction-xsl-fo/>.)
- **fo:block:** Specifies the formatting of paragraphs, titles, headlines, figure captions, and table captions. (For more information, refer to http://w3schools.sinsixx.com/xslfo/xslfo_blocks.asp.htm.)
- **fo:inline:** Applies formatting to text within a block. (For more information, refer to <https://www.webucator.com/xml-training/course/introduction-xsl-fo/>.)
- **fo:table:** Defines a table. Common attributes include table-layout (fixed, auto, or inherit), border-width, and background-color. (For more information, refer to <https://www.webucator.com/xml-training/course/introduction-xsl-fo/>.)

Elements in the xmlns:xsl namespace define transformations of the original switch order XML file. For example:

- **xsl:value-of:** Inserts a text value. The "select" attribute contains an XPath expression. (For more information, refer to https://www.w3schools.com/xml/xsl_value_of.asp.)
- **xsl:if:** Applies a conditional test against the content of the XML file. (For more information, refer to http://www.w3schools.com/xml/xsl_if.asp.)

SWITCHING ORDER

DOCUMENT VERSION: 7

DATE: Mar 18, 2020

SUBSTATION: RAVEN

NAME: 20000002



TIME ISSUED	TIME COMPLETE	ISSUED TO	ISSUED BY	CUSTOM VALUES	STEP	SWITCHING PREPARED BY: DMSALL APPROVED BY: DMSALL APPROVED BY: DMSALL
3/18/2020 5:19 AM		John Smith	DMSALL	DNOMX: 937604.500000 SITE: 6-7612-2722-0-3	1	OPEN 12226-3
3/18/2020 5:19 AM	3/18/2020 5:26 AM	John Smith	DMSALL	DNOMX: 937604.000000 SITE: 8116 West St RAVEN	2	Disconnect Transformer

NO ISOLATING DEVICES

NO TEMPORARY GROUND DEVICES

NO SAFETY DOCUMENTS

Figure 24. Switching Order PDF Document

The default PDF template defined in the `swo2pdf.xml` configuration file includes the following sections:

- **Header:** Defines the header of the PDF document. Contains the following elements:

- Page number at the upper-right corner
- Document type (Switching Order, Daily Log, or Template) at the top
- Company logo at the right
- Document fields at the left

The document fields and company logo are formatted using a table (<fo:table>) with 2 columns and 1 row. Column widths are set by the corresponding <fo:table-column> elements. Column captions are set by the <fo:table-cell> elements under the <fo:table-header> element.

- **Table:** Provides details for switching steps. Contains the following columns:

- Time Issued
- Time Complete
- Issued To
- Issued By
- Custom Values

- Step
- Switching

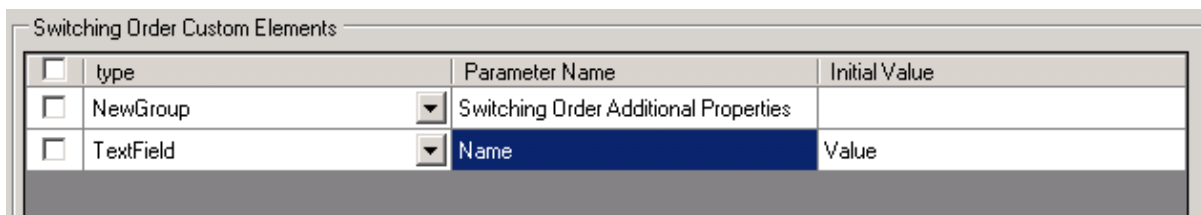
Switching step data is populated from the Switching Order *.xml document using the `<xsl:for-each>` element.

16.1. Adding Header Fields to the PDF Export Template

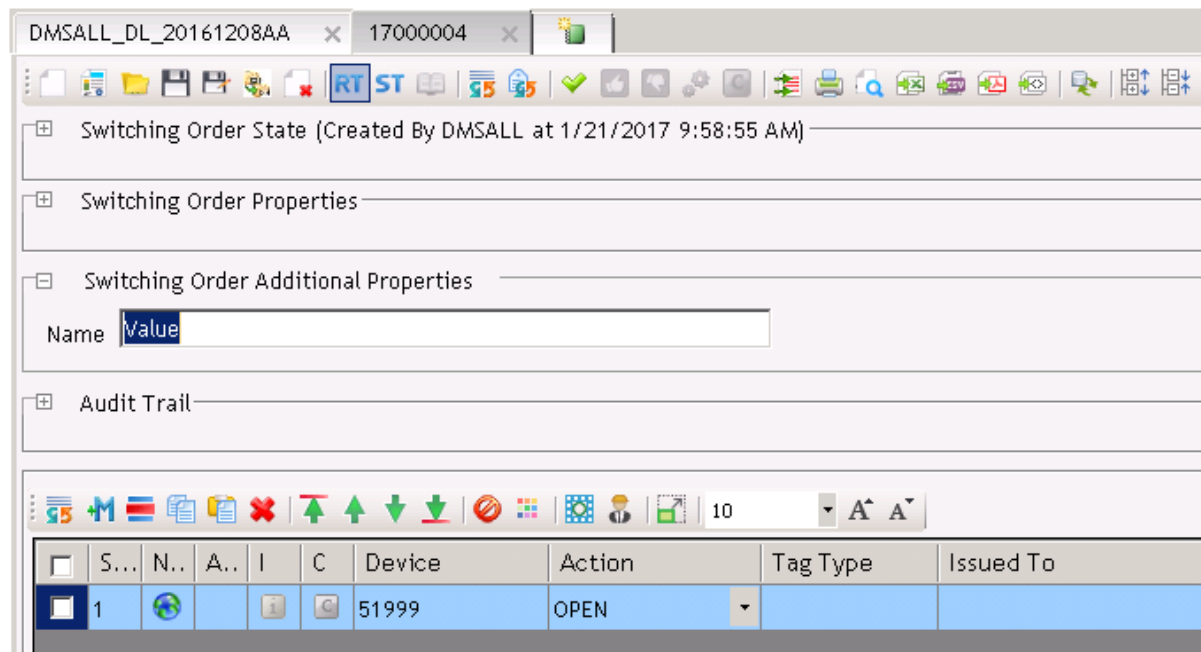
To add a header field in the Switching Order to PDF conversion template:

1. Define a header field in the Switching Order Client Configuration Editor.

For example, define a new text field with Parameter Name set to Name.



2. Create and save a switching order with the new header field.



3. Open the saved switching order XML file and navigate to the new header field.


```

104 </SwitchOrderAuditTrail>
105 </SOAuditTrail>
106 <SOUserFields>
107   <SwoUserFieldInfo>
108     <fieldType>TextField</fieldType>
109     <fieldID>2de95a26-4e1b-4e73-80cd-e74acce551c8</fieldID>
110     <fieldName>Name</fieldName>
111     <fieldValue>Value</fieldValue>
112     <fieldValuesList />
113   </SwoUserFieldInfo>
114 </SOUserFields>
115 <DocType>2</DocType>
116 <searchTags />
117 <lastModifiedUser>DMSALL</lastModifiedUser>

```

4. In the PDF export template file (swo2pdf.xsl), add the Header Custom Field section between the existing NAME and REMARKS sections.

```

112
113
114 <fo:block padding-bottom="3pt">
115   NAME:
116   <fo:inline font-weight="normal">
117     <xsl:value-of select="//name" />
118   </fo:inline>
119 </fo:block>
120
121 <fo:block padding-bottom="3pt">
122   Header Custom Field:
123   <fo:inline font-weight="normal">
124     <xsl:value-of select="//SwoUserFieldInfo[fieldName/text()='Name']/fieldValue" />
125   </fo:inline>
126 </fo:block>
127
128 <fo:block padding-bottom="3pt">
129   REMARKS:
130   <fo:inline font-weight="normal">
131     <xsl:value-of select="substring(//key1, 1, 110)" />
132   </fo:inline>
133 </fo:block>

```

5. In the "Header Custom Field" section, set the <xsl:value-of> element as follows:

```
<xsl:value-of select="//SwoUserFieldInfo[fieldName/text()='Name']/fieldValue" />
```

Where:

- **SwoUserFieldInfo:** The name of the section that defines the new custom field in the switching order XML file.
- **Name:** The name of the custom field.

6. Increase the height of the header table to allocate the new field.

```

<fo:simple-page-master master-name="PAGE"
  page-height="29.7cm" page-width="21cm" margin-top="1cm"
  margin-bottom="2cm" margin-left="0.8cm" margin-right="0.8cm">
  <fo:region-body margin-top="5.1cm" margin-bottom="1.5cm" />
  <fo:region-before region-name="Header" extent = "5.1cm"/>
  <fo:region-after extent="1.5cm" />
</fo:simple-page-master>

```

7. Save your changes and restart the client.
8. Export a switching order document as a PDF and verify the new field.

1 of 1

SWITCHING ORDER

DATE: Jan 21, 2017

SUBSTATION: RAVEN

NAME: NewSwitchingOrder

Header Custom Field: NewValue

REMARKS: Description



TIME ISSUED	TIME COMPLETE	ISSUED TO	ISSUED BY	CUSTOM VALUES	STEP	SWITCHING PREPARED BY: DMSALL APPROVED BY: DMSALL APPROVED BY: DMSALL
1/21/2017 10:43 AM		Andy Johns on	DMSALL	DNOMX: 932344.750000 SITE: 6-7512-2130-0-4	1	OPEN 51621
				DNOMX: 936350.812500 SITE: 8291 West St RAVEN	2	Disconnect Transformer

16.2. Adding Columns to the PDF Export Template

To add a column in the Switching Order to PDF conversion template:

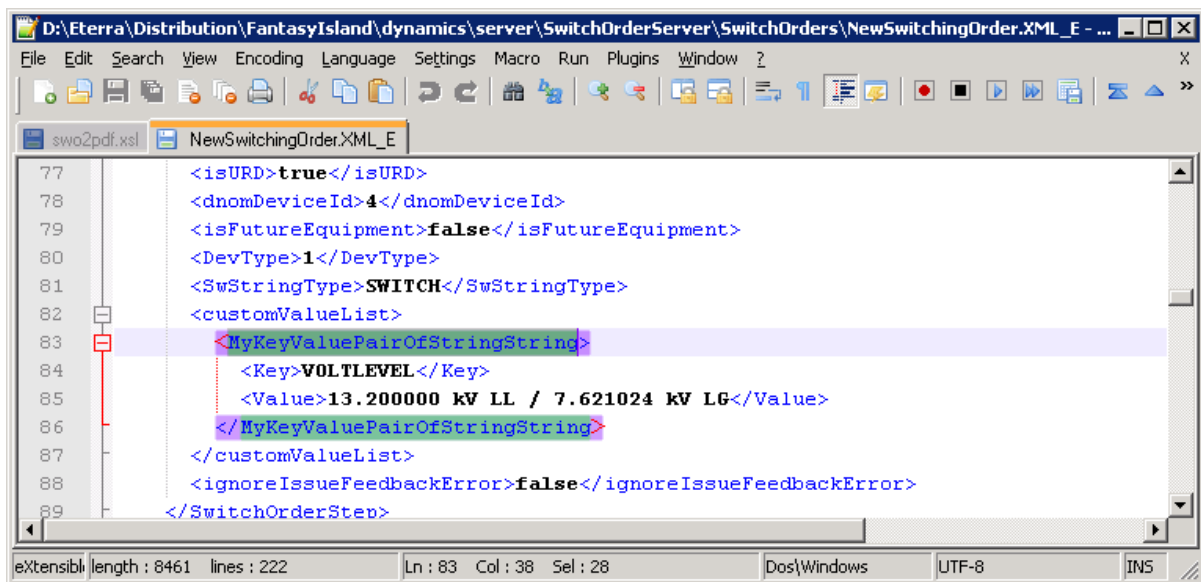
1. Define a switching step custom column in the Switching Order Client Configuration Editor.
For example, define a Voltage Level column with the VOLTLEVEL query field.
2. Disable the "PDF/HTML Exportable & Printable" option to not show the Voltage Level column value in a CUSTOM VALUES column in the exported PDF file.

General	Switch Order Related	Device & State	Tag & Comment	Switch Steps	Right-click Menu	Validation Rules	Step Custom Columns	Step Columns Order	Header Custom Fields
Switch Step Columns									
	Header Name	Serialize Key	Queried Value	RT Visibility	ST Visibility	PDF/HTML Exportable & Printable	Editable		
☐	VOLTAGE LEVEL	VOLTLEVEL	VOLTLEVEL	☒	☒	☐	☒		

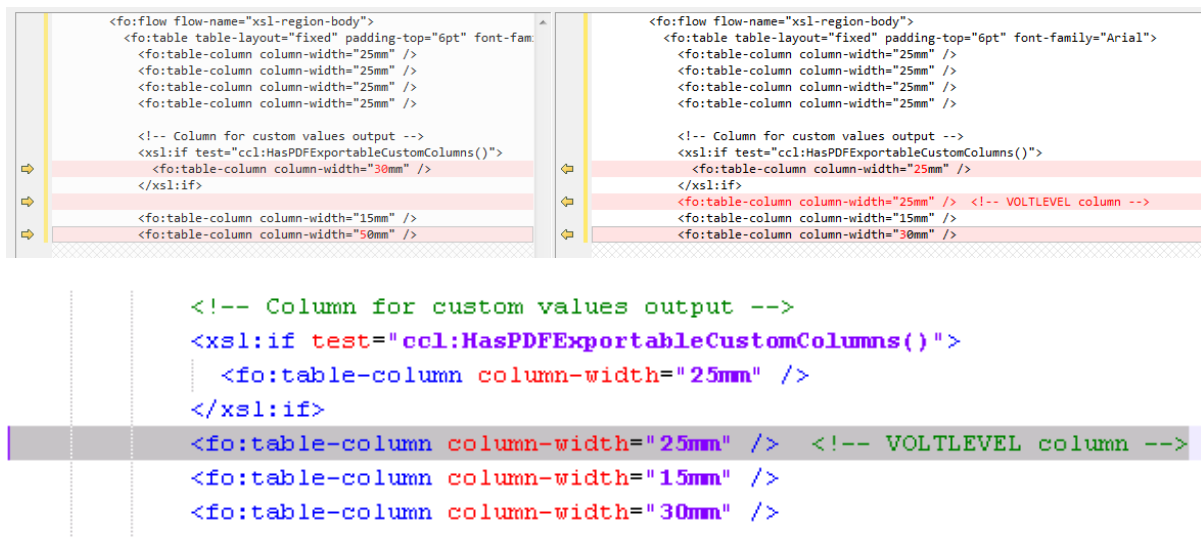
3. Create and save a switching order with the new custom column.

☒	S...	N..	A..	I	C	Device	Action	VOLTAGE LEVEL	Tag Typ
☐	1					5257-5	OPEN	13.200000 kV LL / 7.621024 kV LG	
☒	2					51621	OPEN		

- Open the saved switching order XML file and navigate to the new column field.



- In the PDF export template file (swo2pdf.xml), add a new column in the table definition and modify the widths of the other columns to fit on a page.



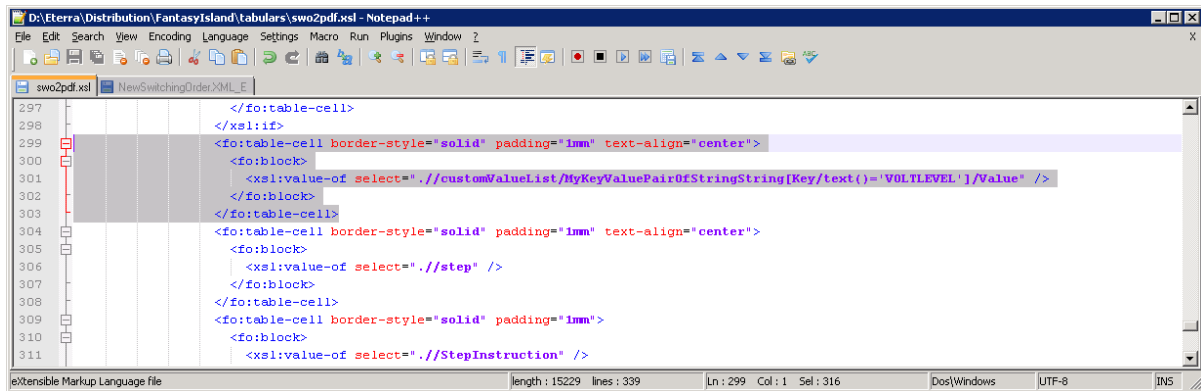
- Add the name for the "VOLTAGE LEVEL" column.

```

<fo:table-cell padding="1mm" border-style="solid">
  <fo:block>ISSUED BY</fo:block>
</fo:table-cell>
<fo:table-cell padding="1mm" border-style="solid">
  <fo:block>VOLTAGE LEVEL</fo:block>
</fo:table-cell>

```

7. Add the value for the "VOLTAGE LEVEL" column.



Set the <xsl:value-of> element for the column as follows:

```

<xsl:value-of
select="//customValueList/MyKeyValuePairOfStringString[Key/text()='VOLTLEVEL']/Value" />

```

Where:

- **MyKeyValuePairOfStringString:** The name of the section that defines the new column in the switching order XML file.
- **Name:** The key of the custom column.

8. Save your changes and restart the client.

9. Export a switching order document as a PDF and verify the new field.

SWITCHING ORDER

DATE: Jan 21, 2017

SUBSTATION: RAVEN

NAME: NewSwitchingOrder

Header Custom Field: NewValue

REMARKS: Description



TIME ISSUED	TIME COMPLETE	ISSUED TO	ISSUED BY	VOLTAGE LEVEL	STEP	SWITCHING PREPARED BY: DMSALL APPROVED BY: DMSALL APPROVED BY: DMSALL
1/21/2017 11:33 AM		Frank Myers	DMSALL	13.200000 kV LL / 7.621024 kV LG	1	OPEN 5257-5
1/21/2017 10:43 AM		Andy Johnson	DMSALL		2	OPEN 51621